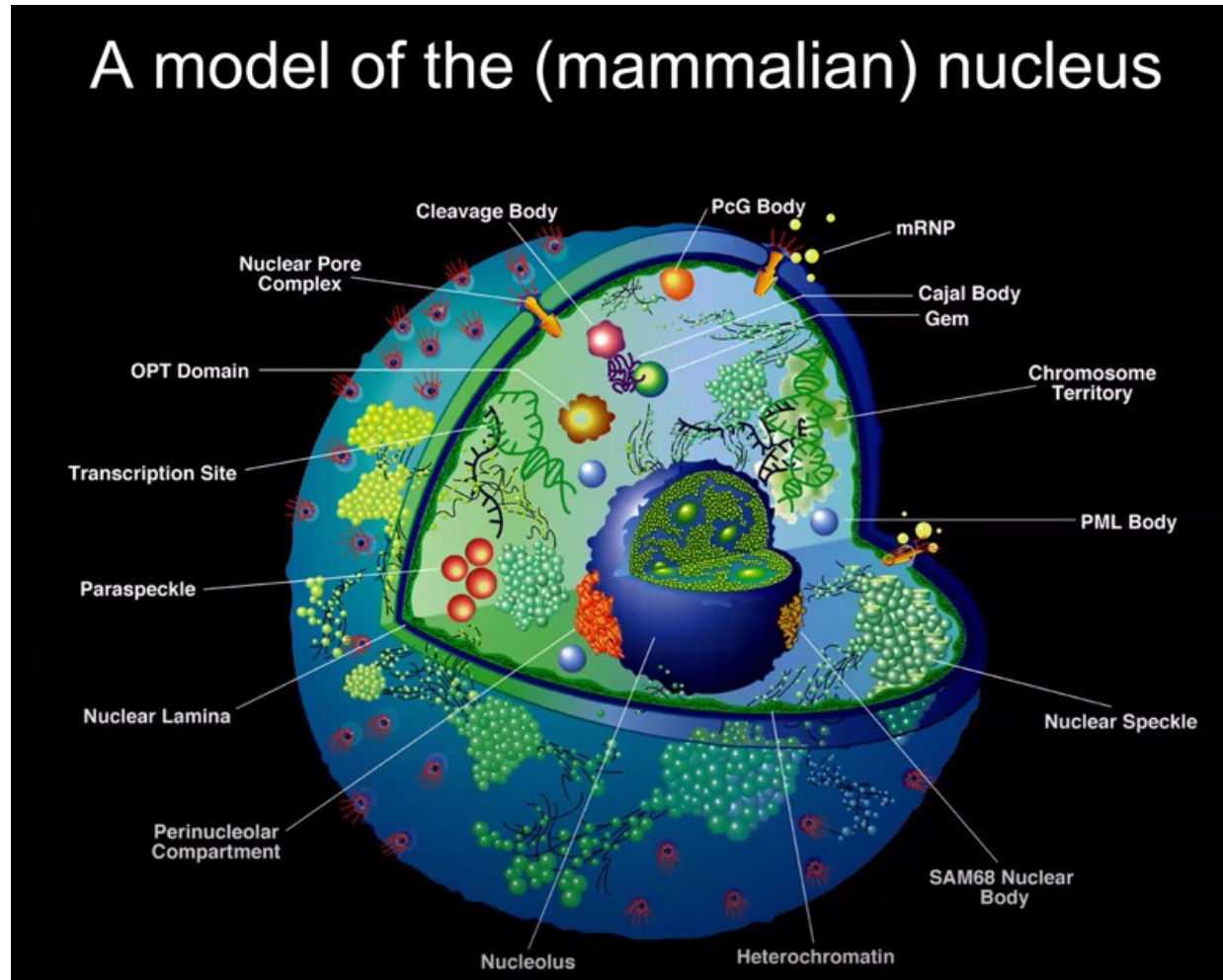


The interaction techniques for 3D Genome

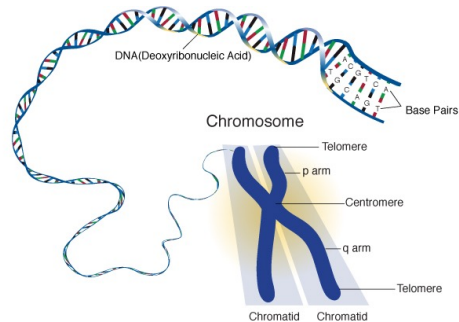
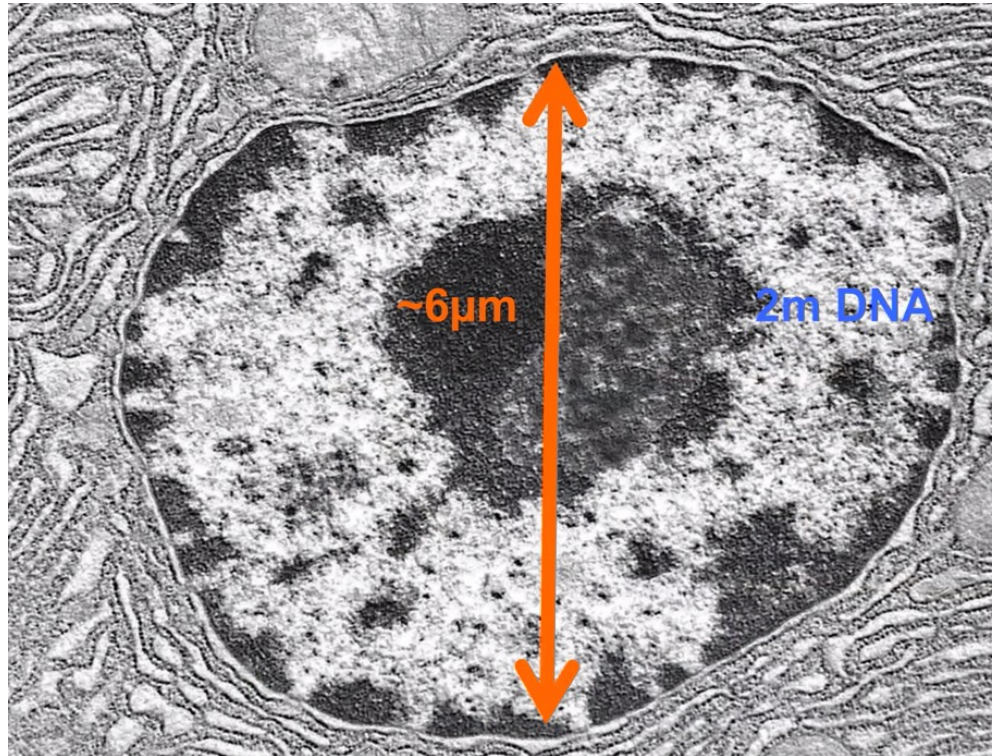
Chapter 13

Background



Spector, d.l. J. Cell Science, 2001

Background



Background

nucleus



diameter: $6\mu\text{m}$
thread length: 2m
thread thickness: 2.5nm

tennis ball



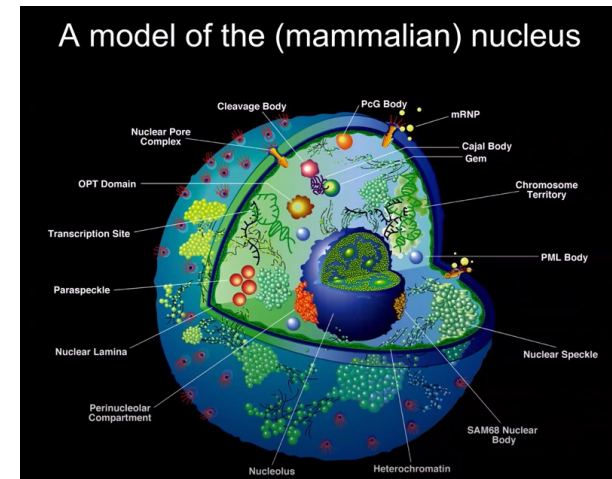
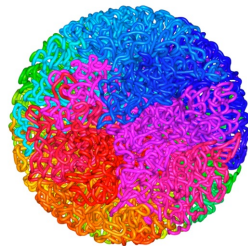
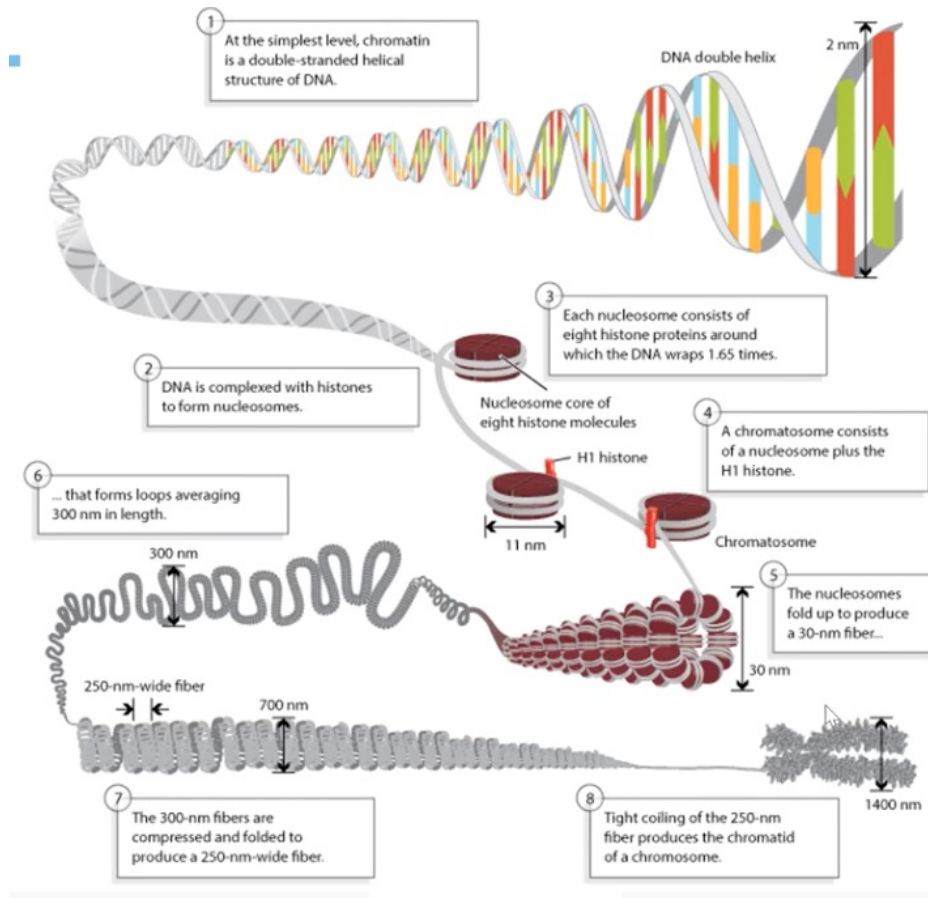
6.7cm
 $\sim 20\text{km}$ (8 x MIT - Harvard)
 $\sim 20\mu\text{m}$

giga-inflatable



" 6m "
 $2,000\text{km}$ NY-Miami
 2mm

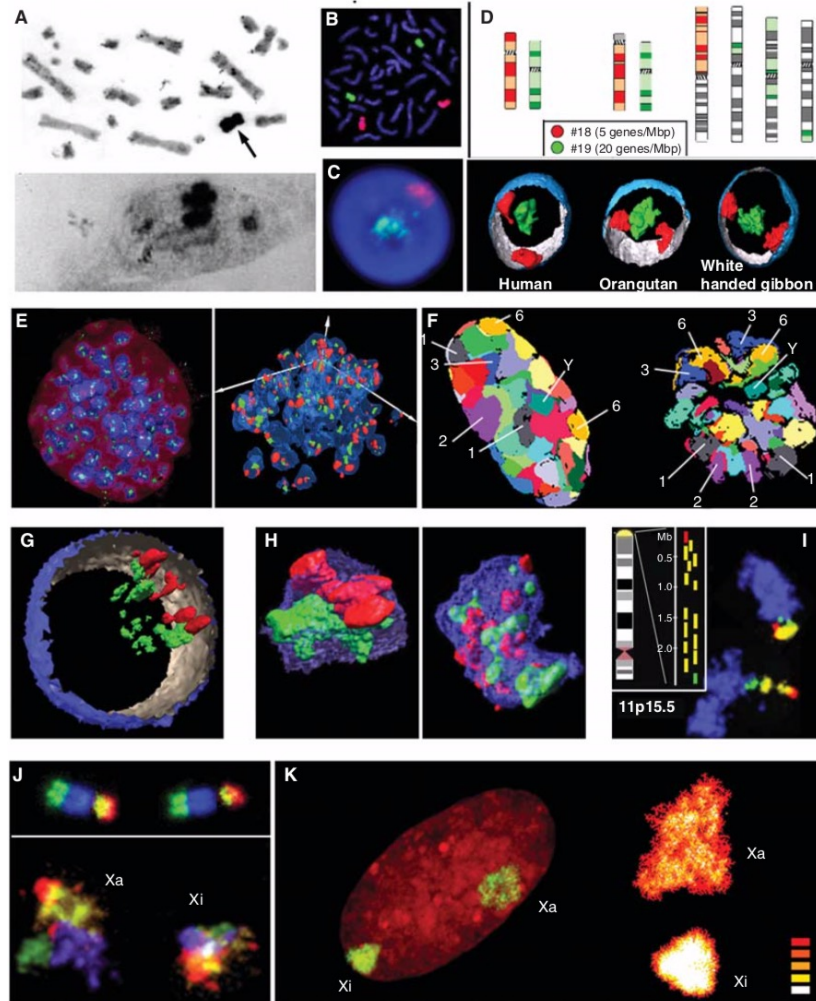
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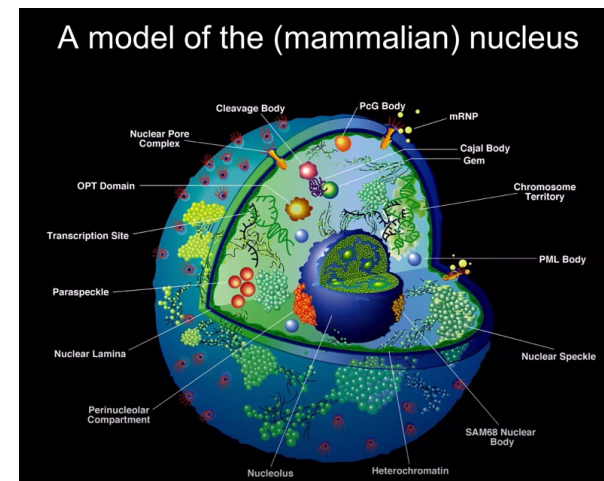
Spector, d.I. J. Cell Science, 2001



Background



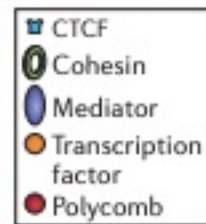
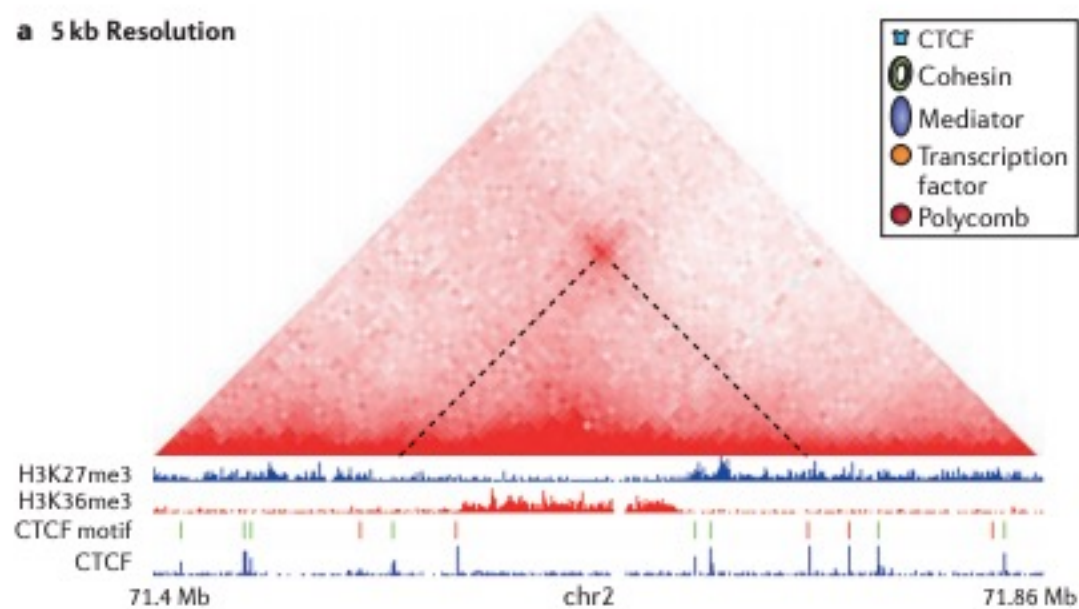
T. Cremer and C. Cremer, Cold Spring Harb Perspect Biol 2010



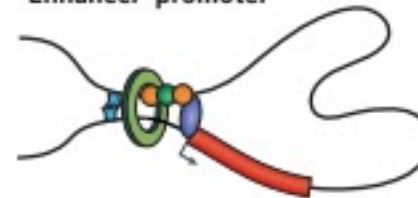
Spector, d.l. J. Cell Science, 2001

Background

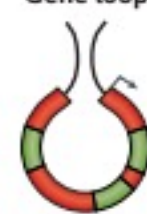
a 5 kb Resolution



Enhancer-promoter



Gene loop



Architectural loop



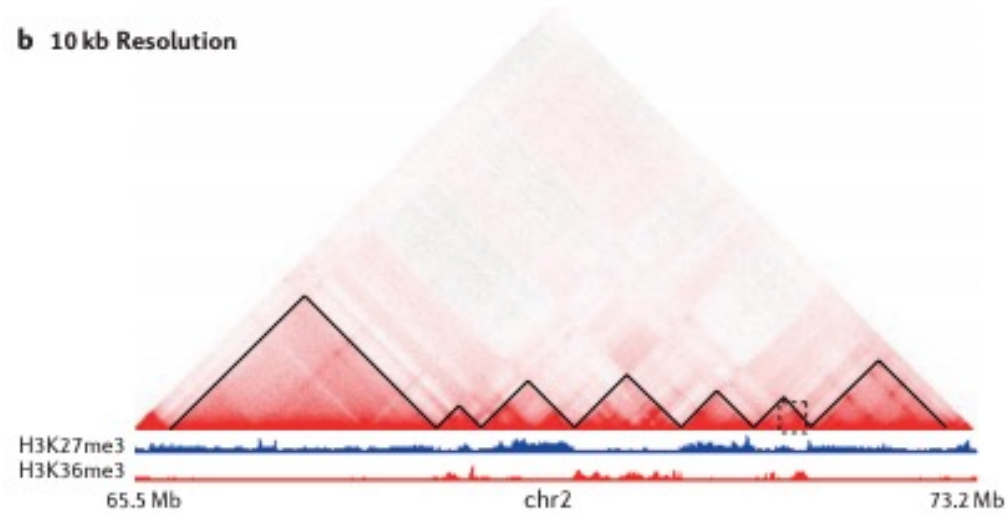
Polycomb-mediated



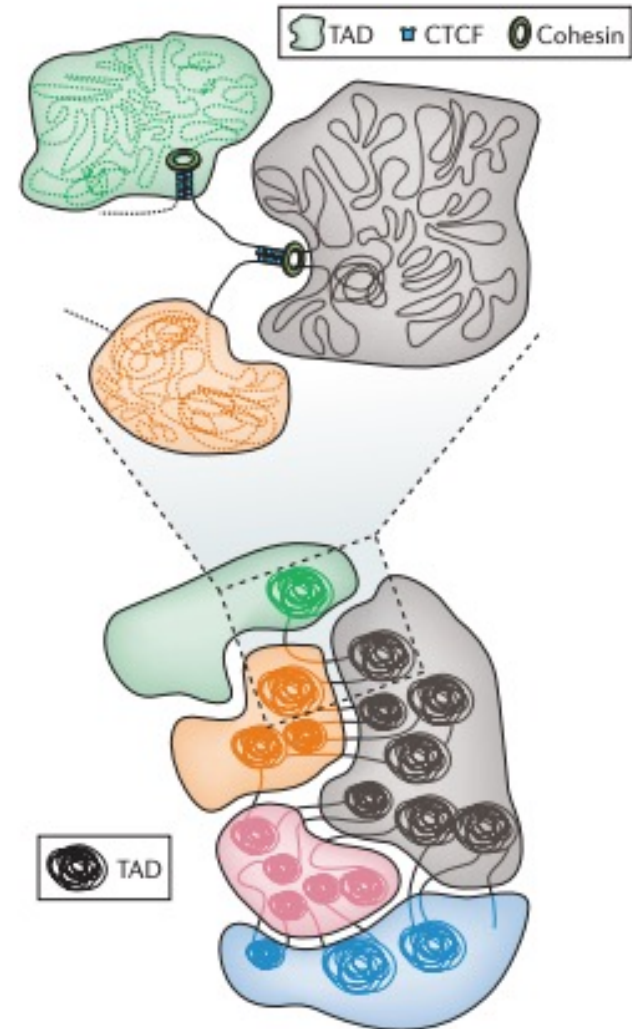
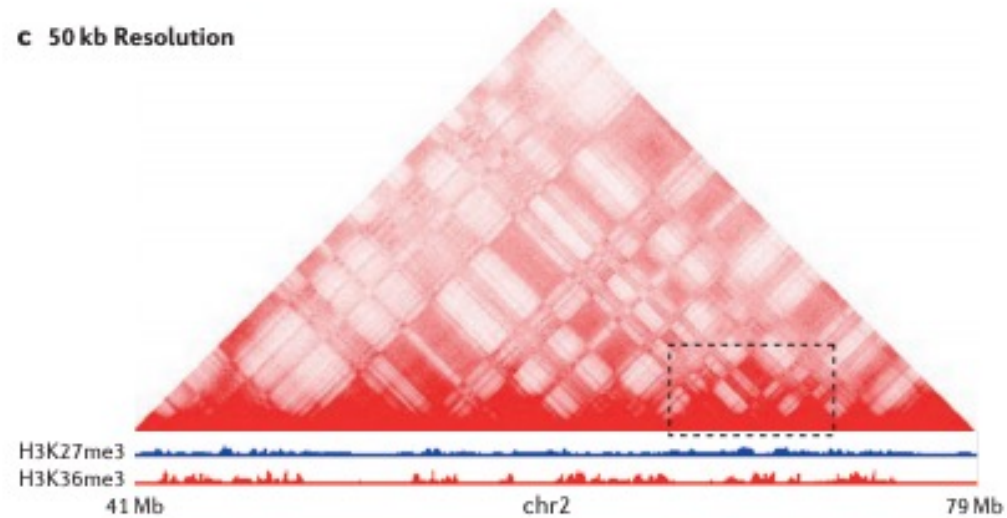
Boyan Bonev and Giacomo Cavalli, Nature Reviews Genetics, 2016

Background

b 10 kb Resolution

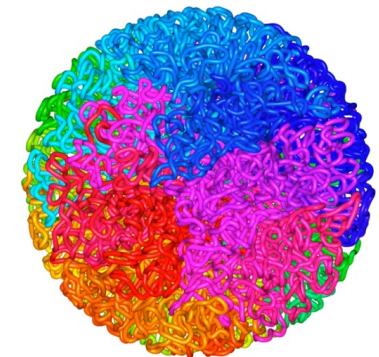
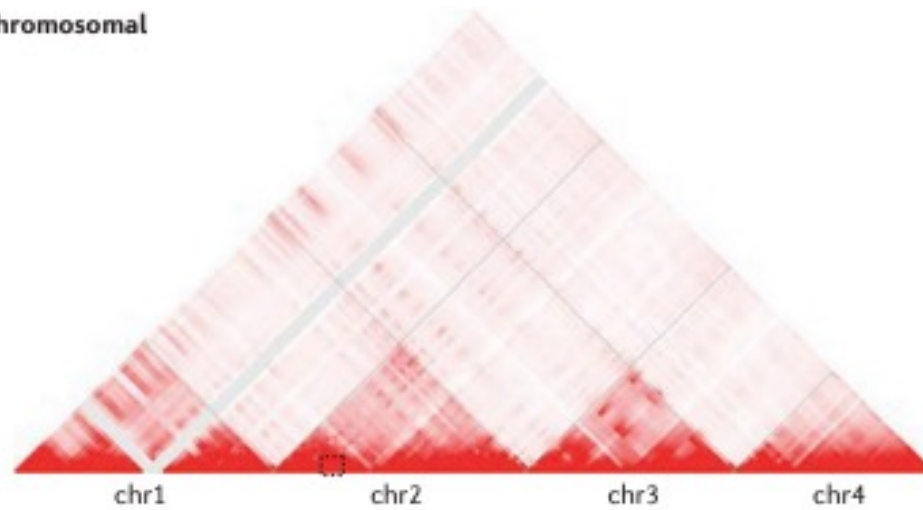


c 50 kb Resolution

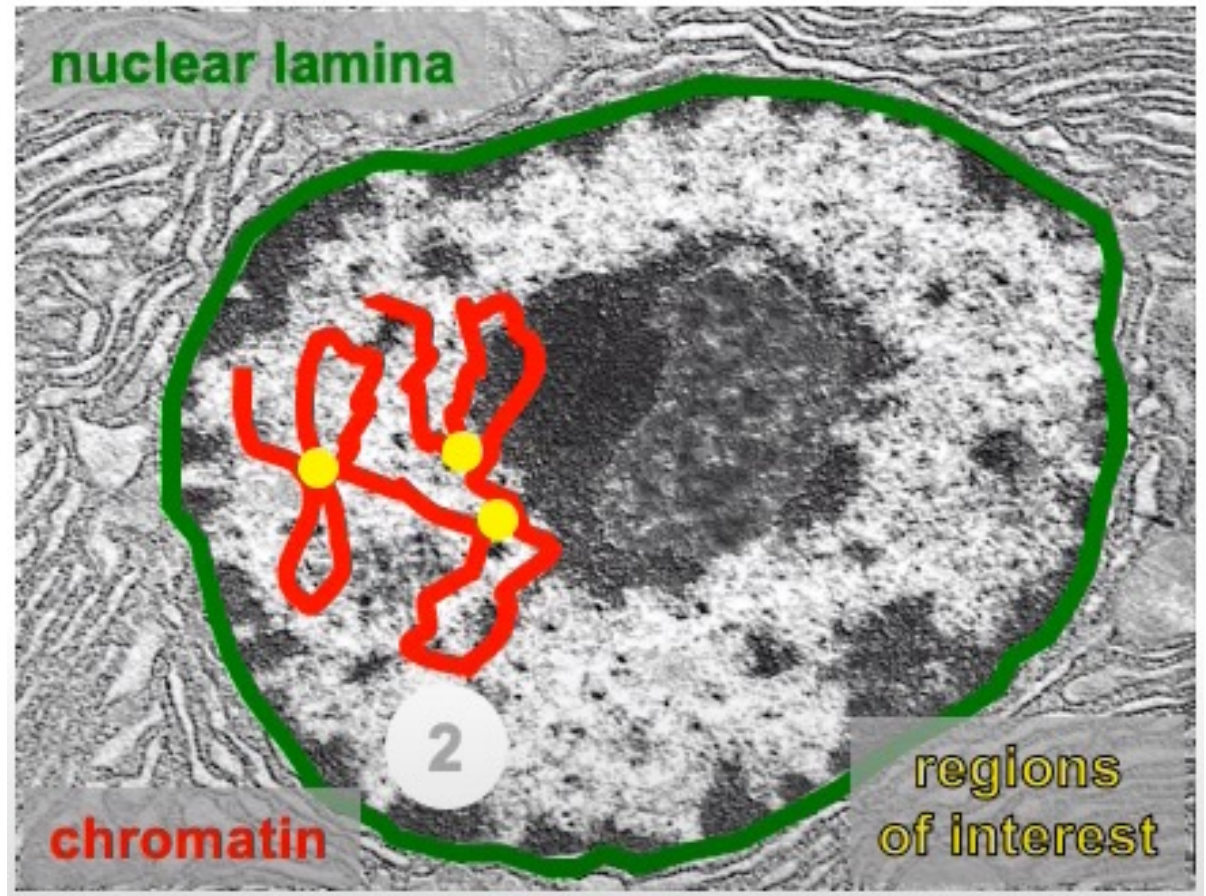
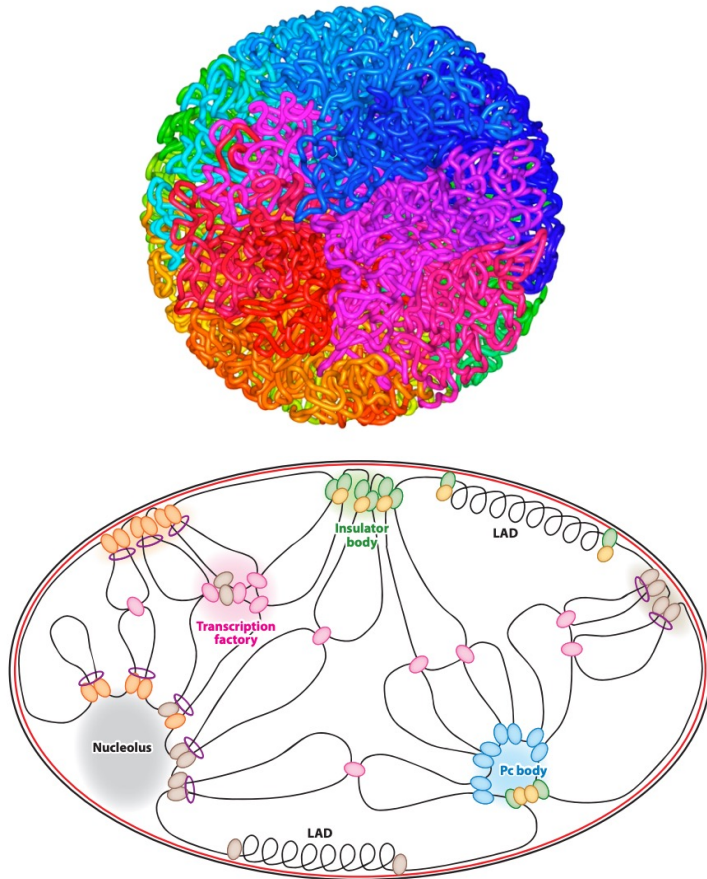


Background

d Interchromosomal



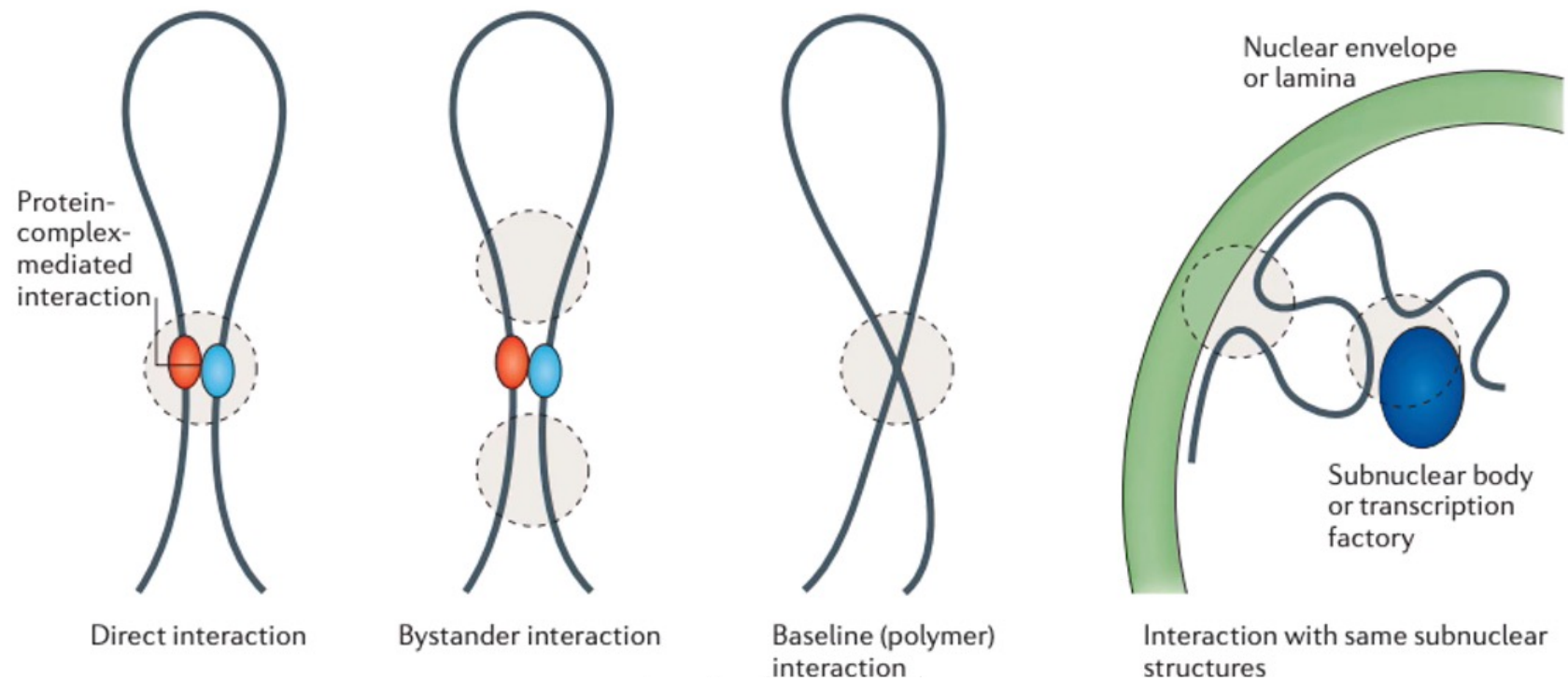
Background



Van Bortle K, Corces VG, Nature Reviews Genetics, 2012

- Measure **interactions** between genomic loci.
- Trick: “**cross-link**” DNA so that regions in **close proximity** are “stuck” together. Then, identify these regions.

Background



Nature Reviews Genetics, 2013

Main Techniques

3C

4C

Hi-C

Main contents

Principle

Experimental design

Procedure

Results

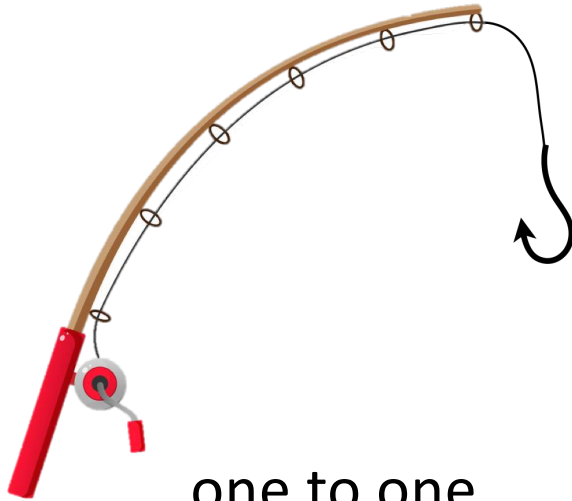
Visualization

3C Chromosome Conformation Capture

Part one

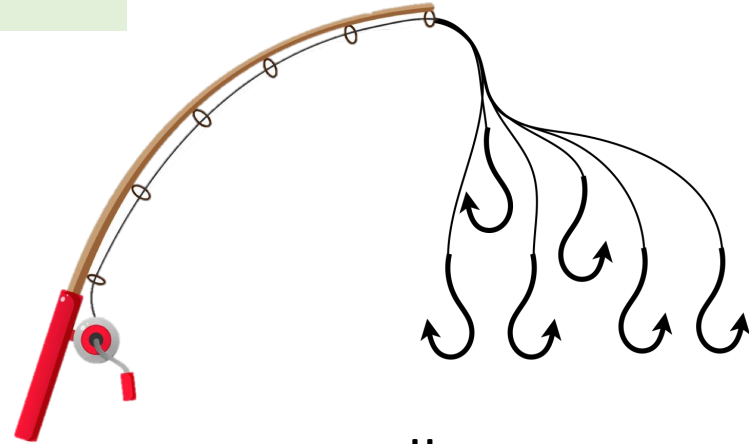
14

3C



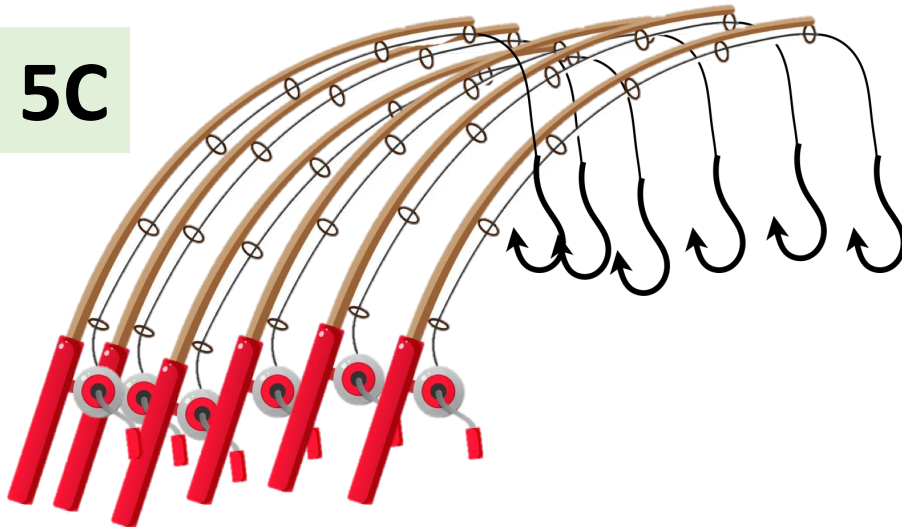
one to one

4C



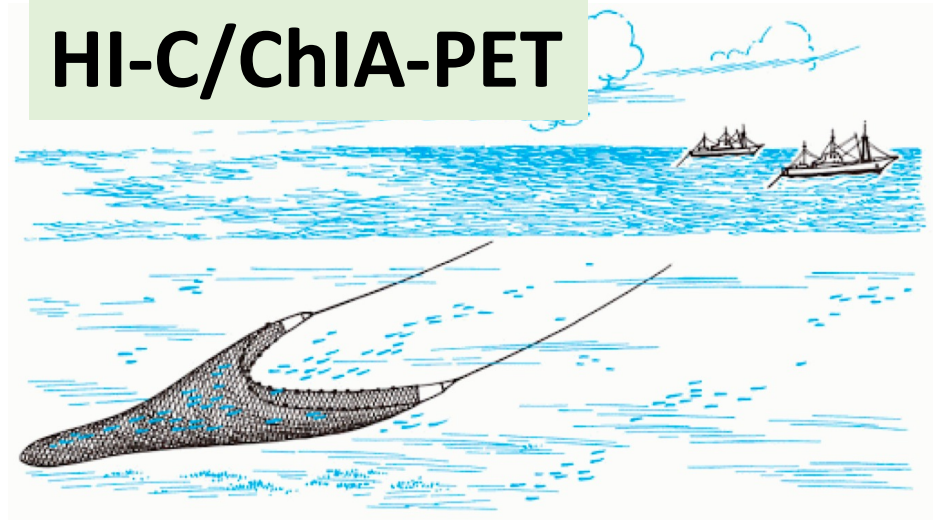
one to all

5C



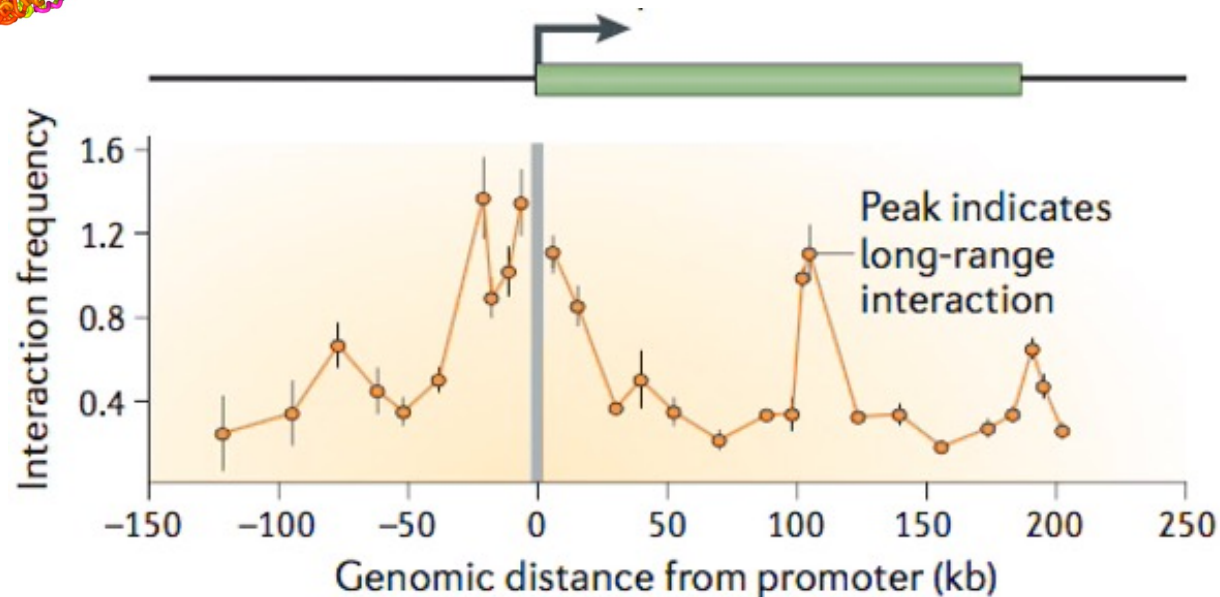
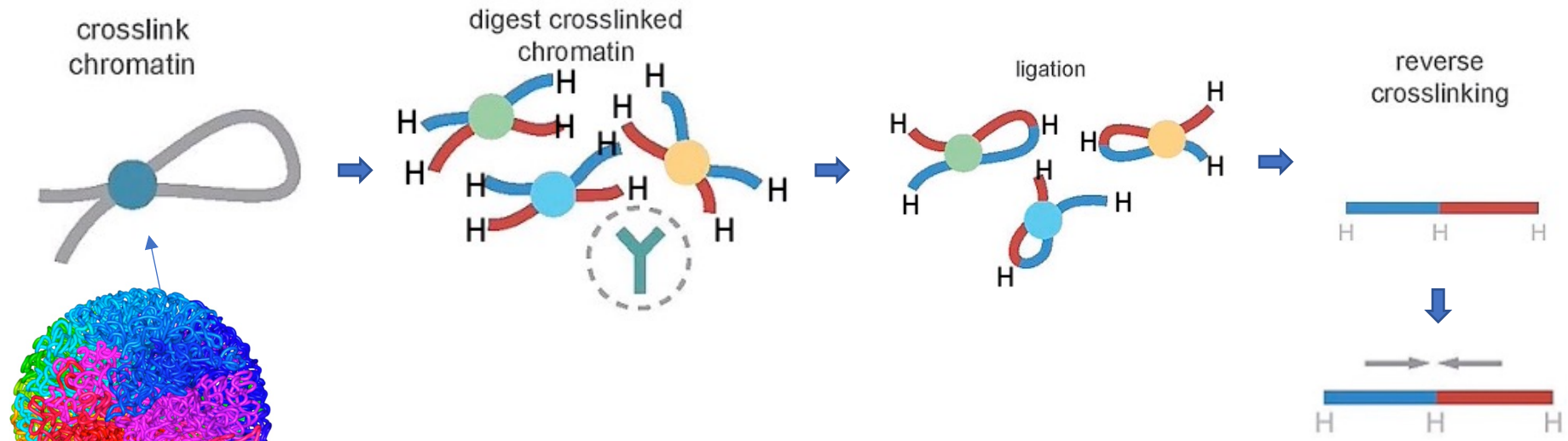
many to many

HI-C/ChIA-PET



all to all

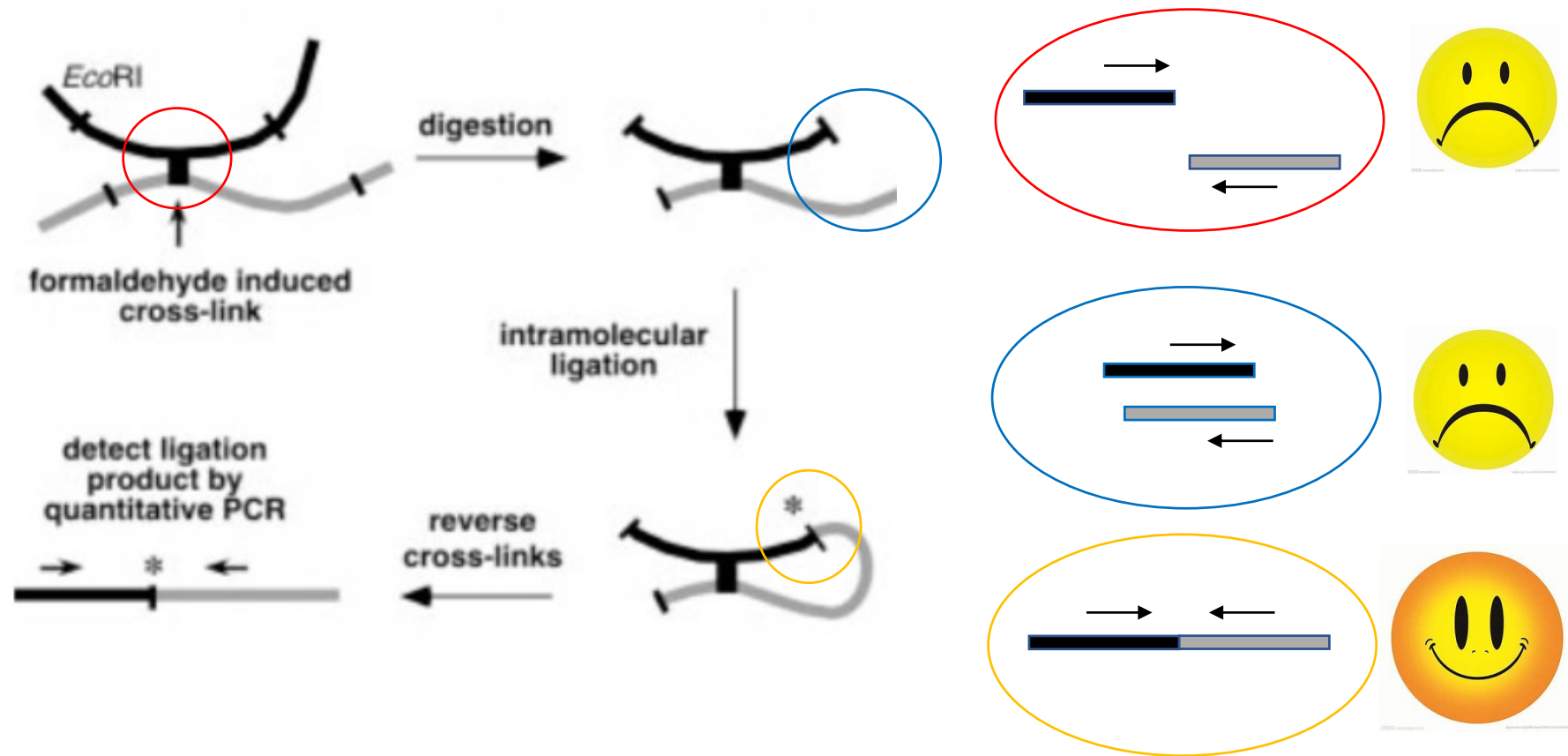
Chromosome Conformation Capture (3C)



- Design locus-specific primers near restriction sites for the interacting loci of study

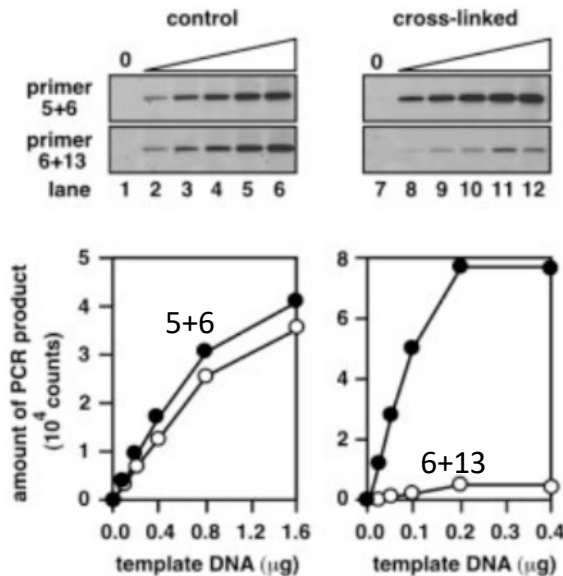
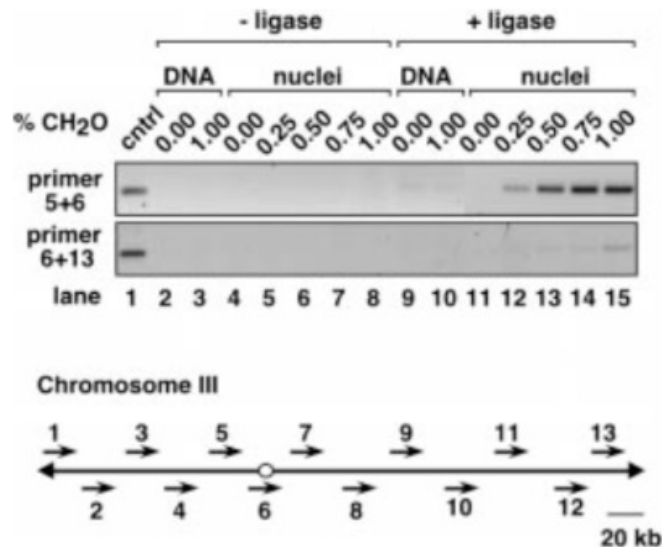
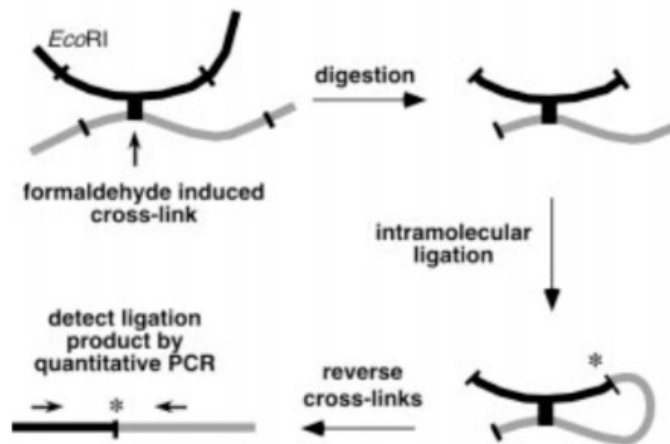
Principle__

Chromosome Conformation Capture (3C)

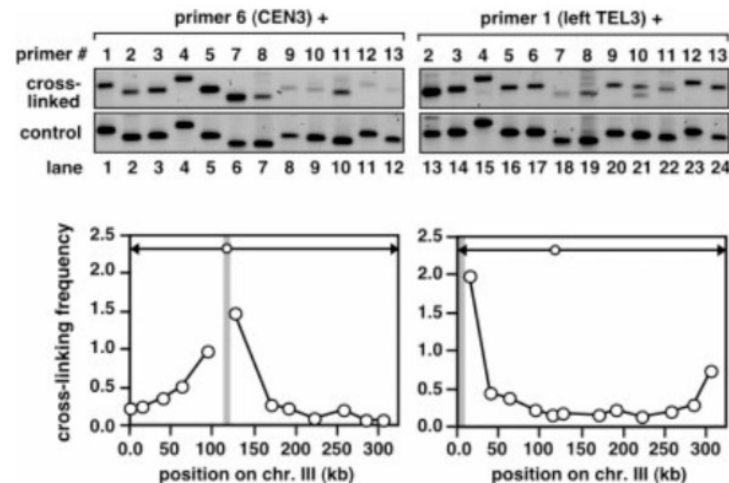
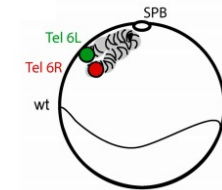


QC&Results

Chromosome Conformation Capture (3C)



➤ control: naked DNA-digestion--ligation

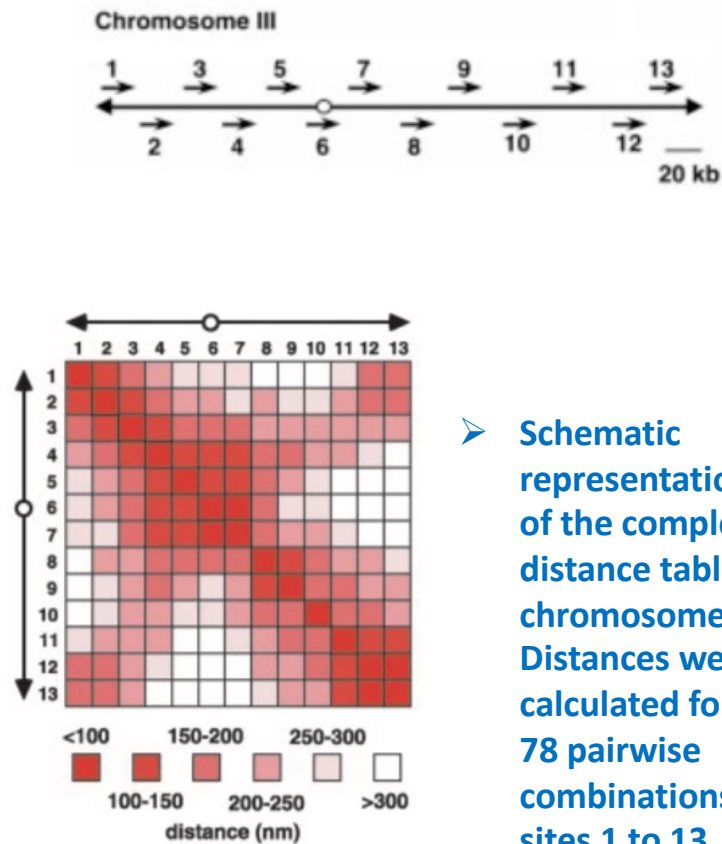


➤ chromosome ends are in close spatial proximity.

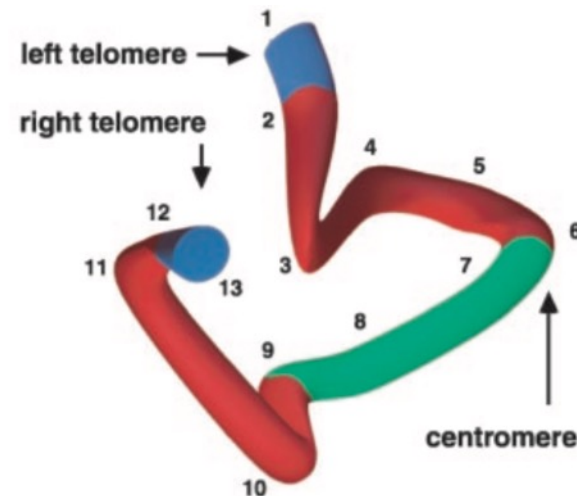
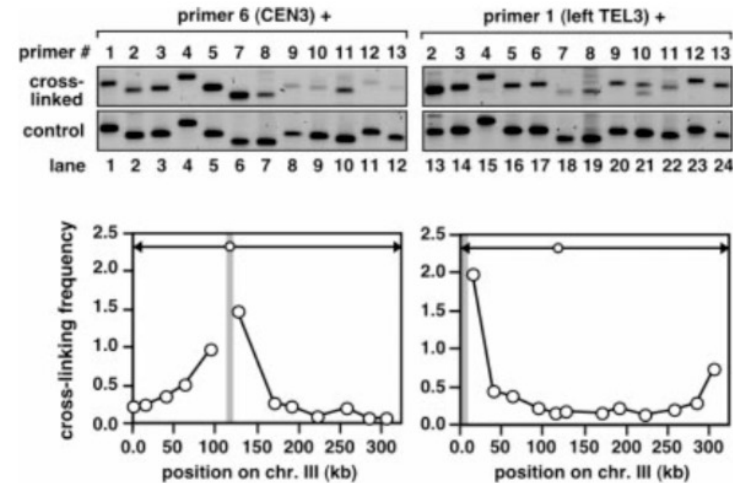
➤ This is expected because yeast telomeres are known to occur in clusters.

Results&Visulization

Chromosome Conformation Capture (3C)

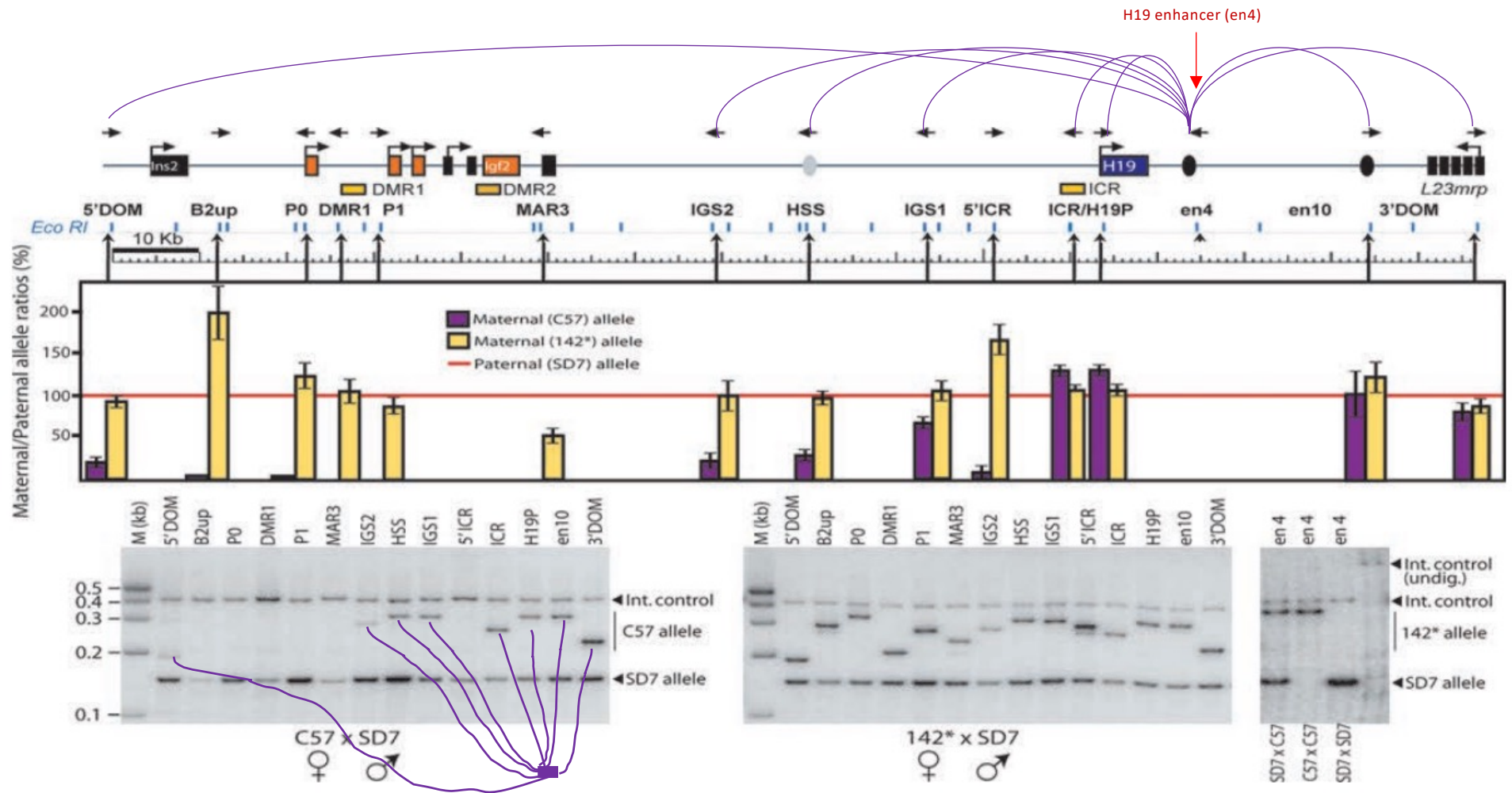


➤ Schematic representation of the complete distance table of chromosome III. Distances were calculated for all 78 pairwise combinations of sites 1 to 13.

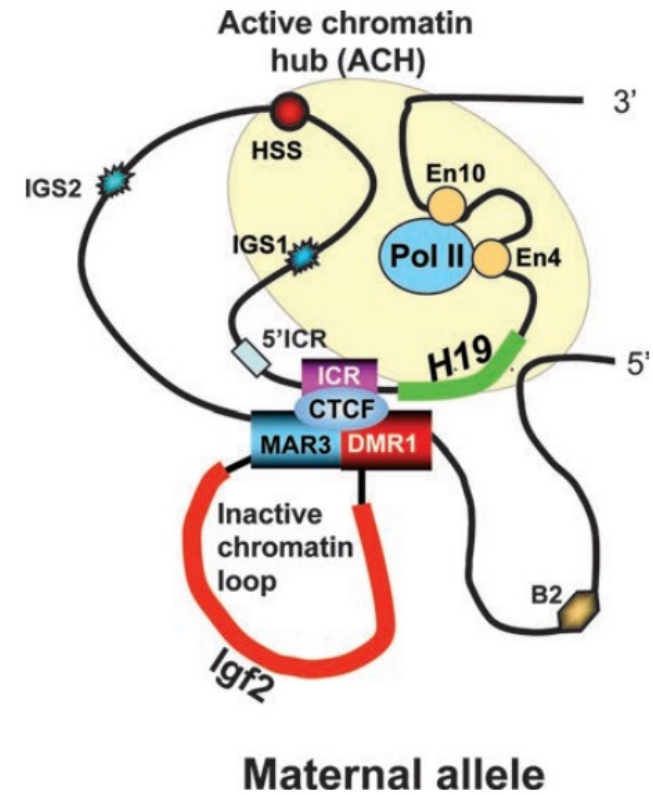
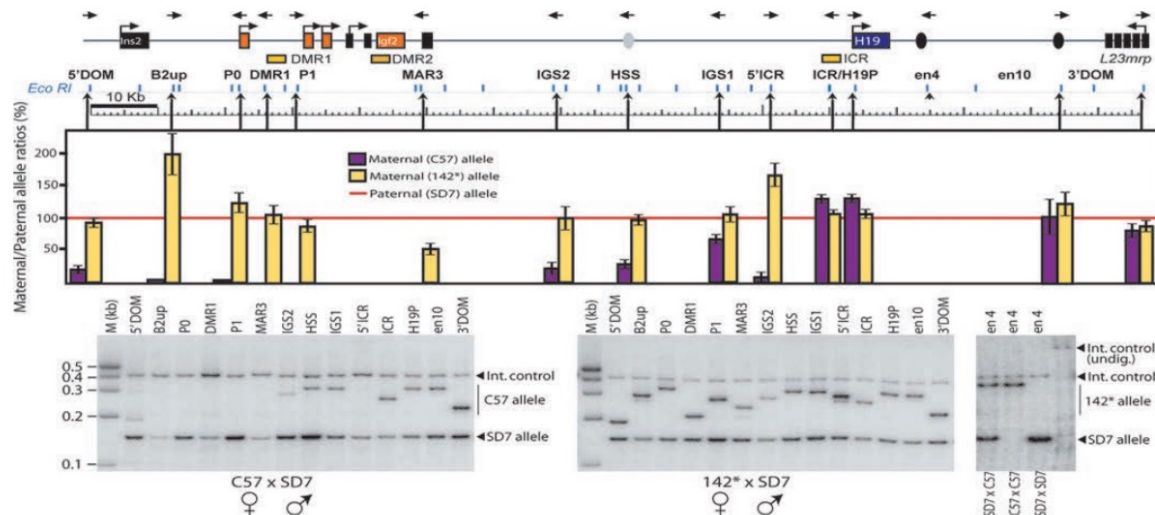


➤ Population-average 3D model of chromosome III

3C application

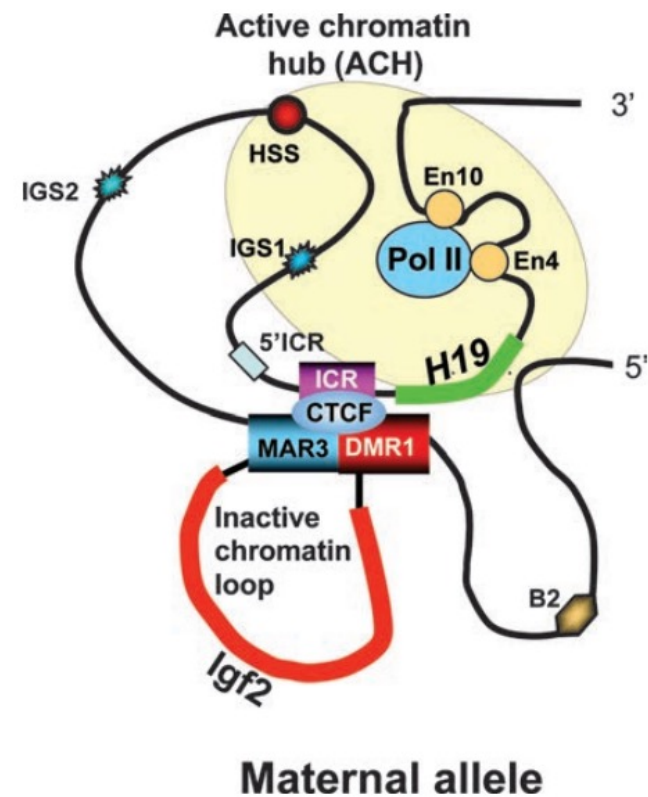
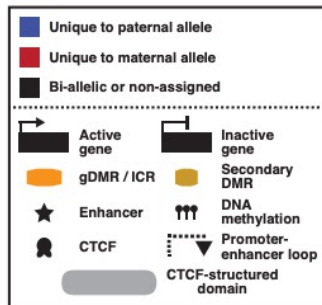
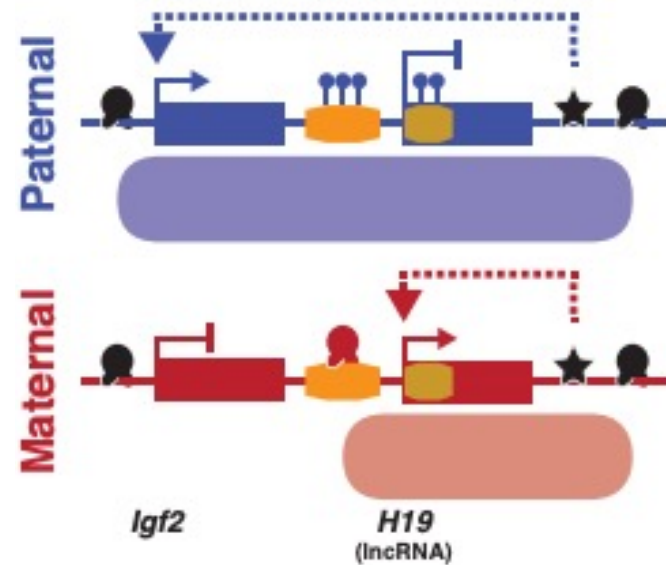


3C application



3C application

Igf2-H19 domain



Sreenivasulu Kurukuti et al., PNAS 2006



4C

Part two

4C



Primary digestion by BglII



Primary ligation



Second digestion by CviQI

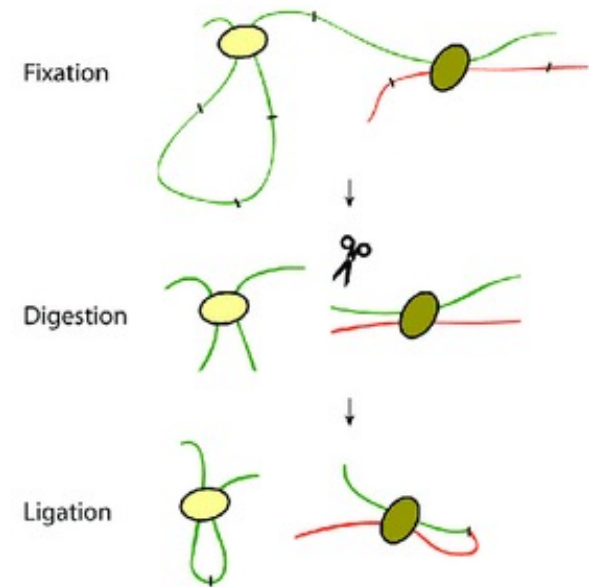
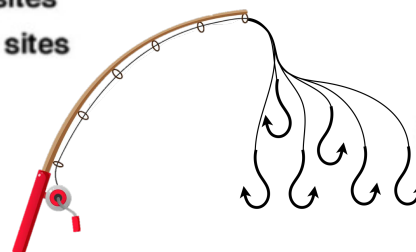


Inverse PCR



Next-generation sequencing

■ BglII sites
■ CviQI sites



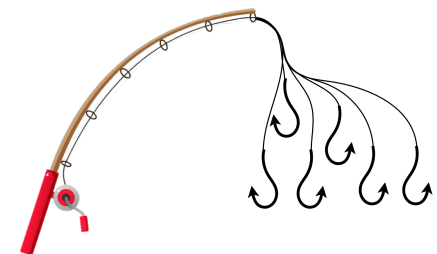
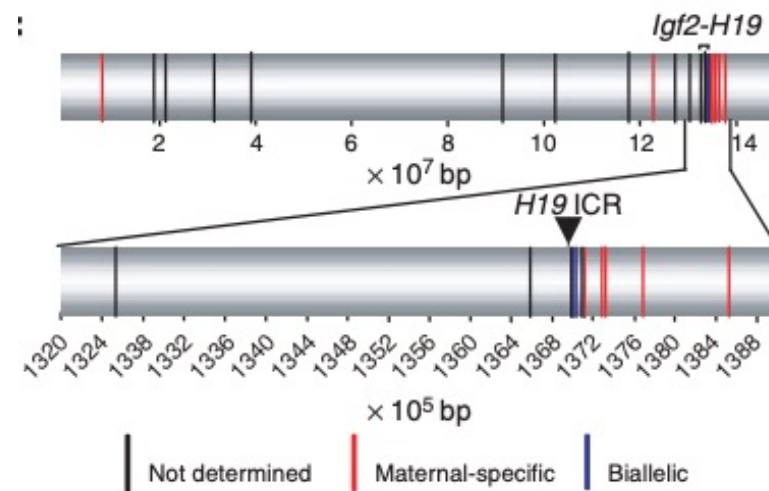
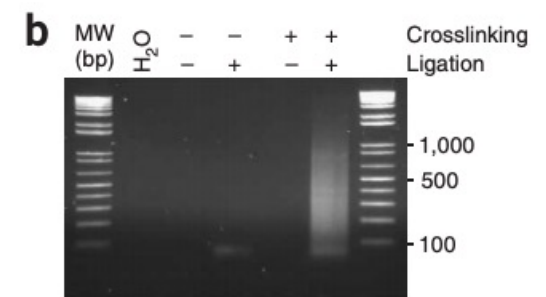
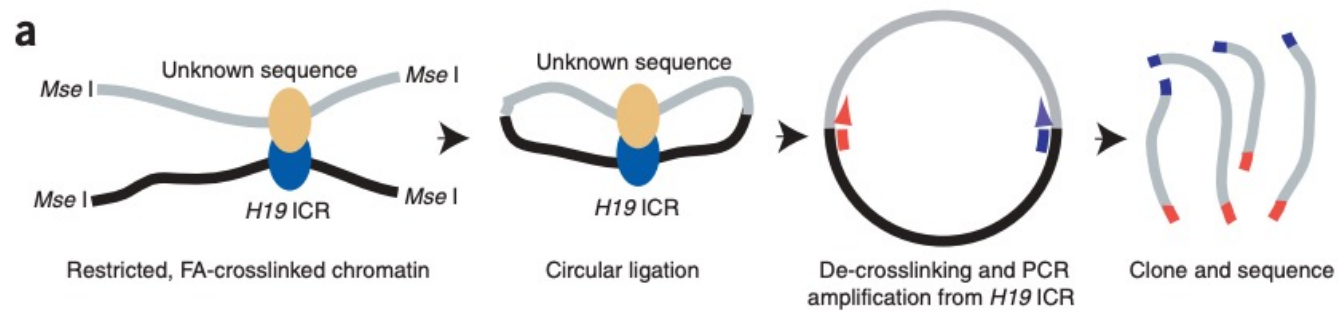
DNA purification

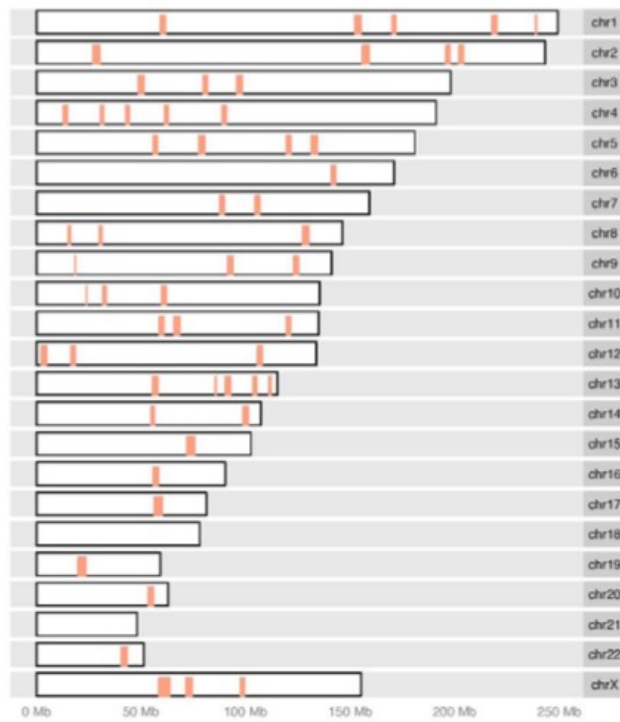
Secondary digestion

Circularization

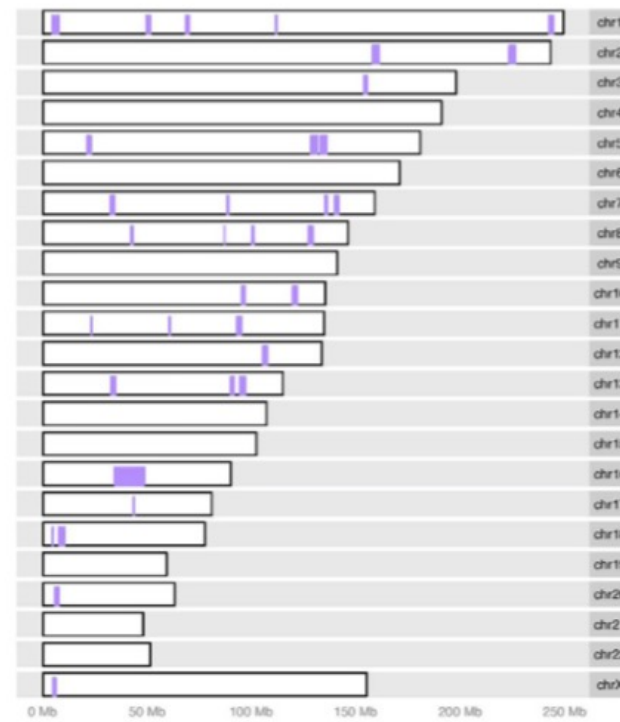
Inverse PCR

Sequencing

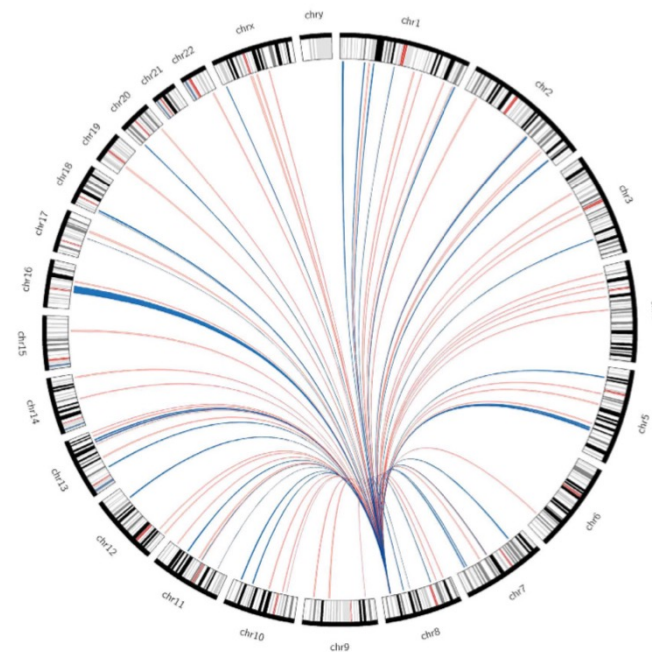




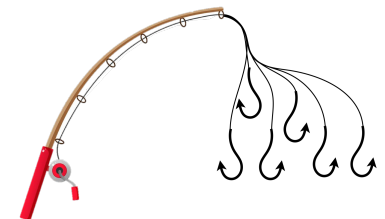
LNCaP



C4-2B



● LNCaP
● C4-2B



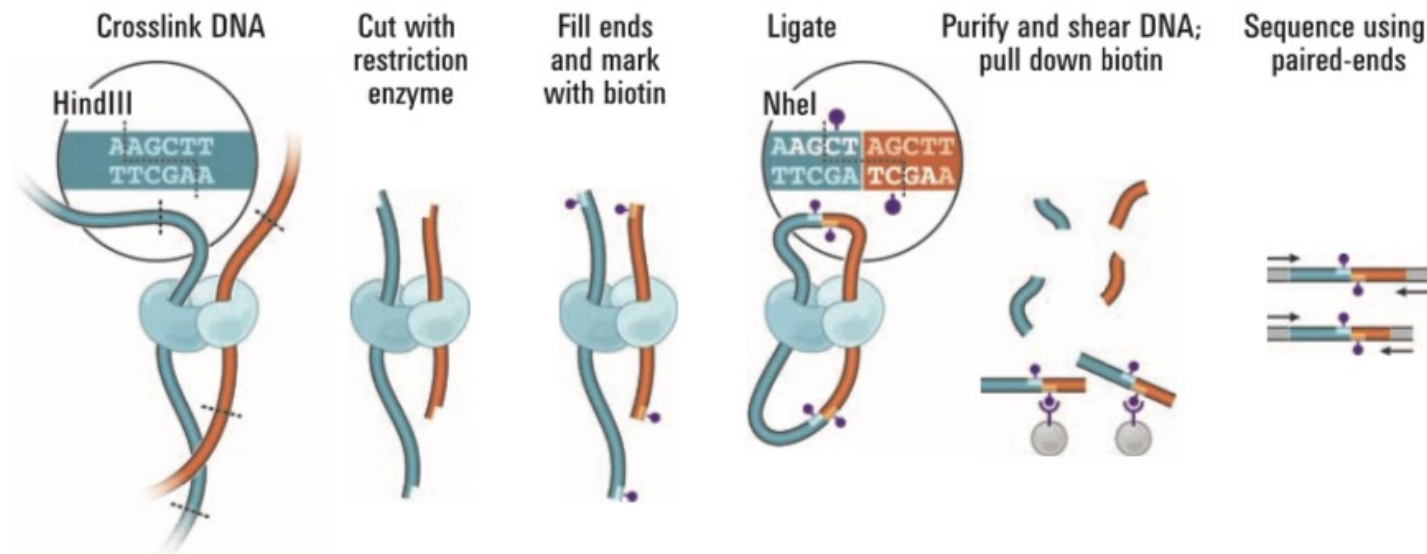
HiC

Part three

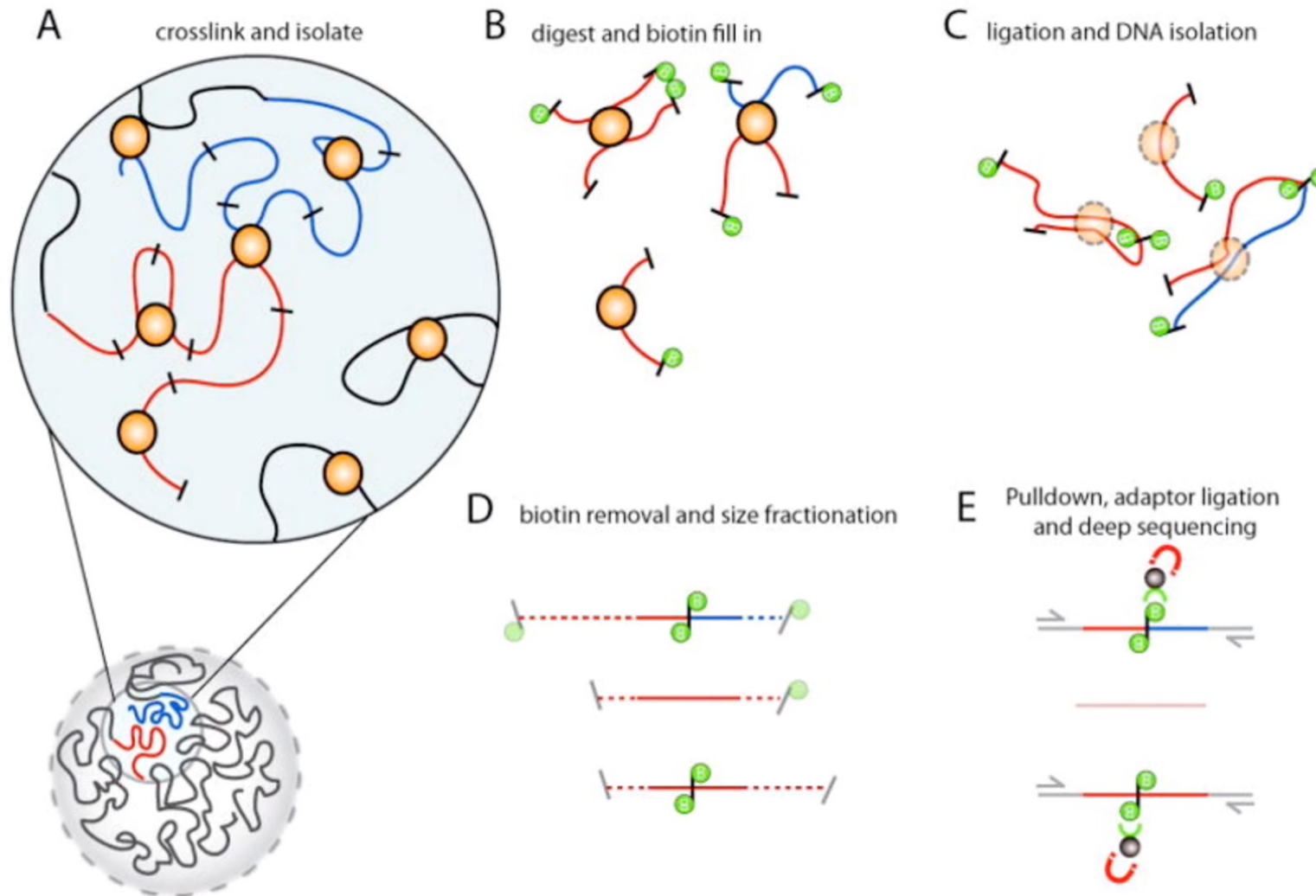
Comprehensive Mapping of Long-Range Interactions Reveals Folding Principles of the Human Genome

Erez Lieberman-Aiden,^{1,2,3,4*} Nynke L. van Berkum,^{5*} Louise Williams,¹ Maxim Imakaev,² Tobias Ragoczy,^{6,7} Agnes Telling,^{6,7} Ido Amit,¹ Bryan R. Lajoie,⁵ Peter J. Sabo,⁸ Michael O. Dorschner,⁸ Richard Sandstrom,⁸ Bradley Bernstein,^{1,9} M. A. Bender,¹⁰ Mark Groudine,^{6,7} Andreas Gnirke,¹ John Stamatoyannopoulos,⁸ Leonid A. Mirny,^{2,11} Eric S. Lander,^{1,12,13†} Job Dekker^{5†}

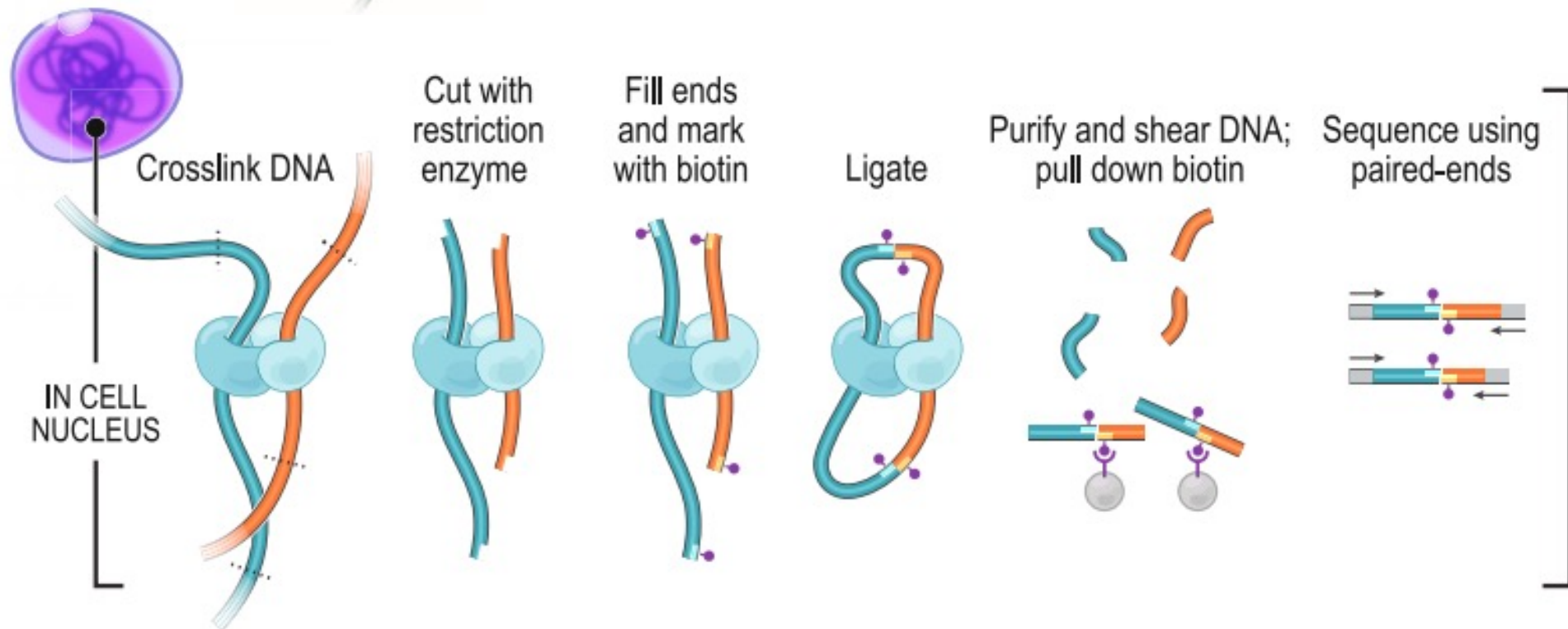
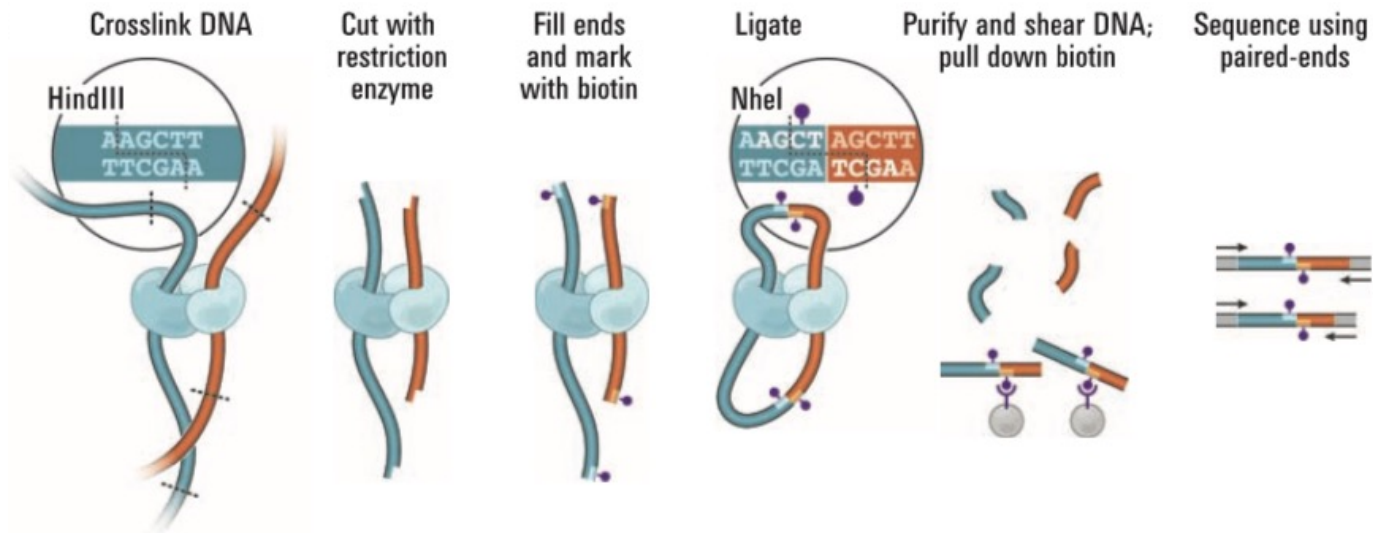
We describe Hi-C, a method that probes the three-dimensional architecture of whole genomes by coupling proximity-based ligation with massively parallel sequencing. We constructed spatial proximity maps of the human genome with **Hi-C** at a resolution of 1 megabase. These maps confirm the presence of chromosome territories and the spatial proximity of small, gene-rich chromosomes. We identified an additional level of genome organization that is characterized by the spatial segregation of open and closed chromatin to form two genome-wide compartments. At the megabase scale, the chromatin conformation is consistent with a fractal globule, a knot-free, polymer conformation that enables maximally dense packing while preserving the ability to easily fold and unfold any genomic locus. The fractal globule is distinct from the more commonly used globular equilibrium model. Our results demonstrate the power of Hi-C to map the dynamic conformations of whole genomes.



HiC procedure



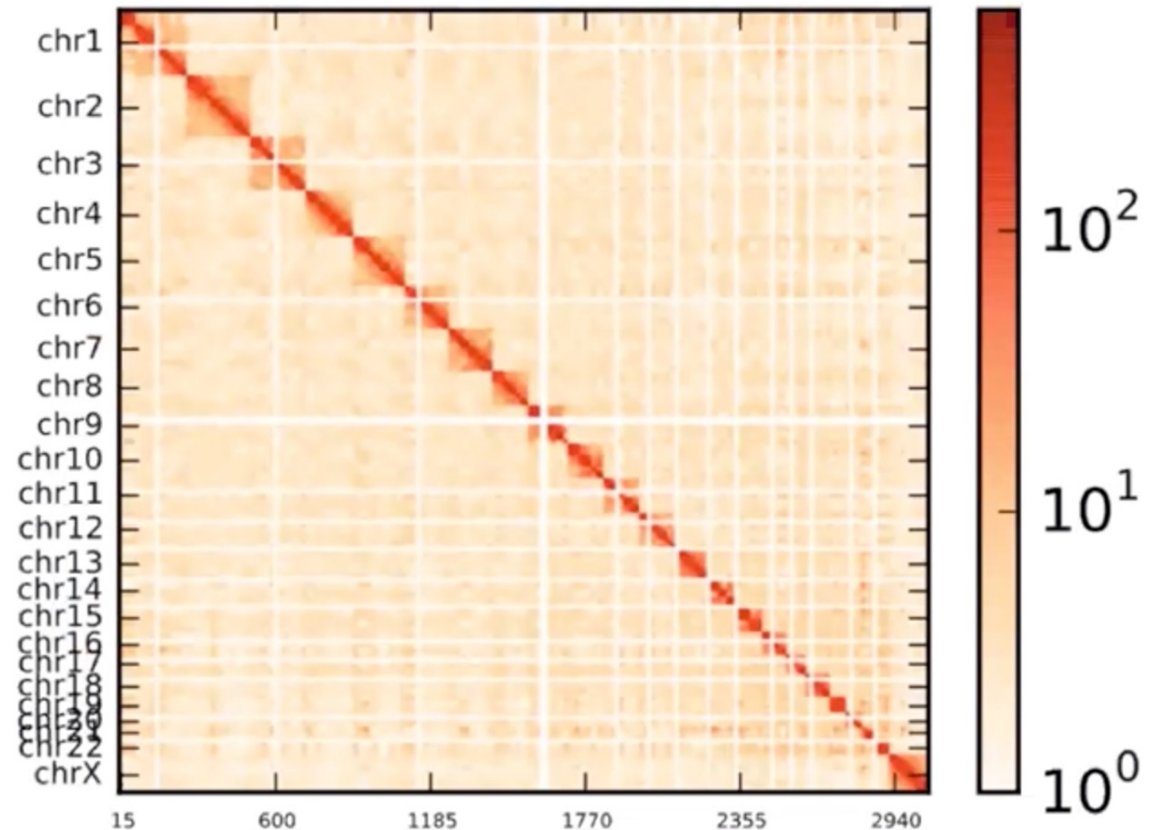
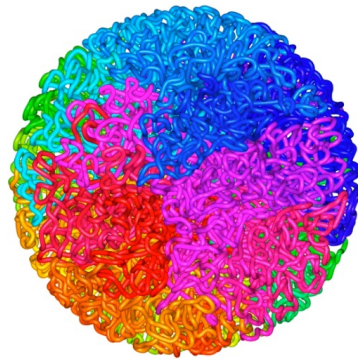
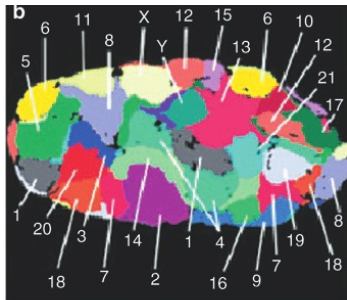
HiC

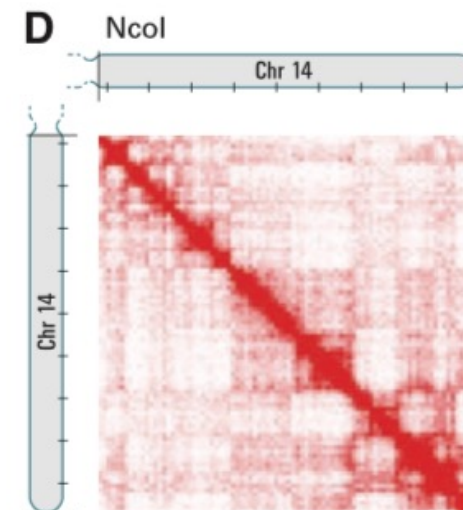
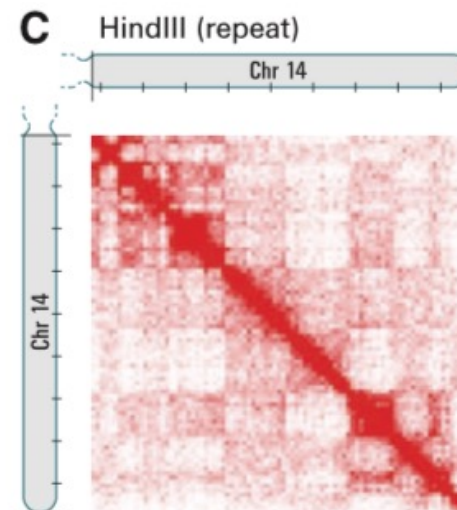
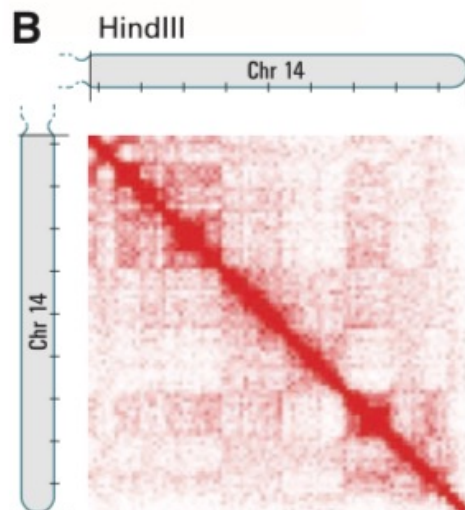
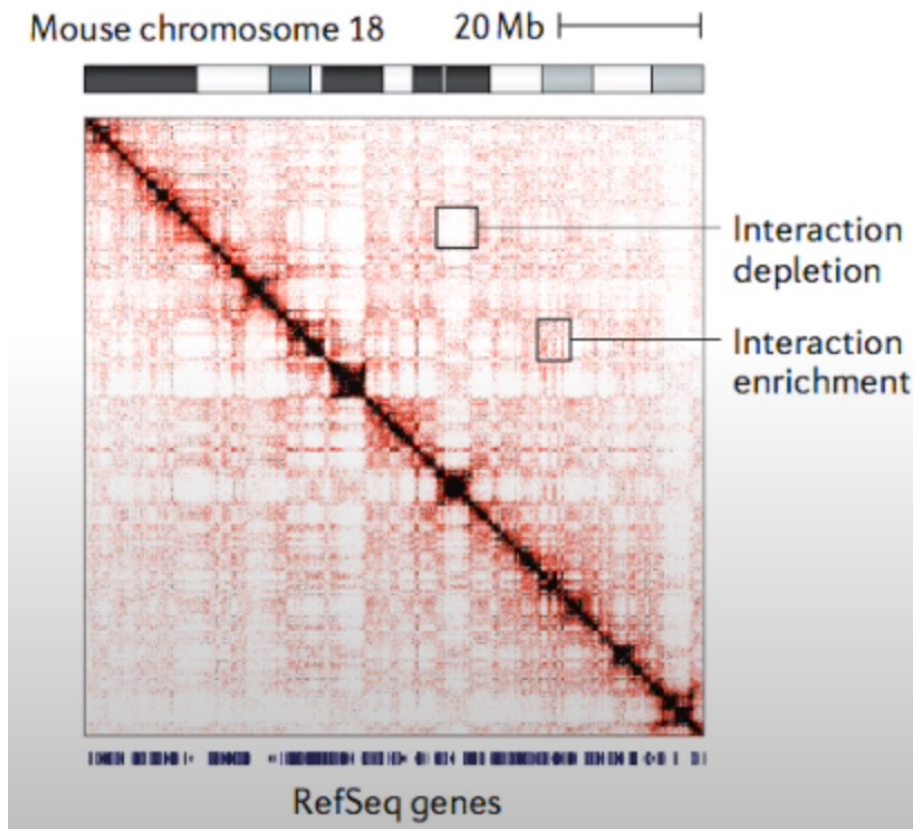


In situ HiC

HiC data visualization_Juicebox

- More interactions within chromosomes
- More interactions in shorter distances

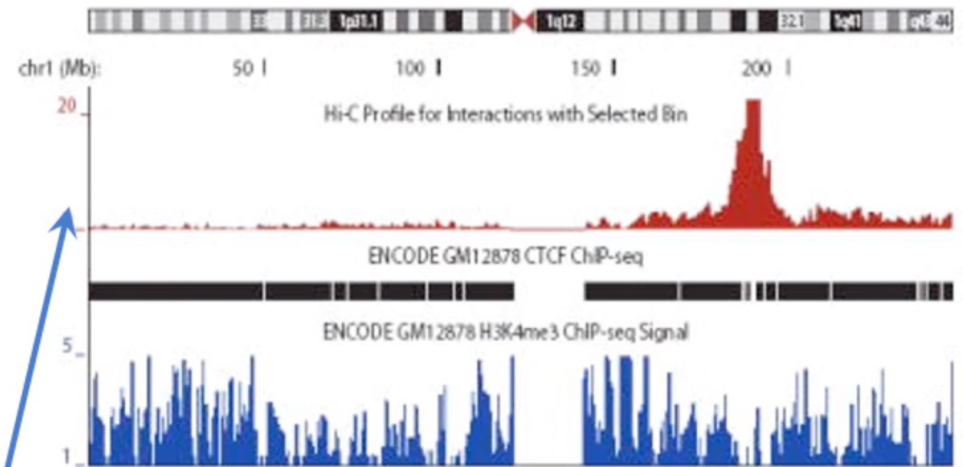
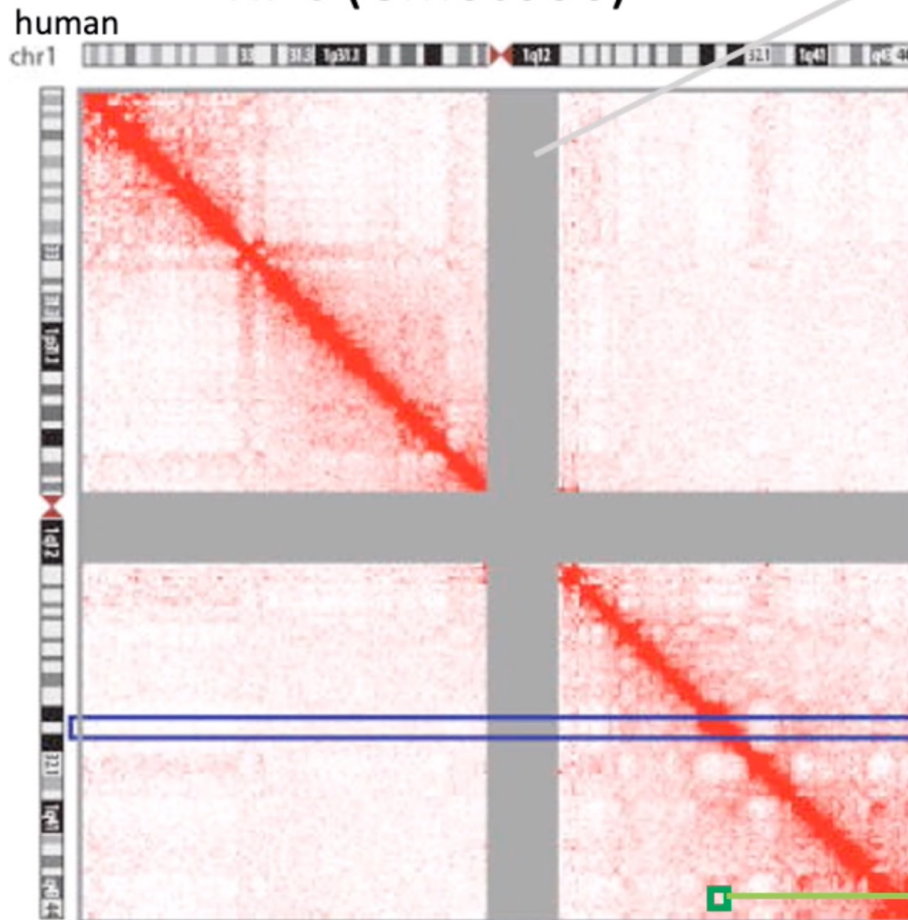




HiC data visualization Juicebox

Hi-C vs 3C vs 4C
Hi-C (GM06990)

Grey area: non-mappable regions of the genome



4C: like extracting one row out of the Hi-C diagram
(and possibly increasing sequencing depth)

3C: like extracting a small box out of the Hi-C diagram
(and increasing sequencing depth/accuracy)

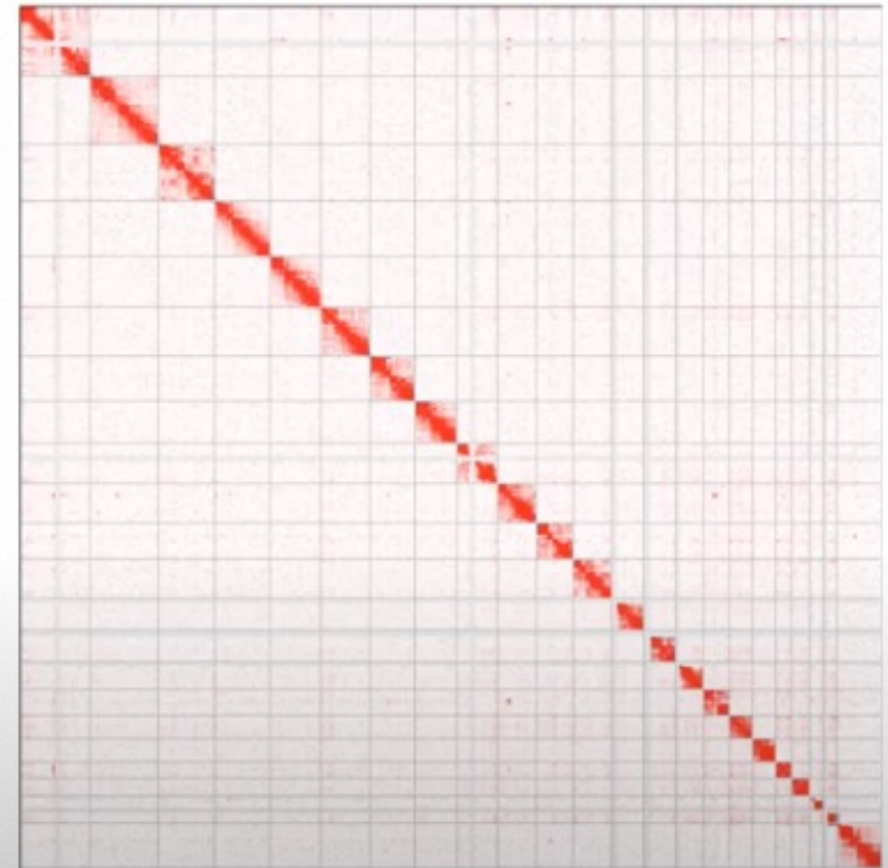
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3874846/>

HiC data visualization Juicebox

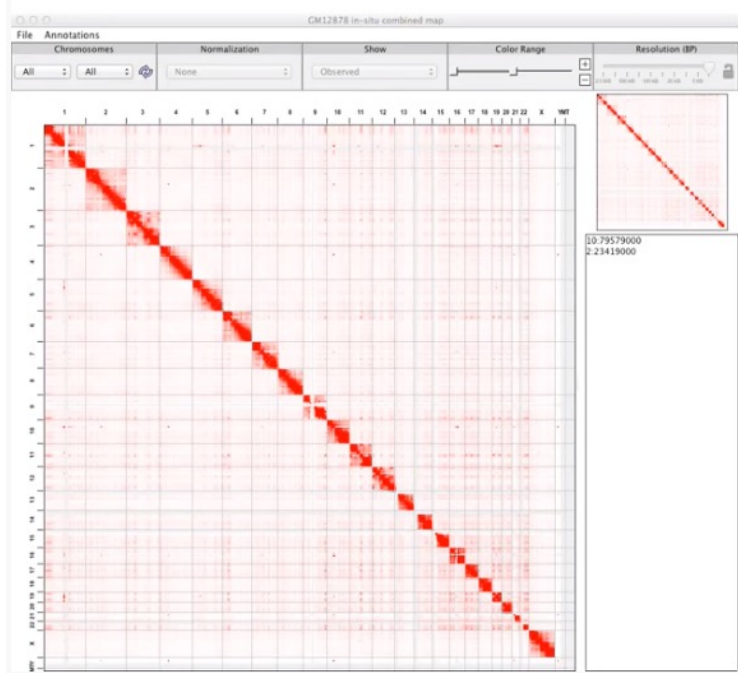
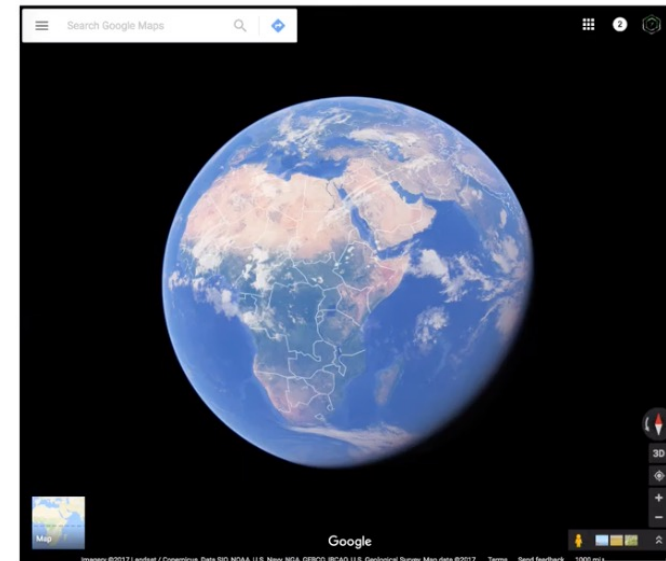
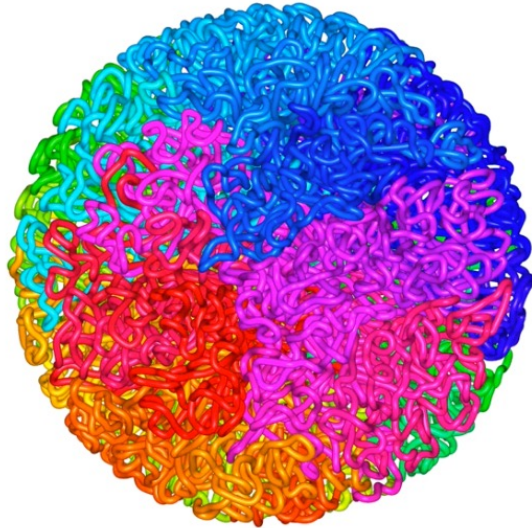
Hi-C

GENERATES GENOME-
WIDE CONTACT MAPS

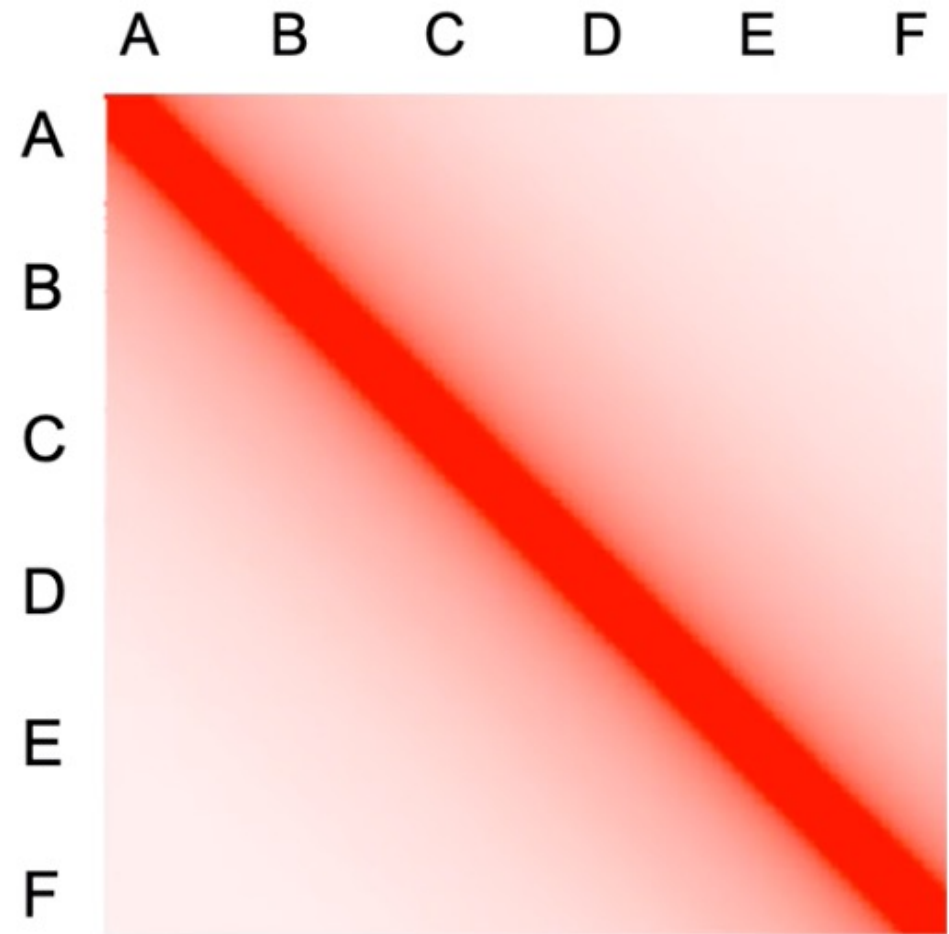
Genome



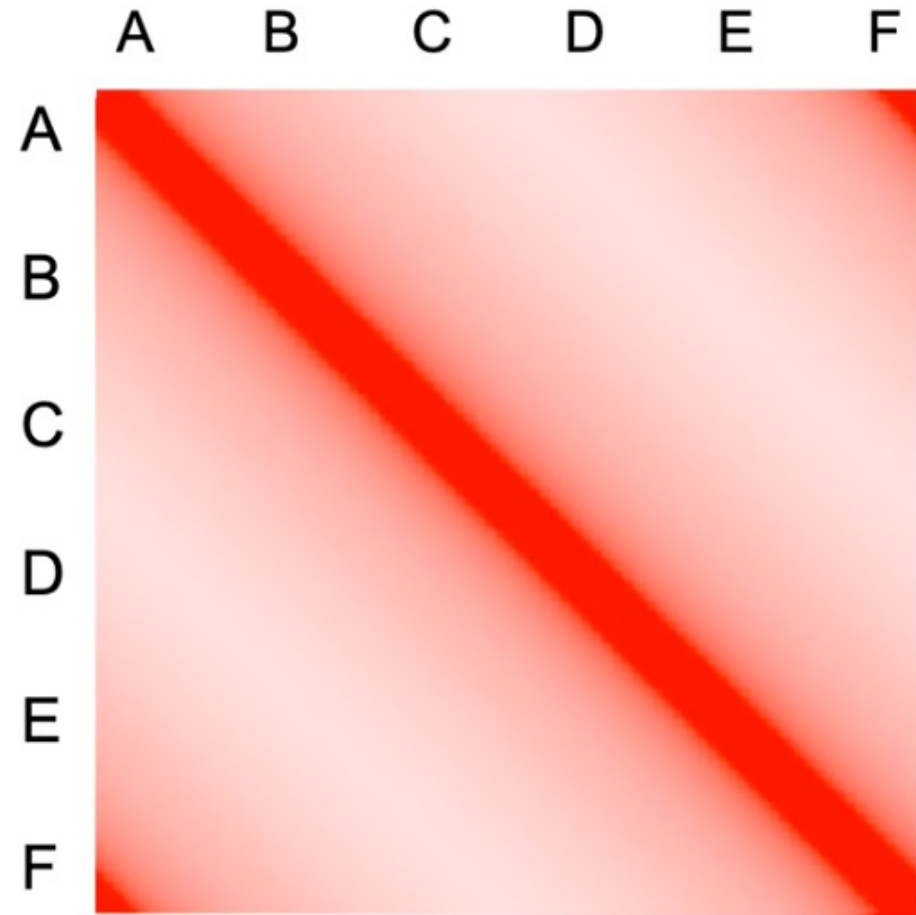
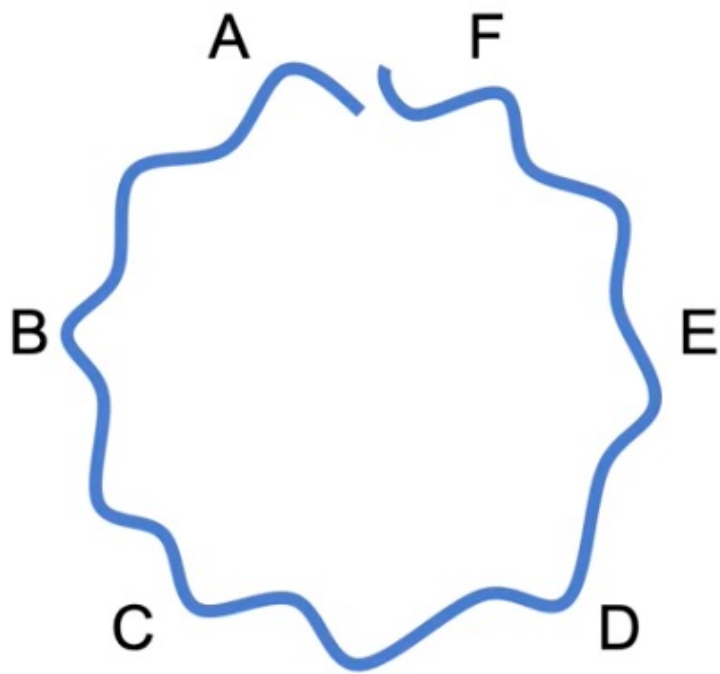
HiC data visualization Juicebox

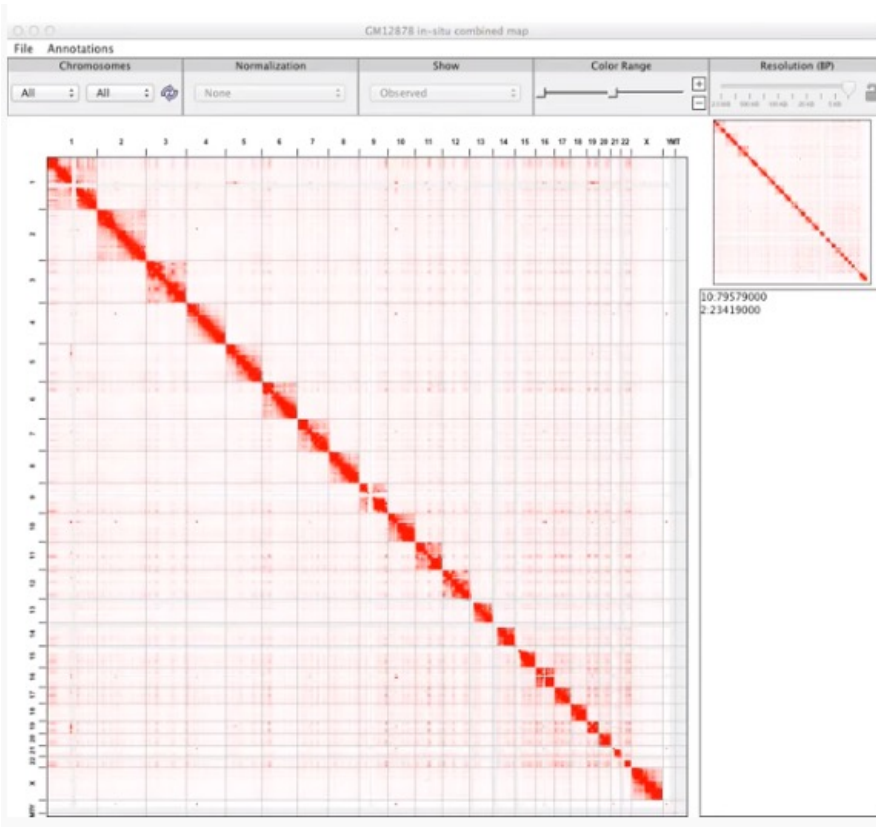


What might a chromosome look like?

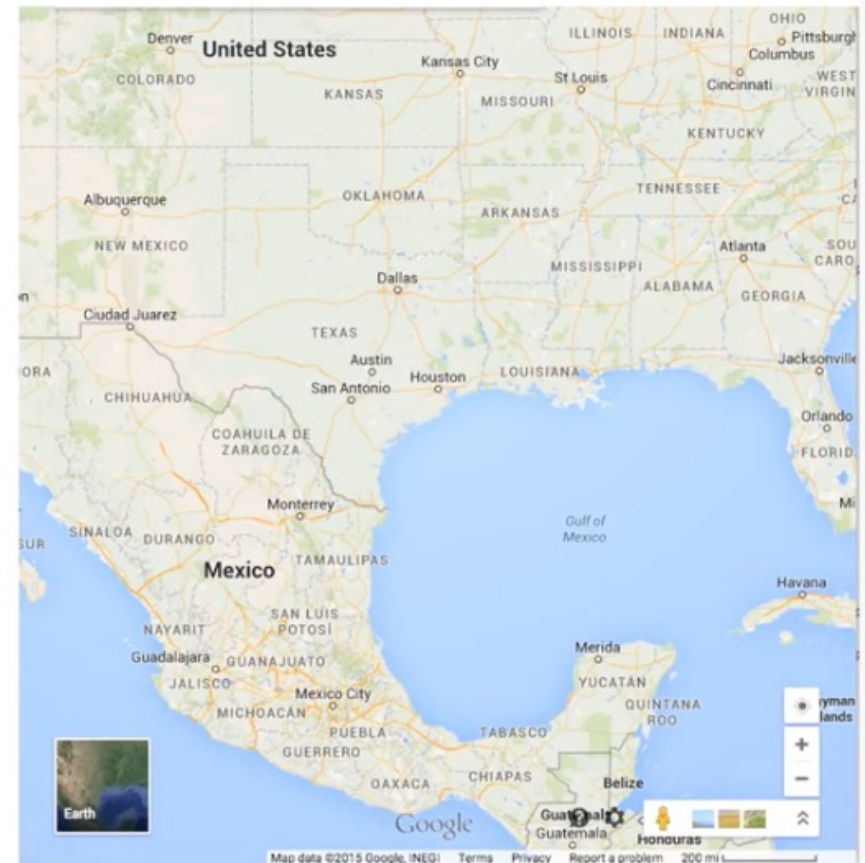
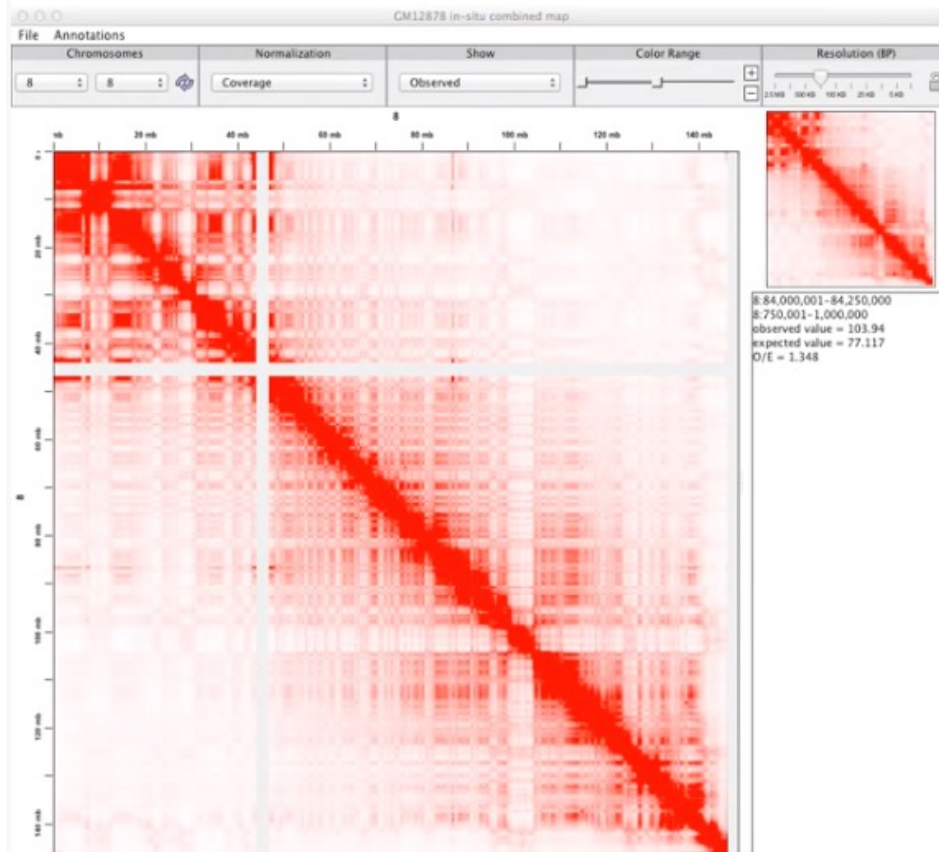


What might a chromosome look like?

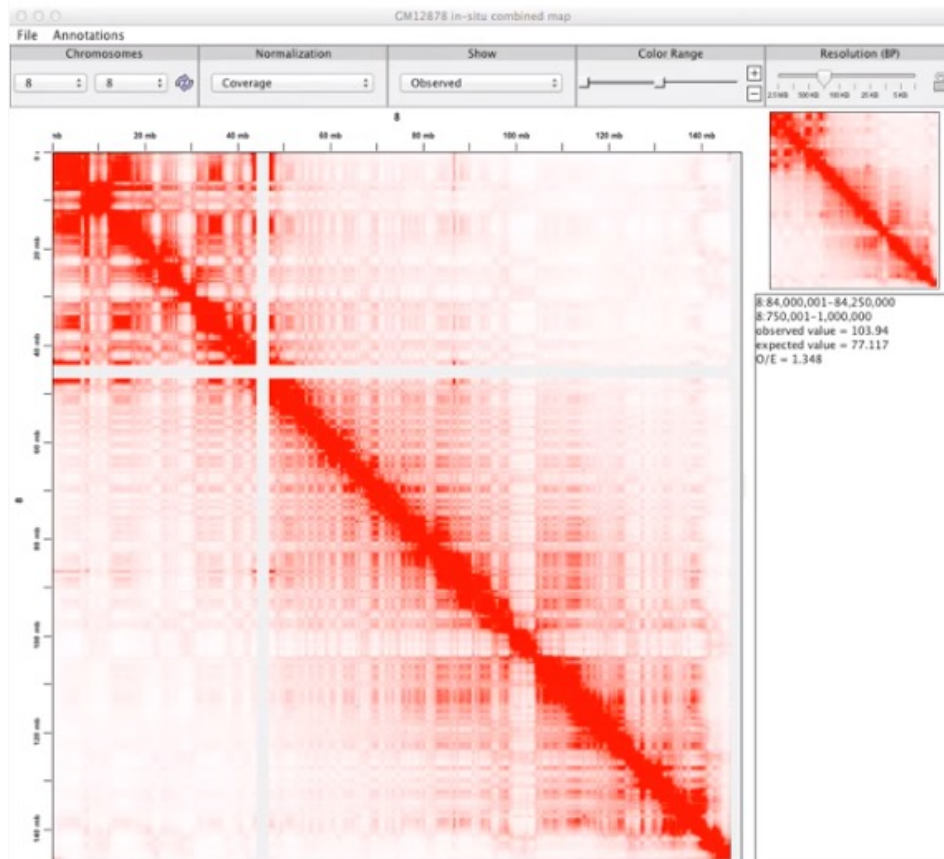




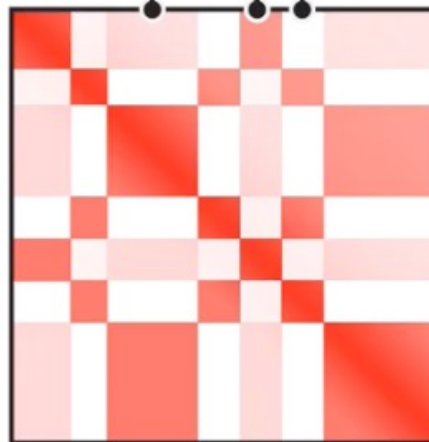
Chromosome



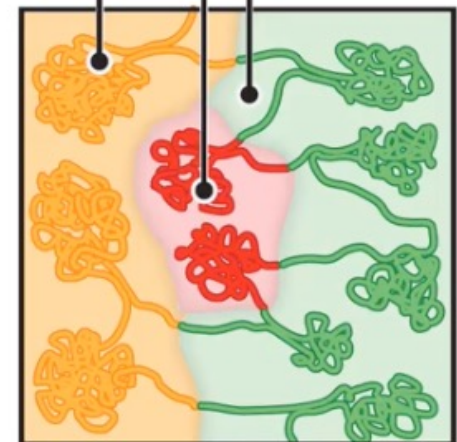
Checkered Pattern = Subcompartments



Subcompartments

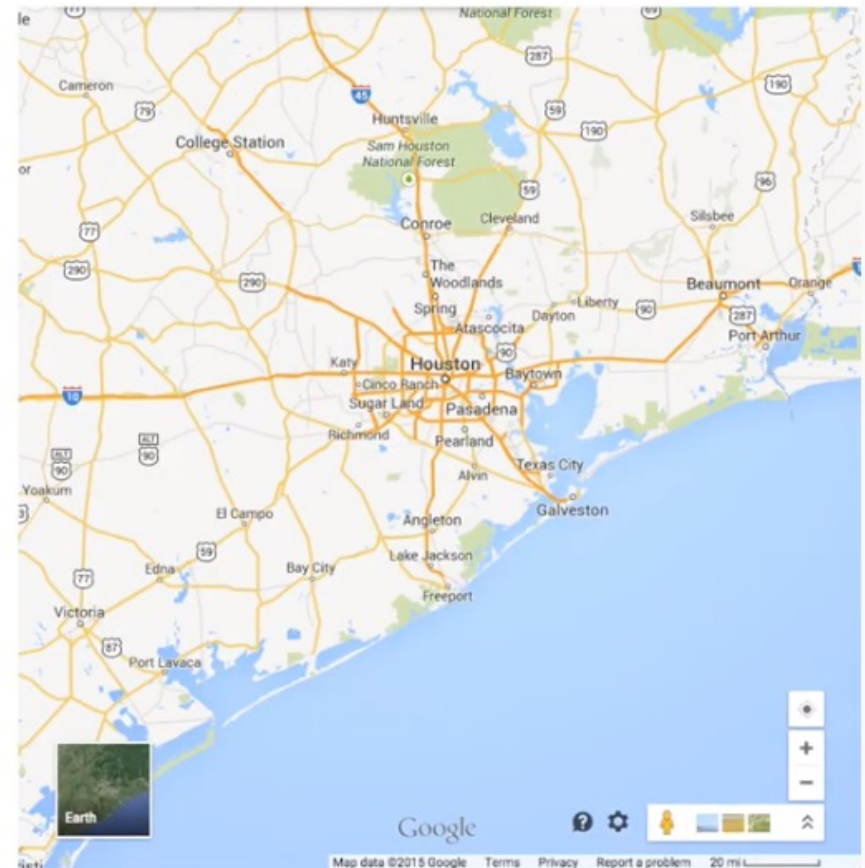
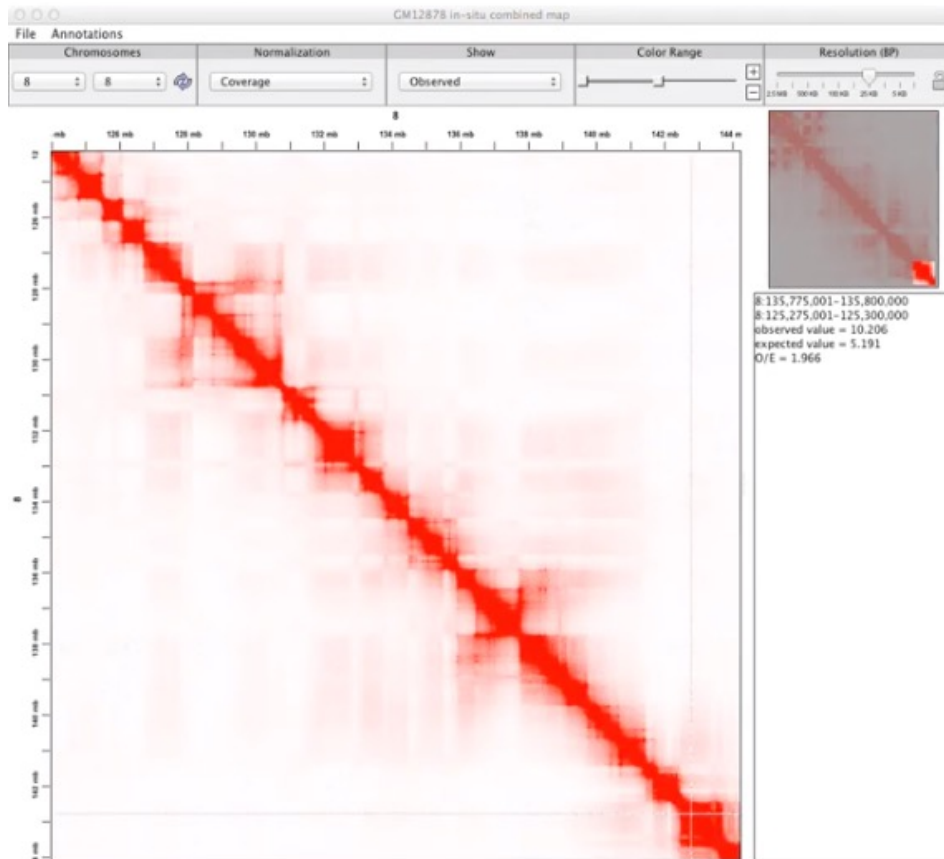


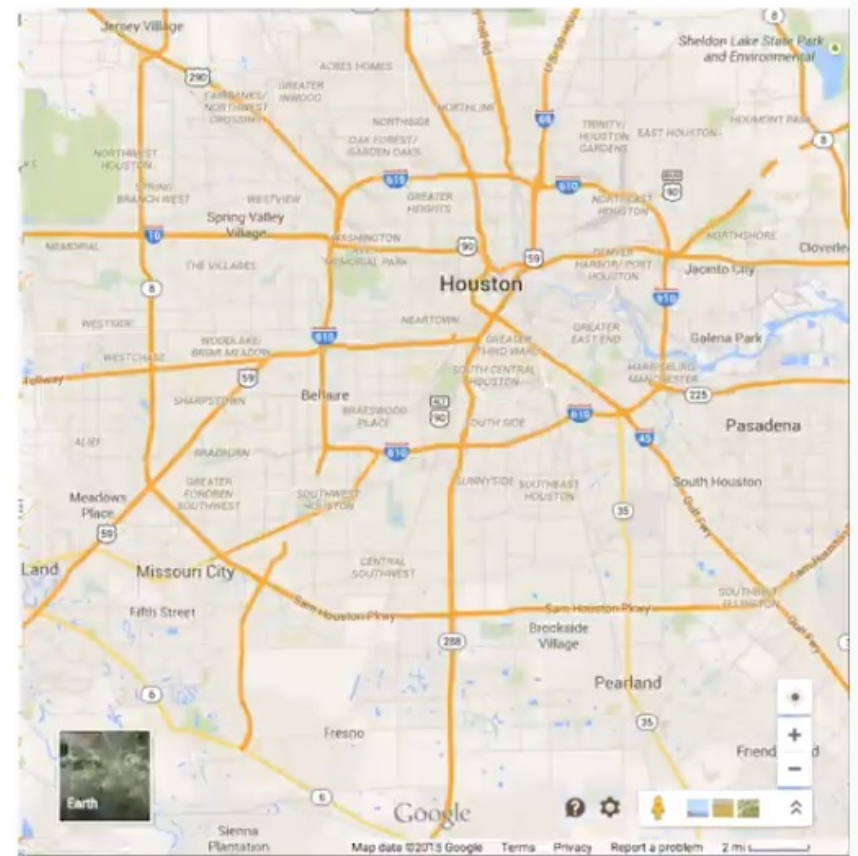
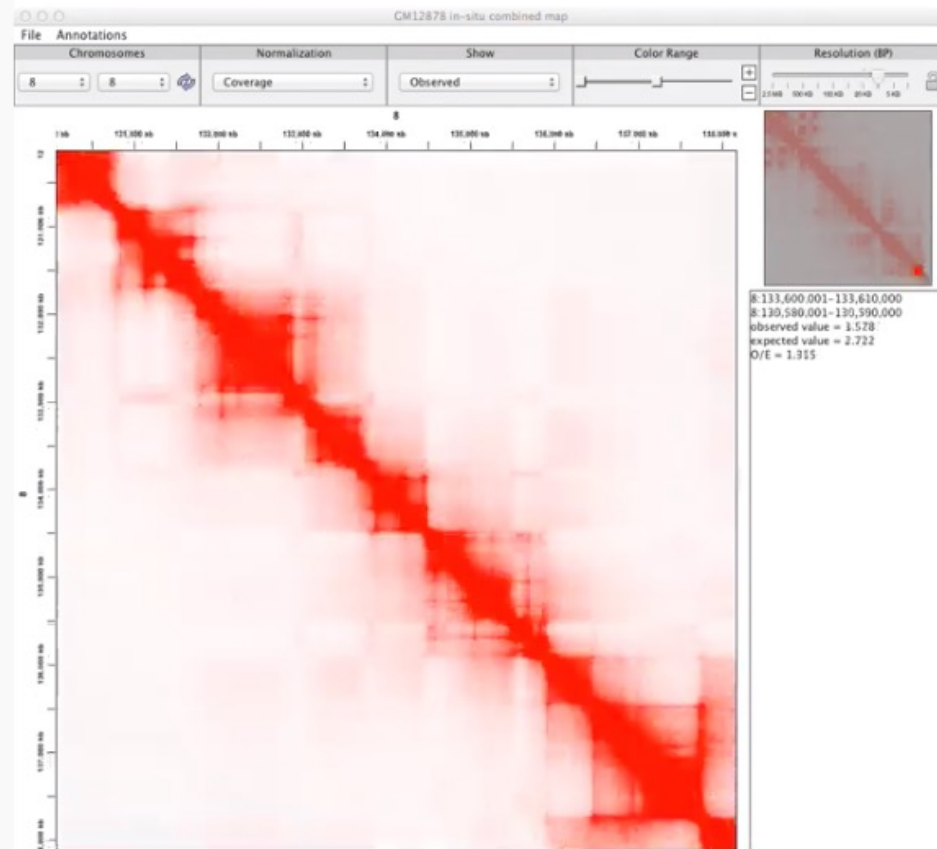
Subcompartments



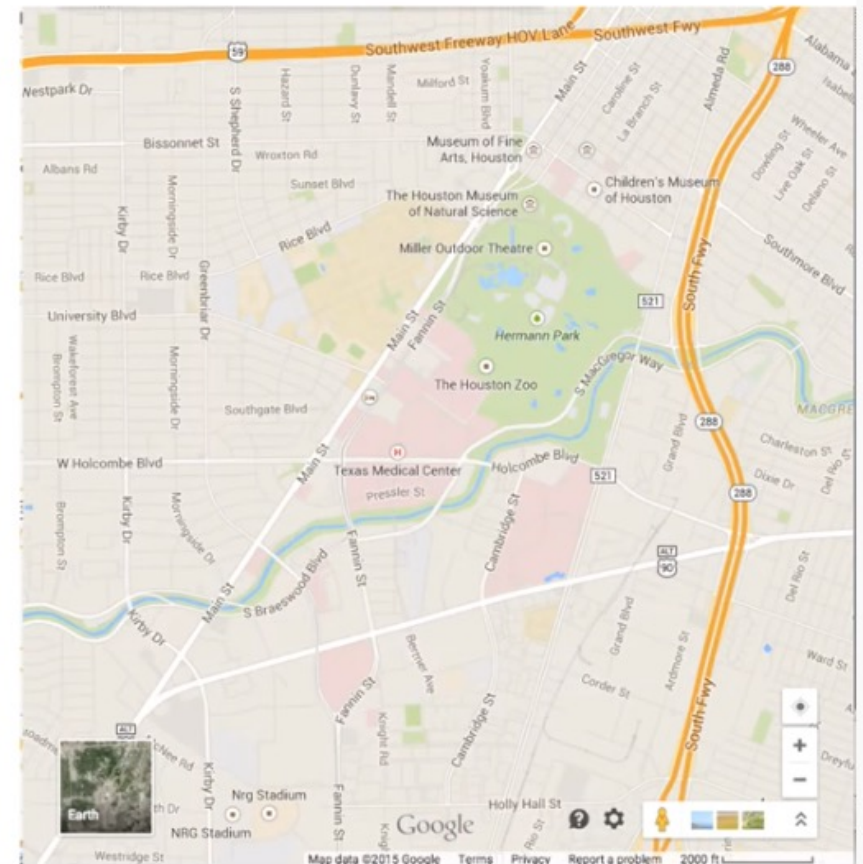
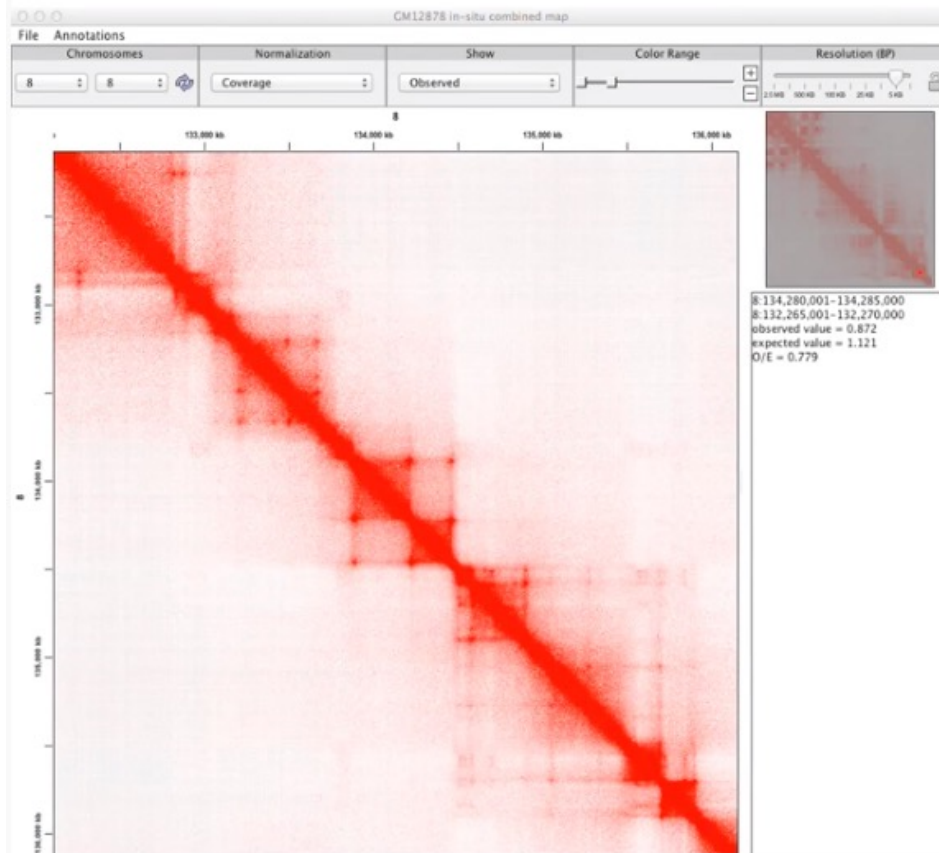
Rao & Huntley et al. *Cell* 2014

Squares and Peaks

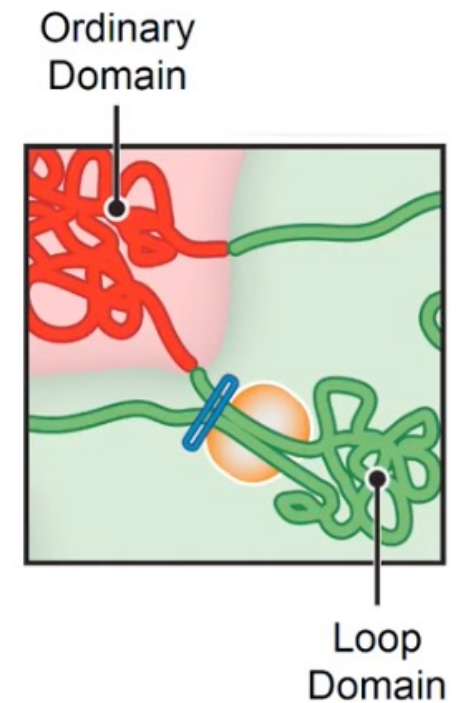
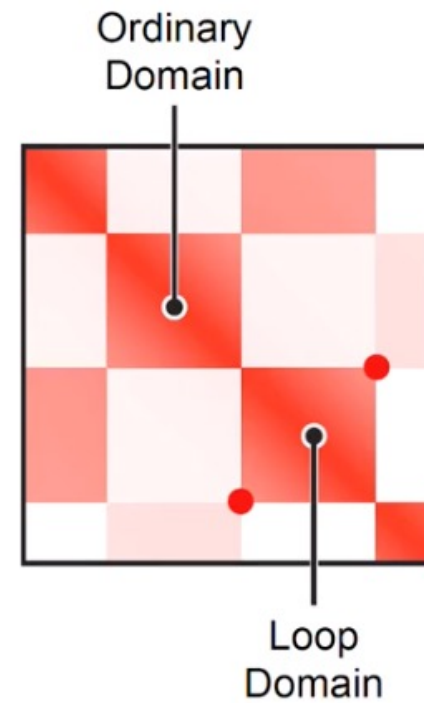
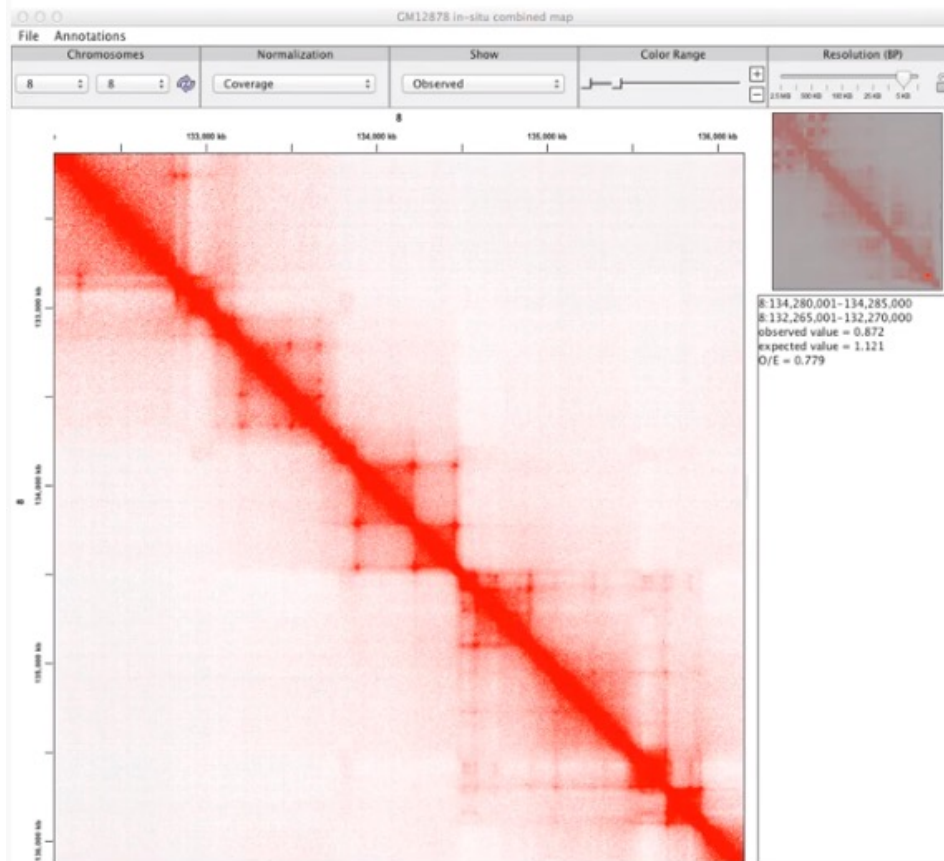




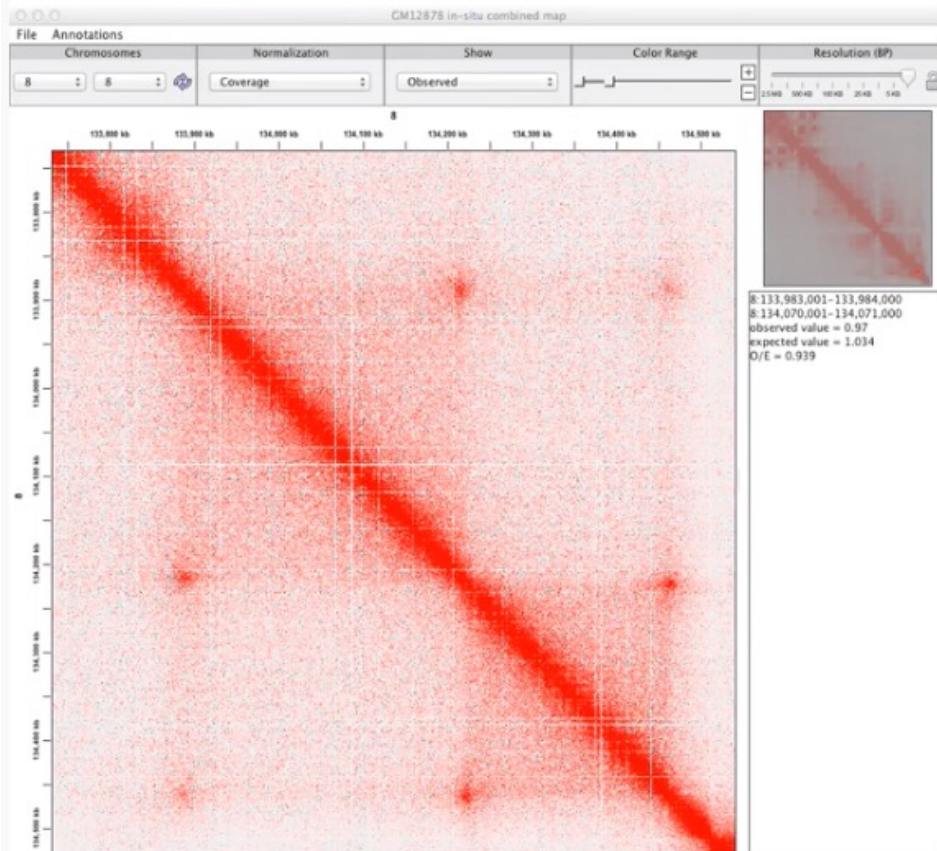
Squares and Peaks



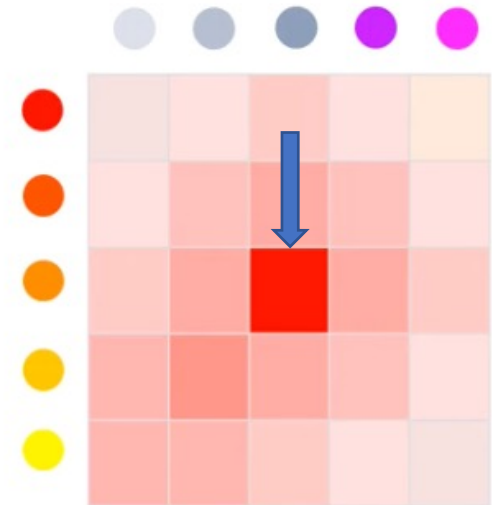
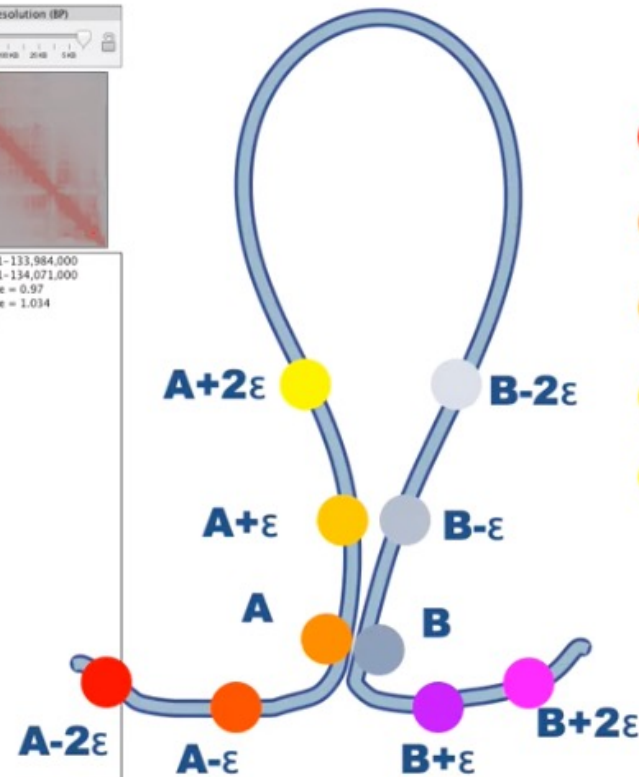
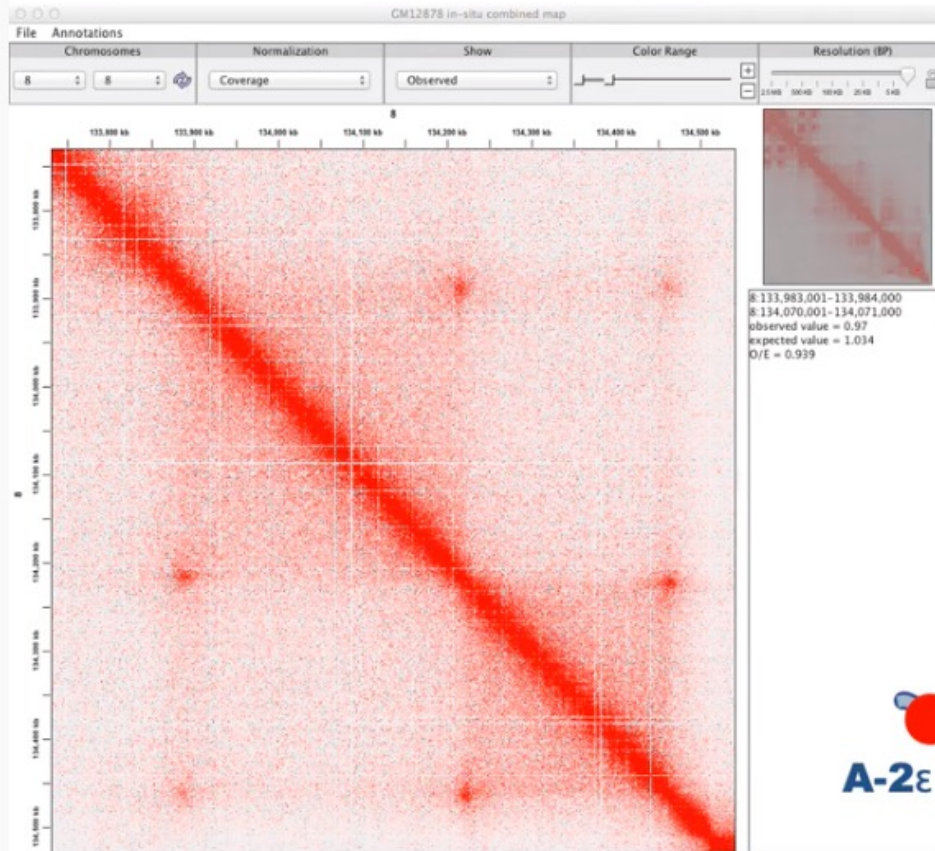
Domains and Loops



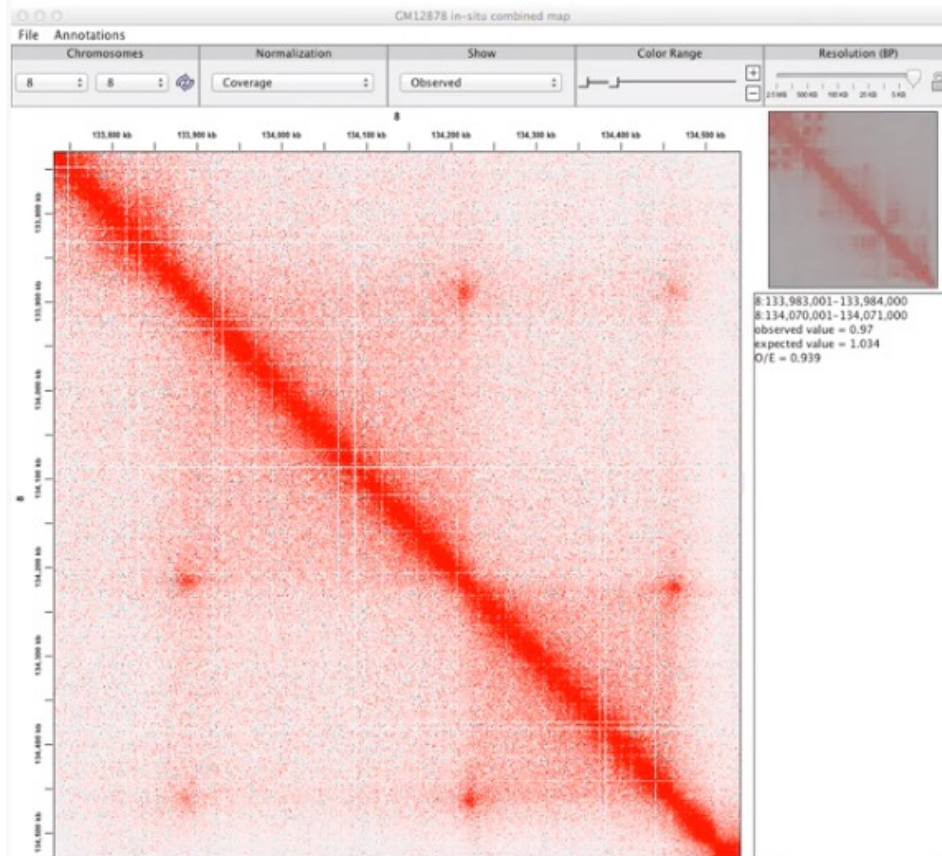
Zooming In (one more time)



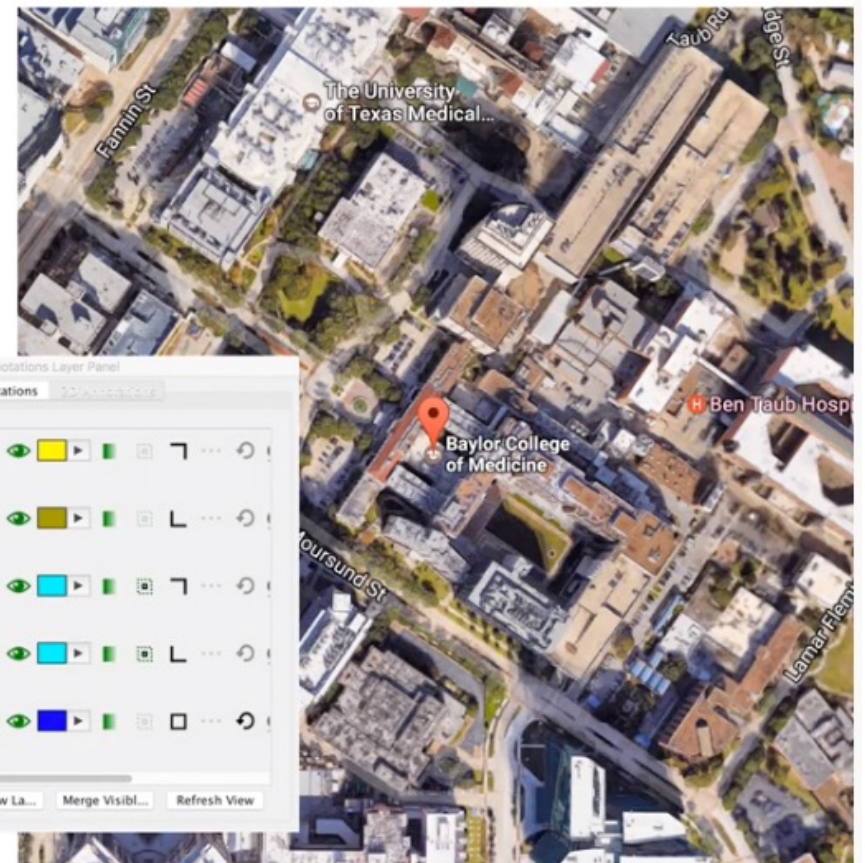
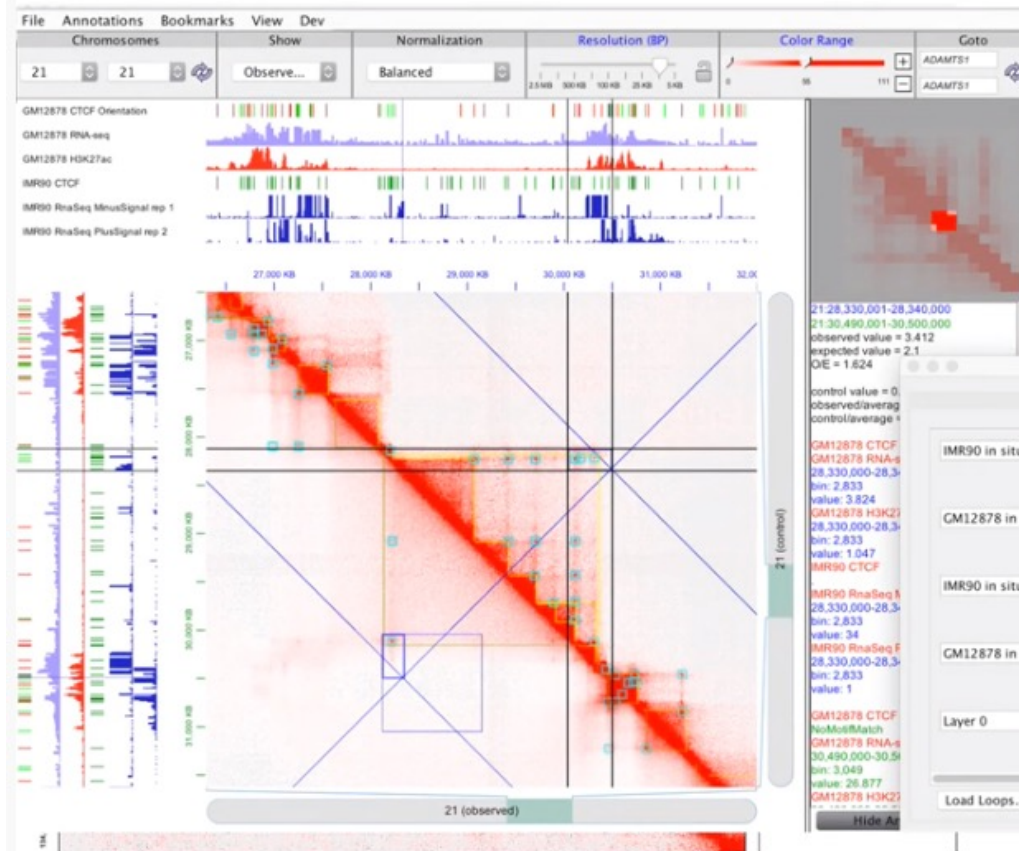
Contact Map Peaks = Chromatin Loops

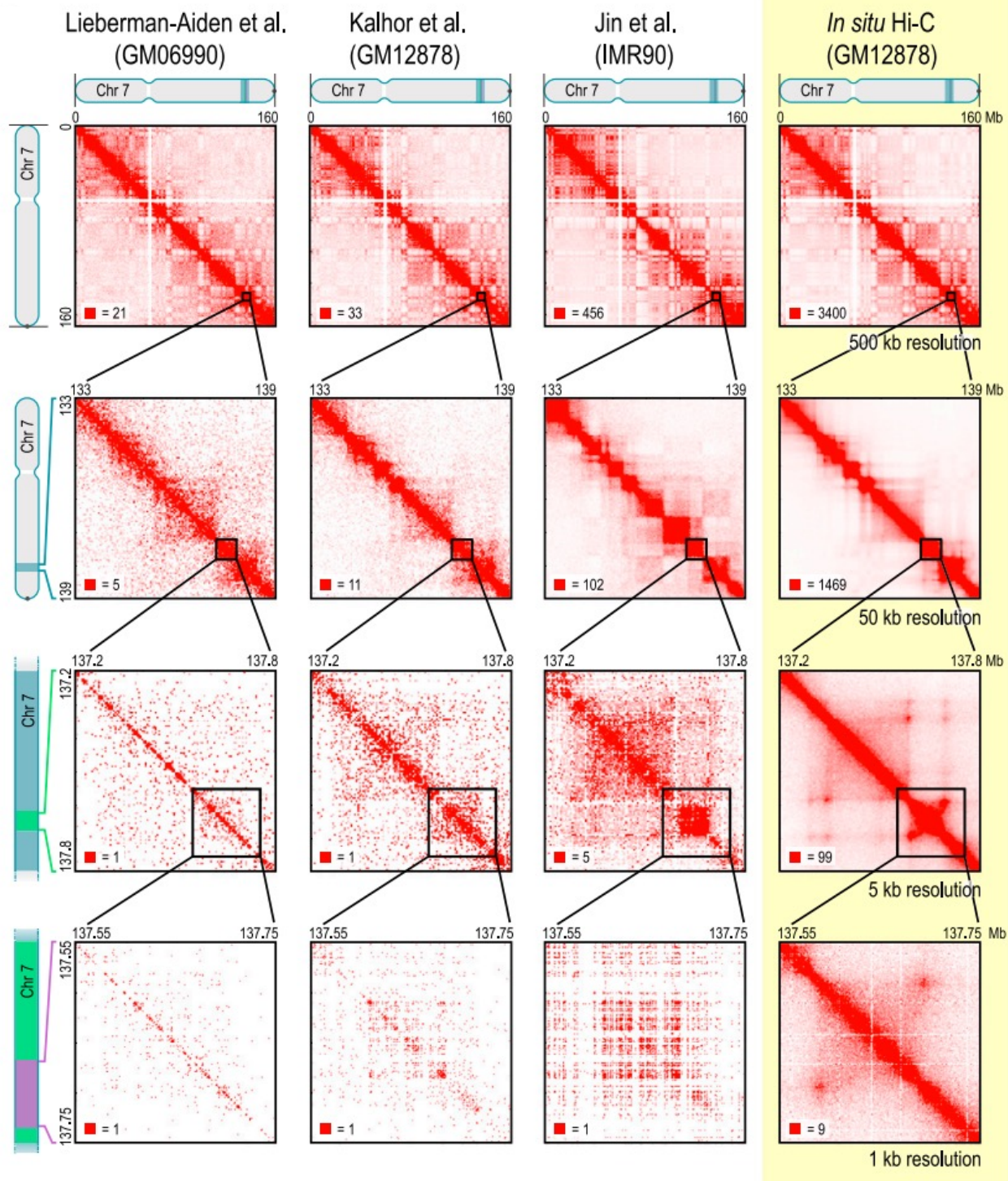


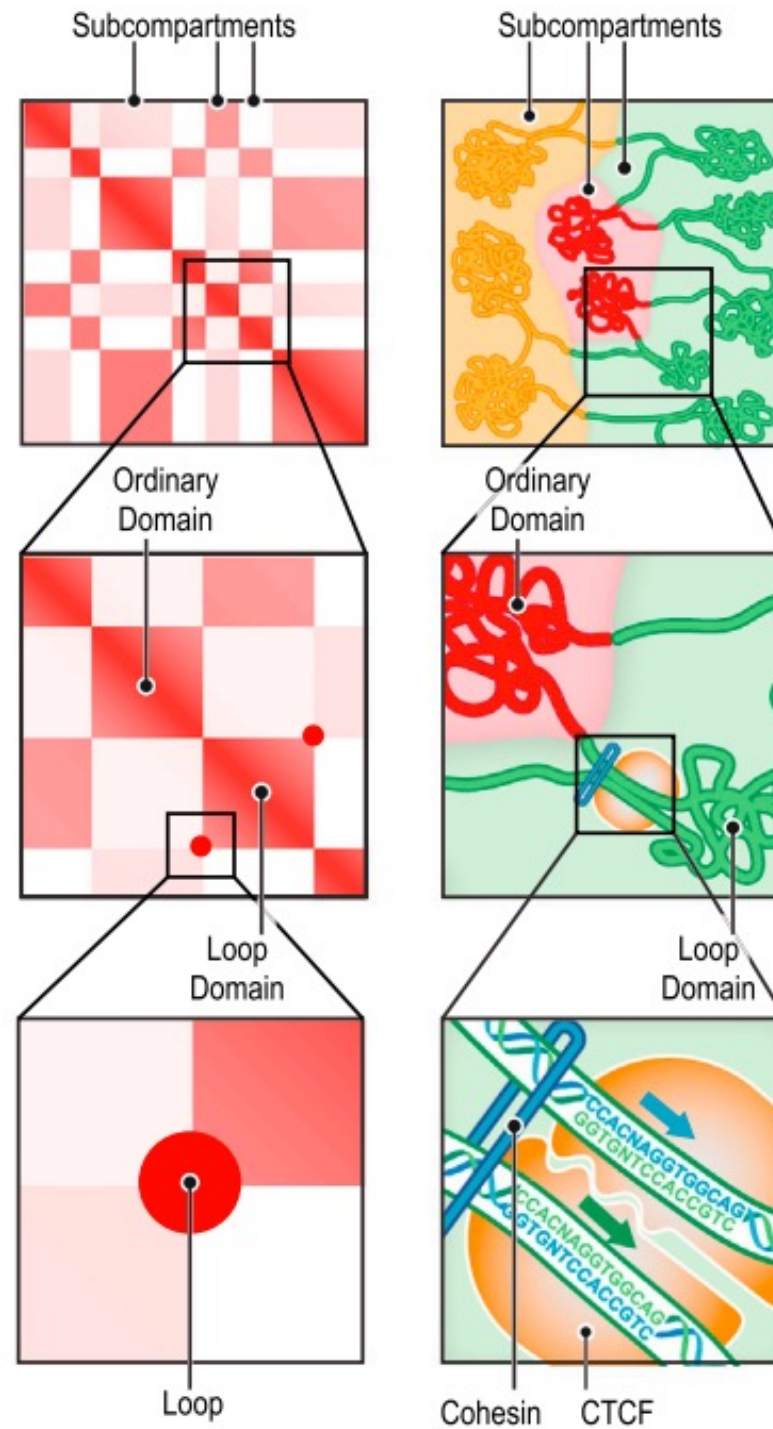
Annotations

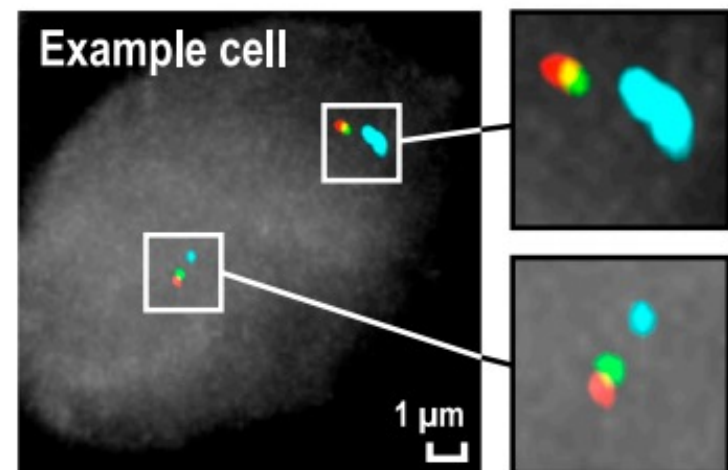
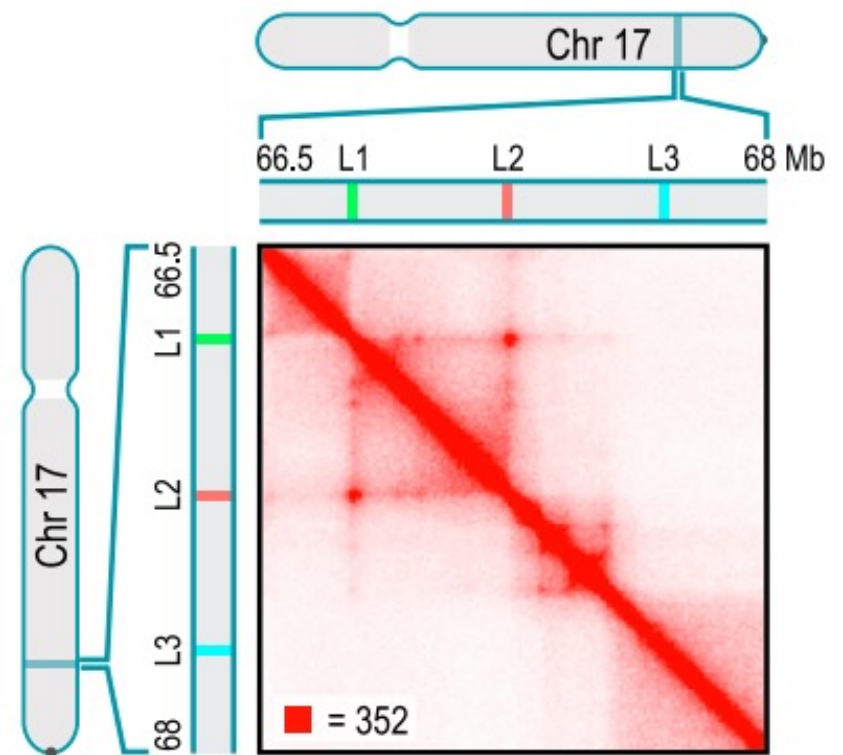
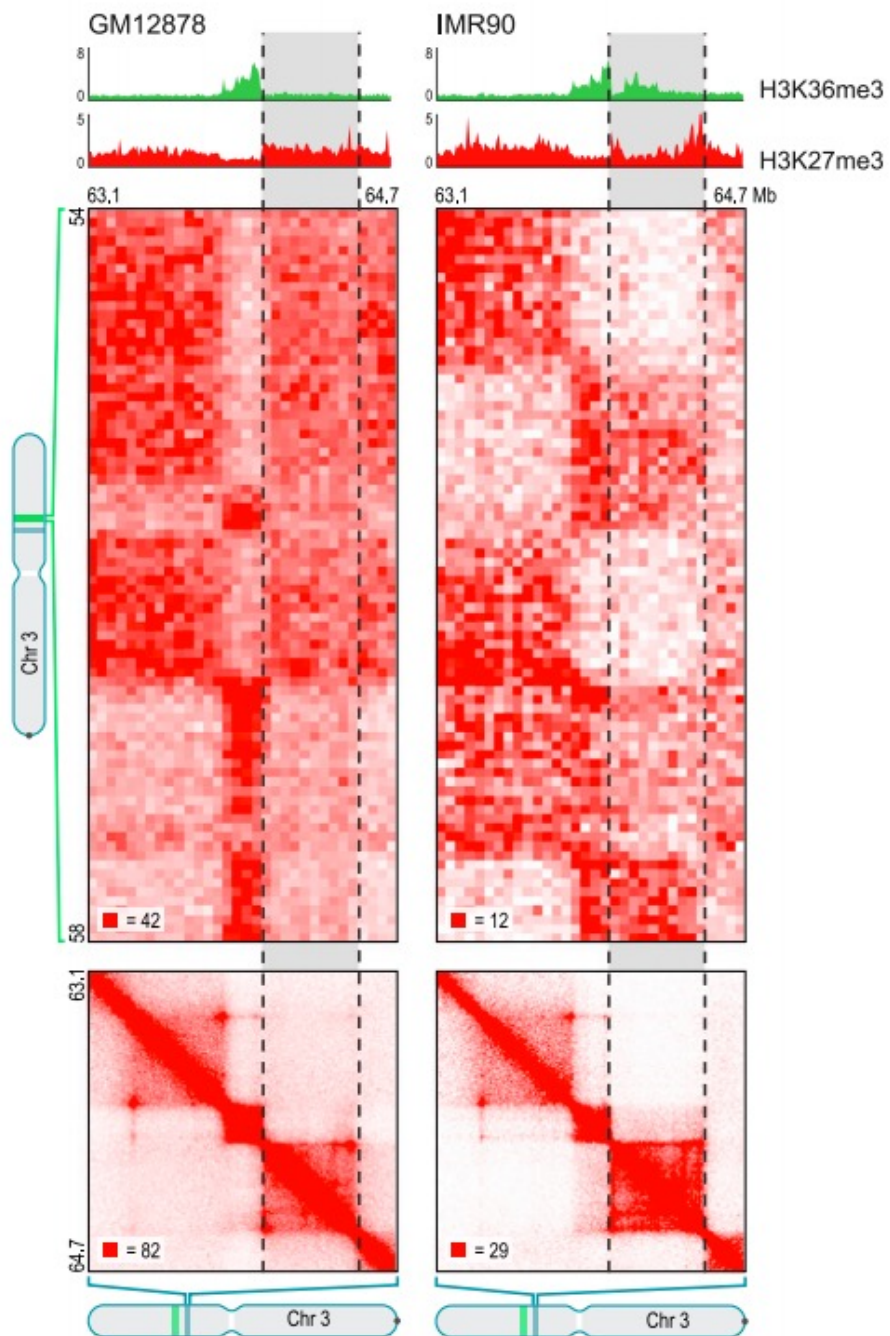


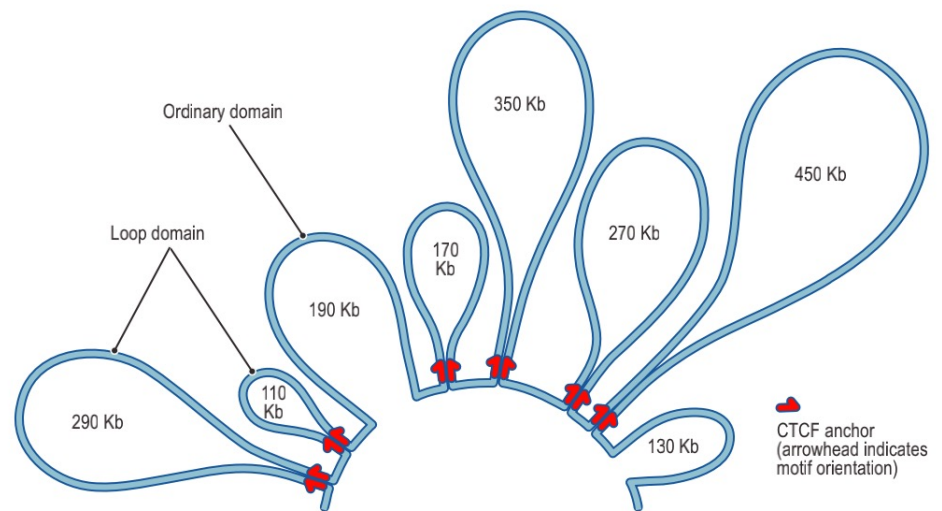
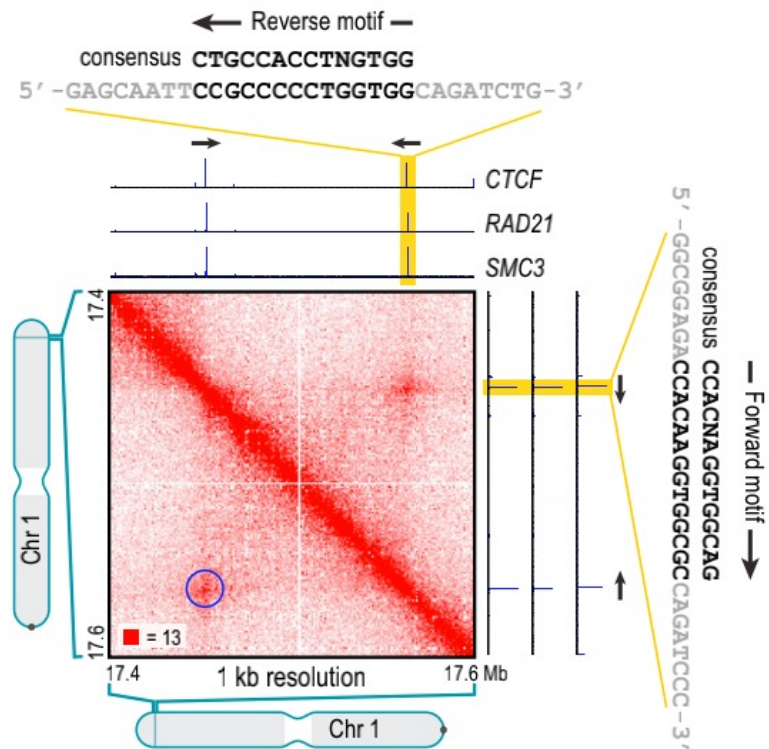
Annotations



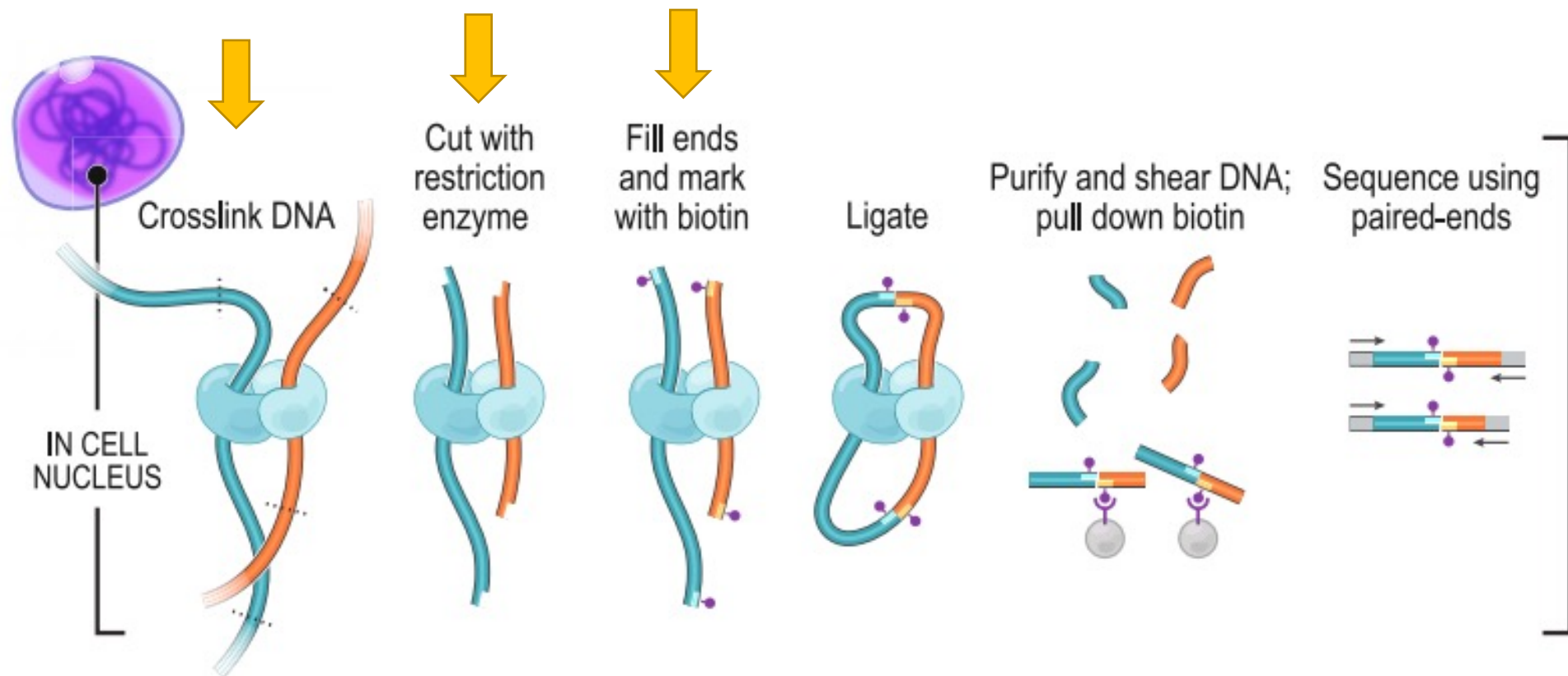




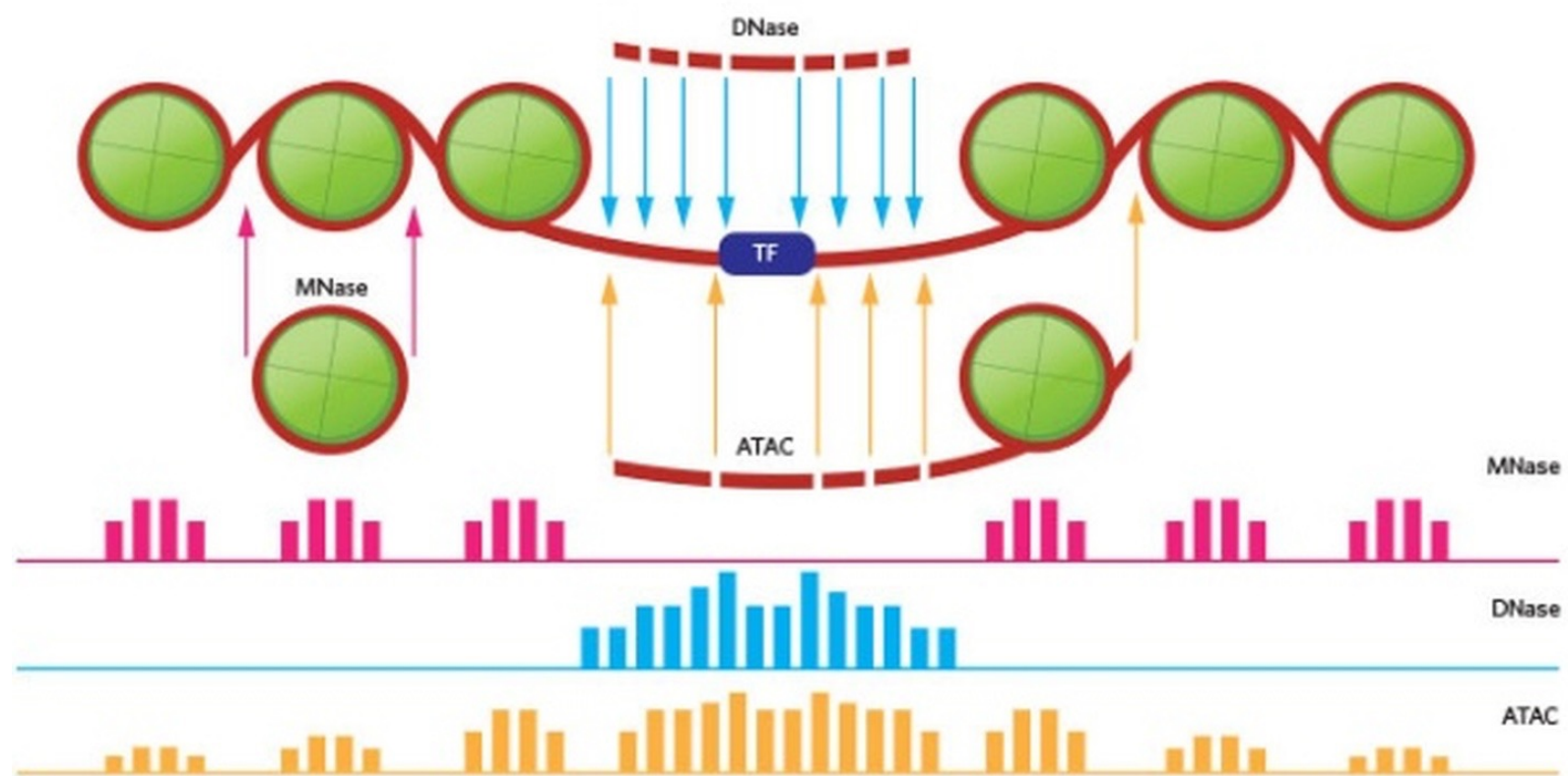




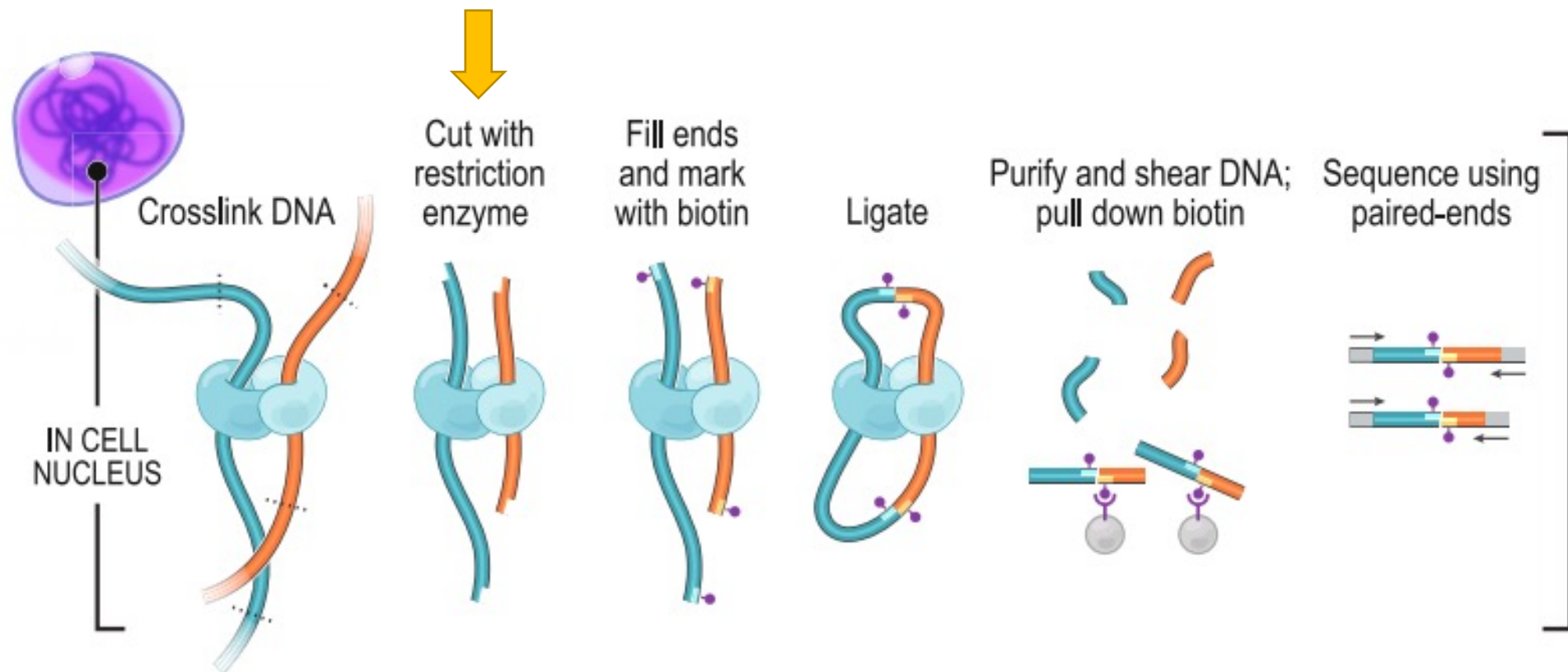
HiC method optimization and development



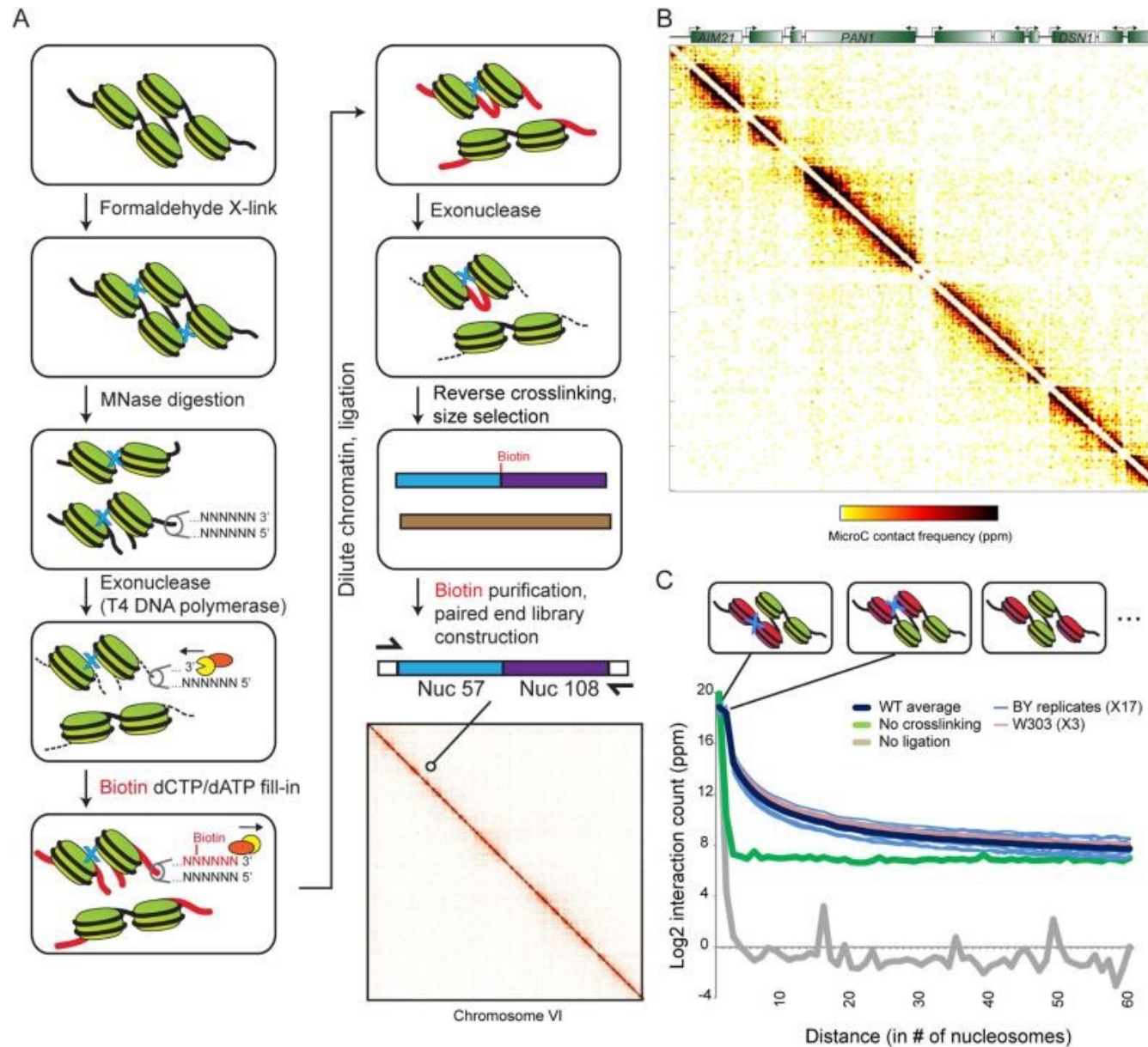
- Low resolution to high resolution
- Reduce cell to single cell



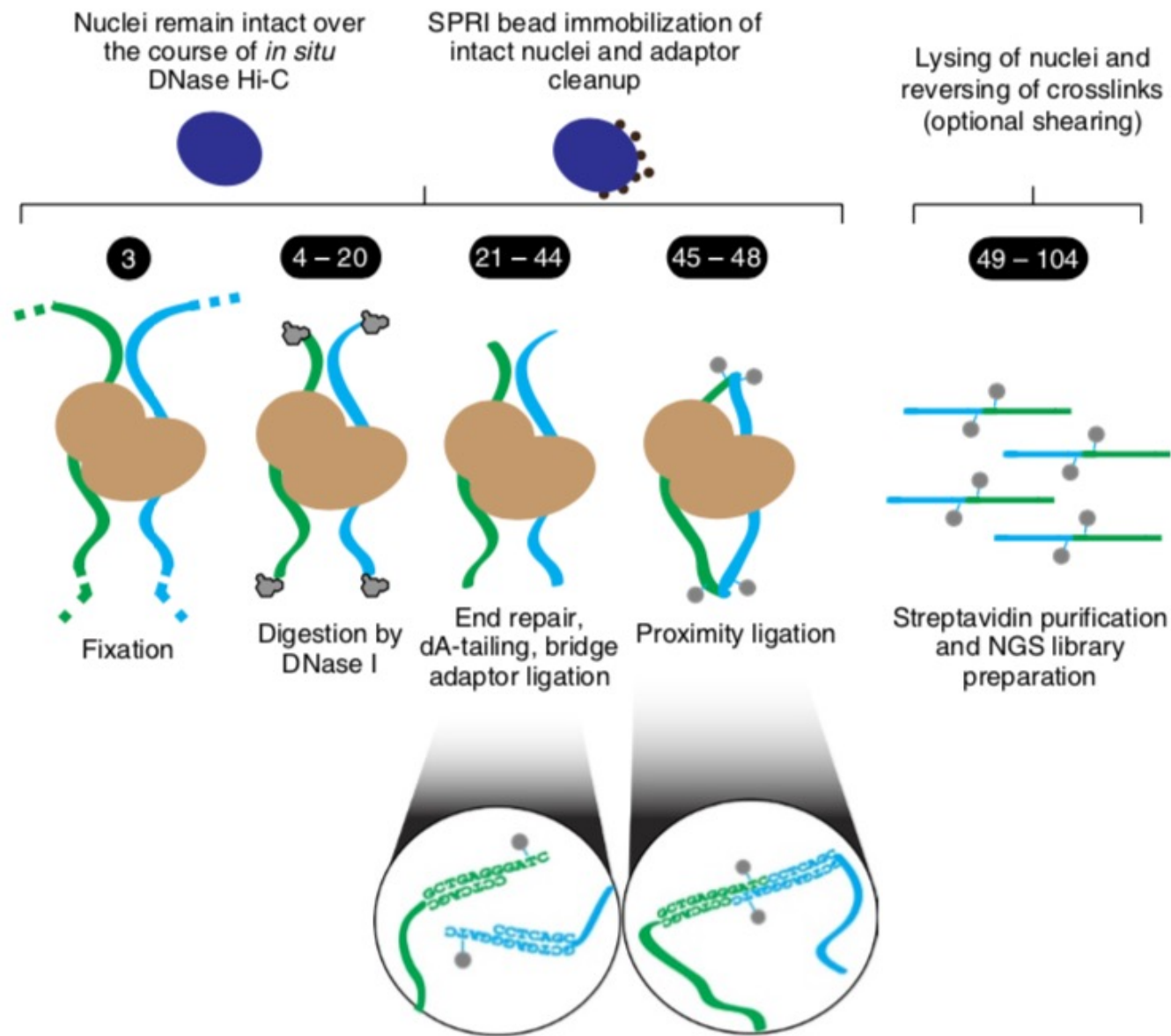
HiC method optimization and development



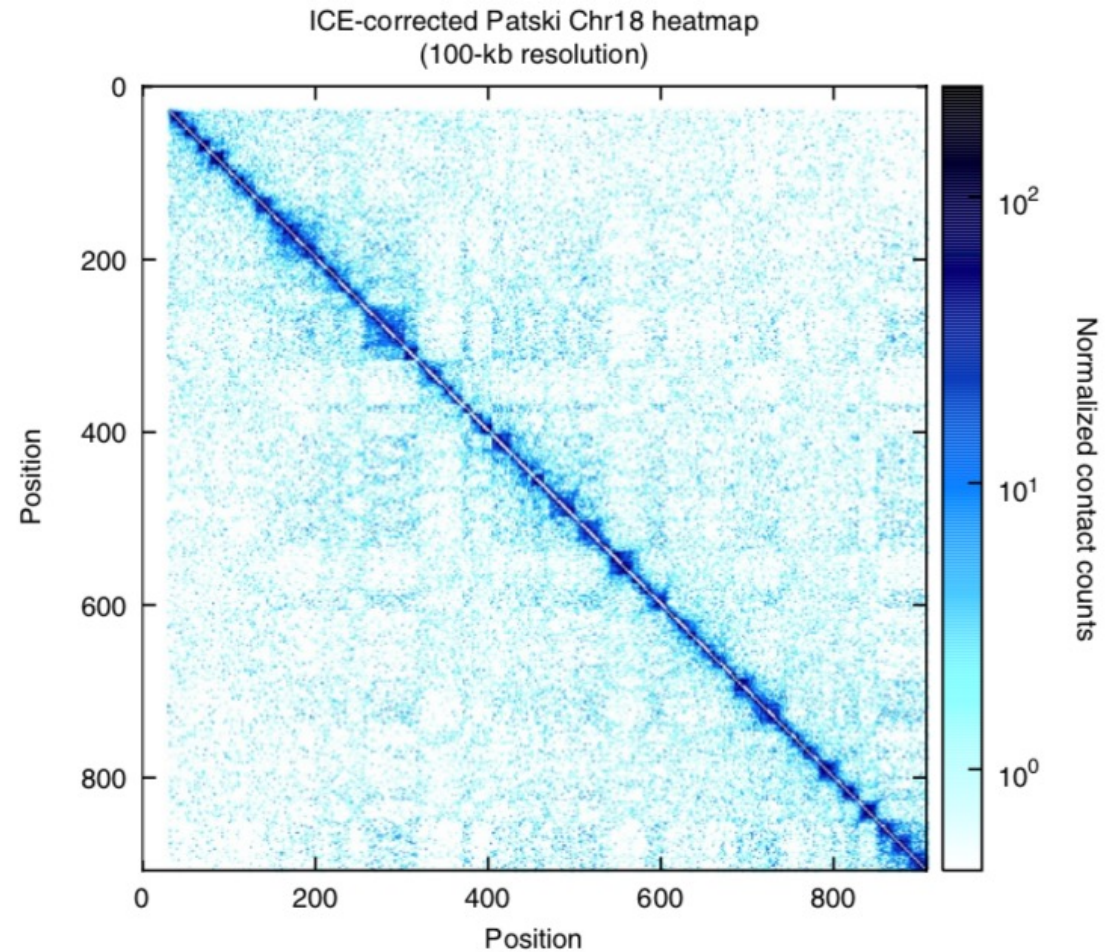
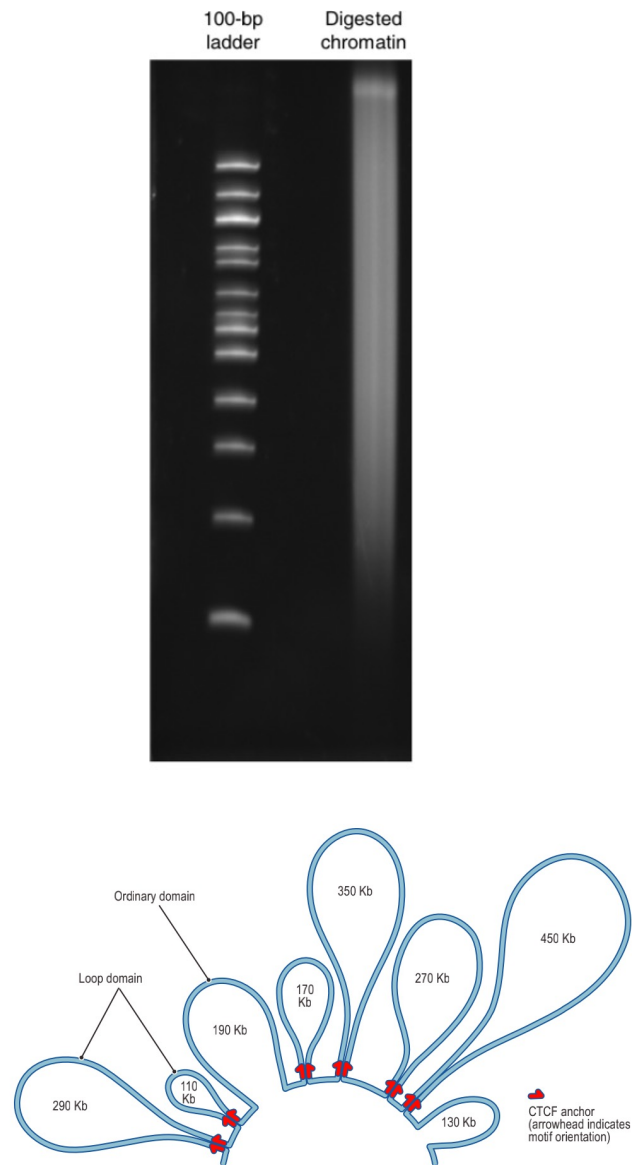
micro-C



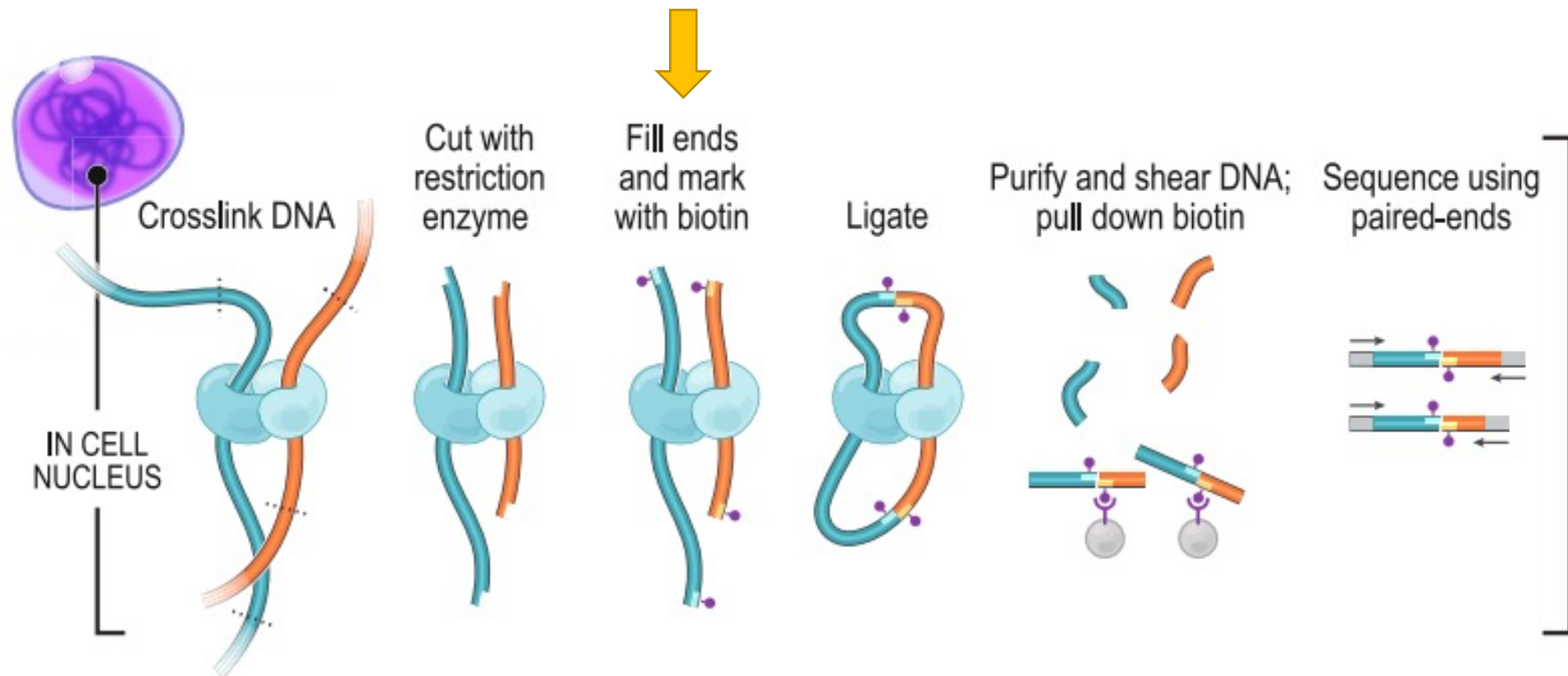
In situ DNase Hi-C



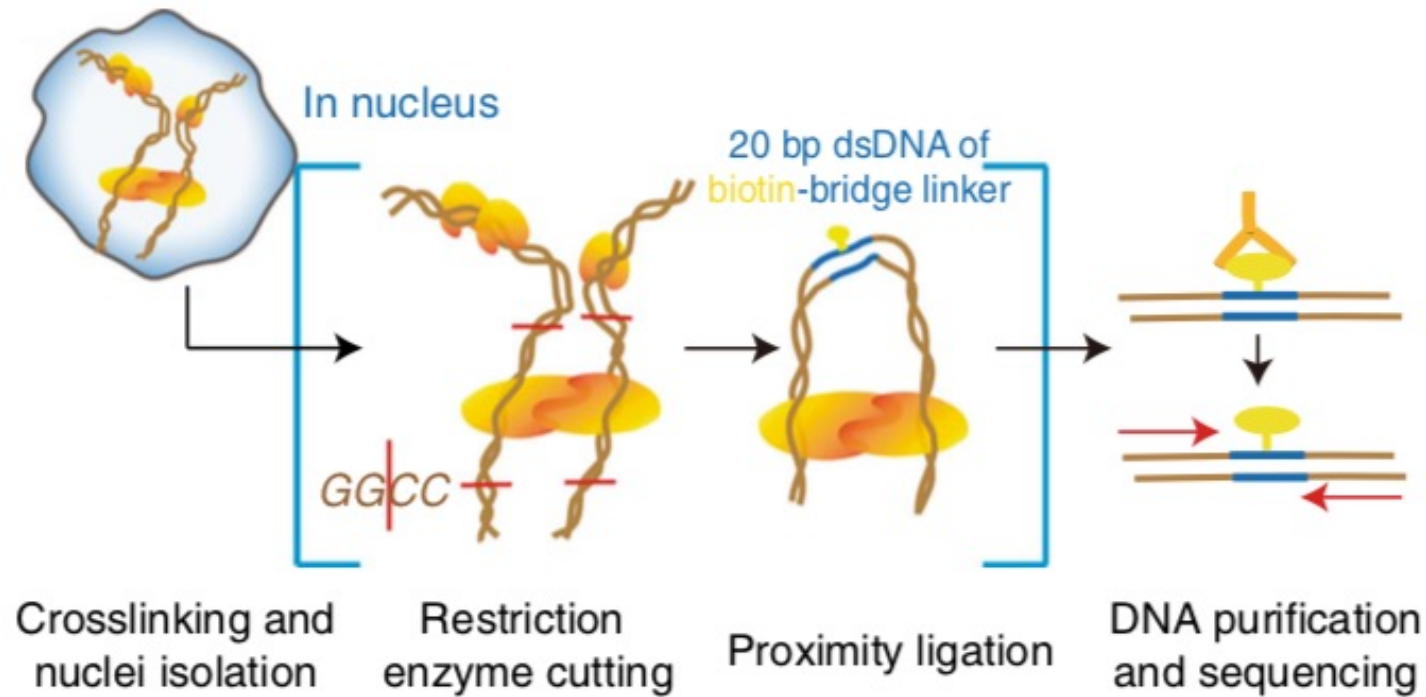
In situ DNase Hi-C

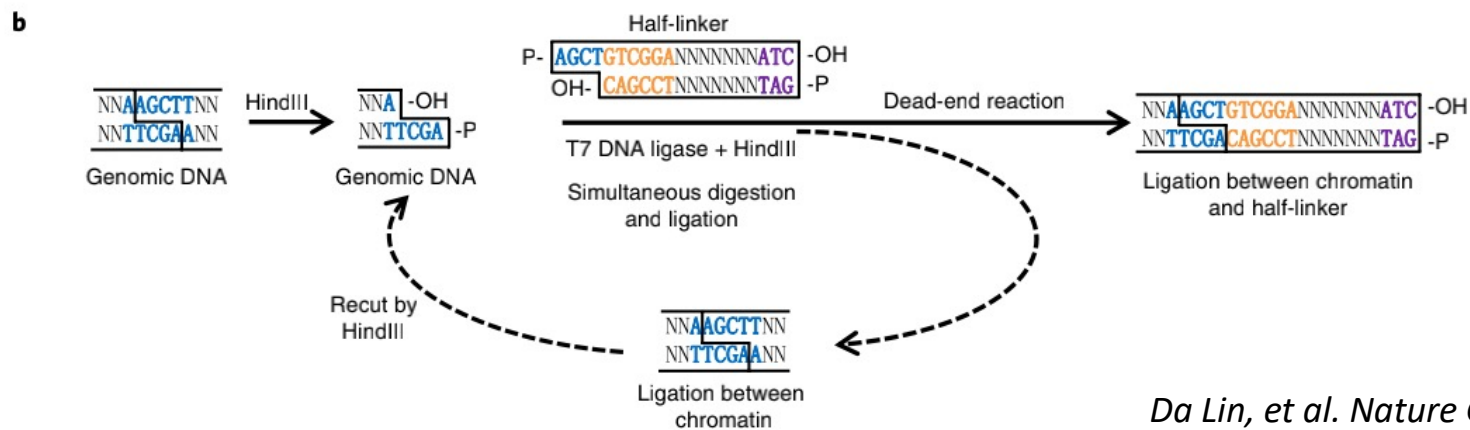
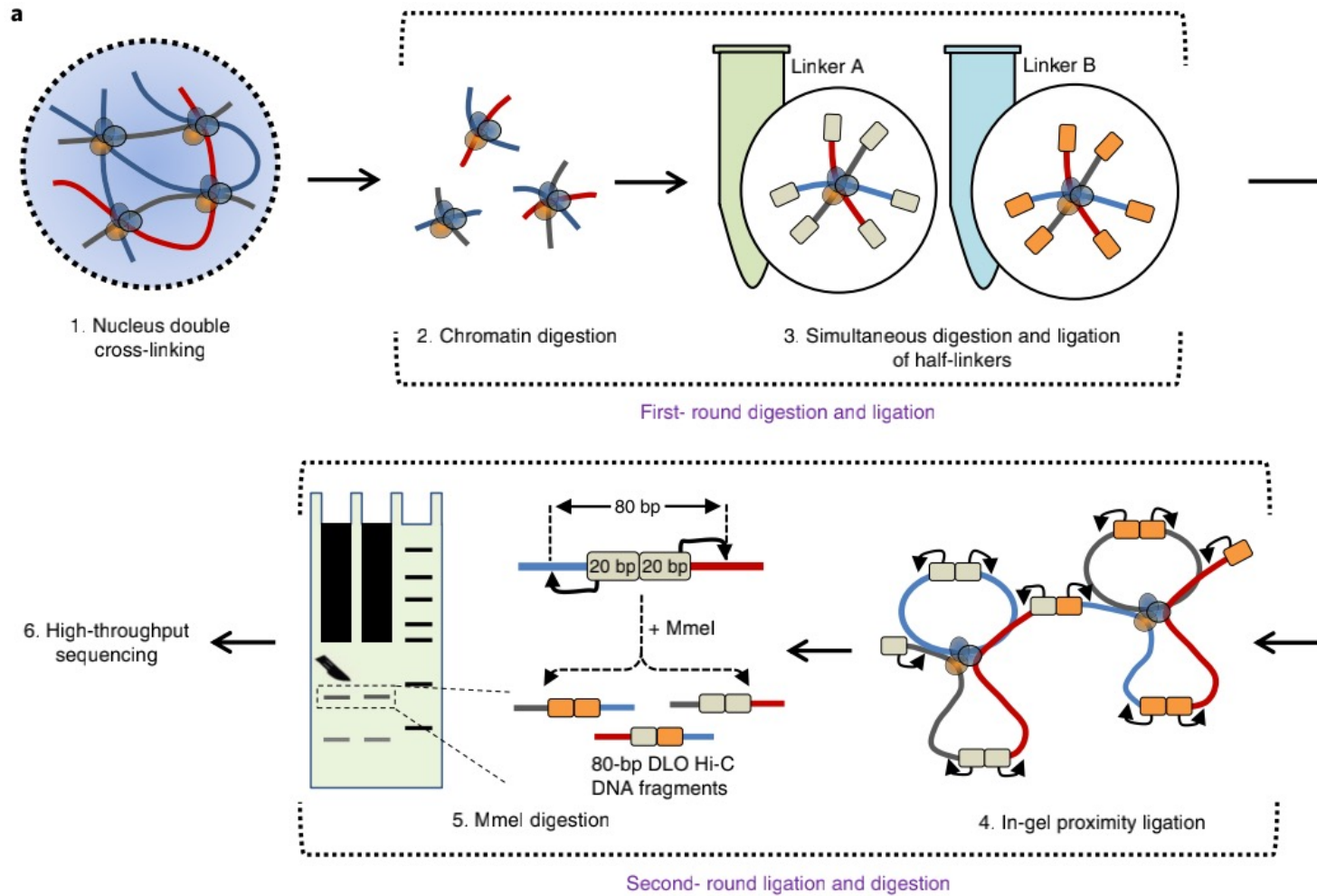


HiC method optimization and development

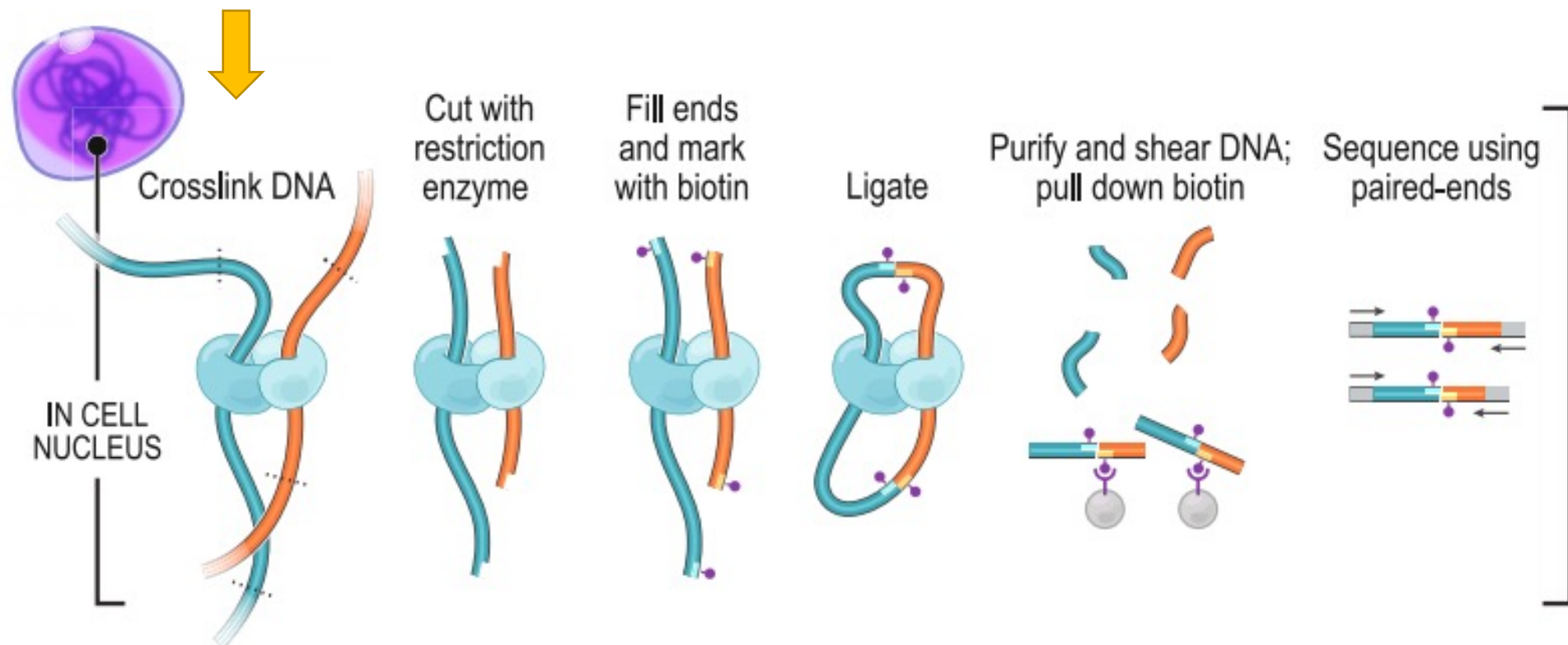


BL-Hi-C (Bridge Linker-Hi-C)

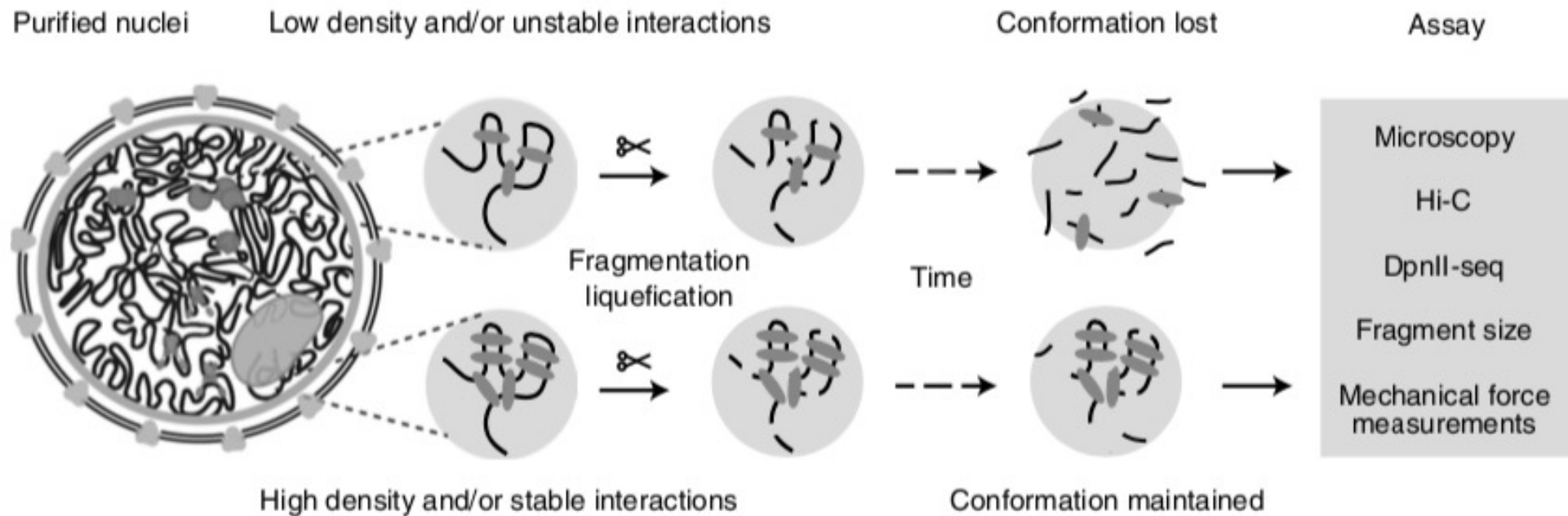




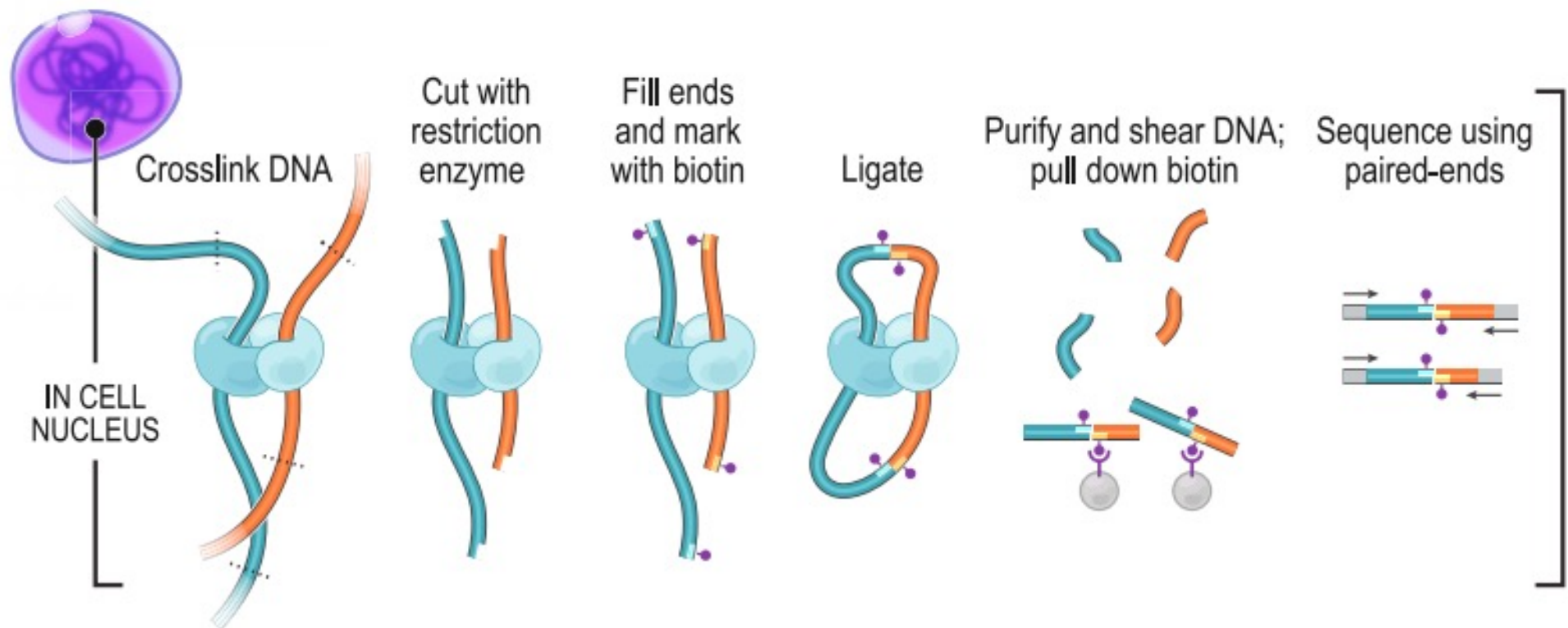
HiC method optimization and development



Liquid chromatin Hi-C



HiC method optimization and development

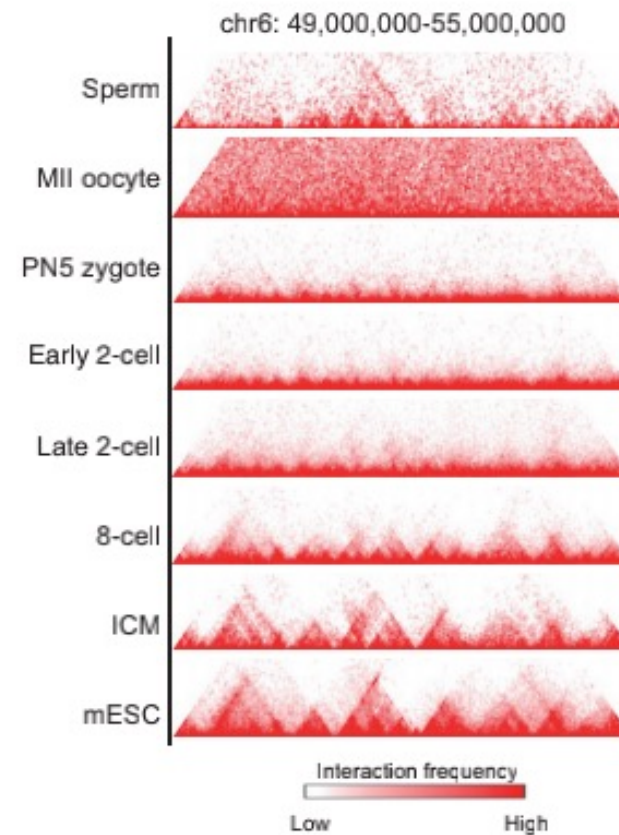
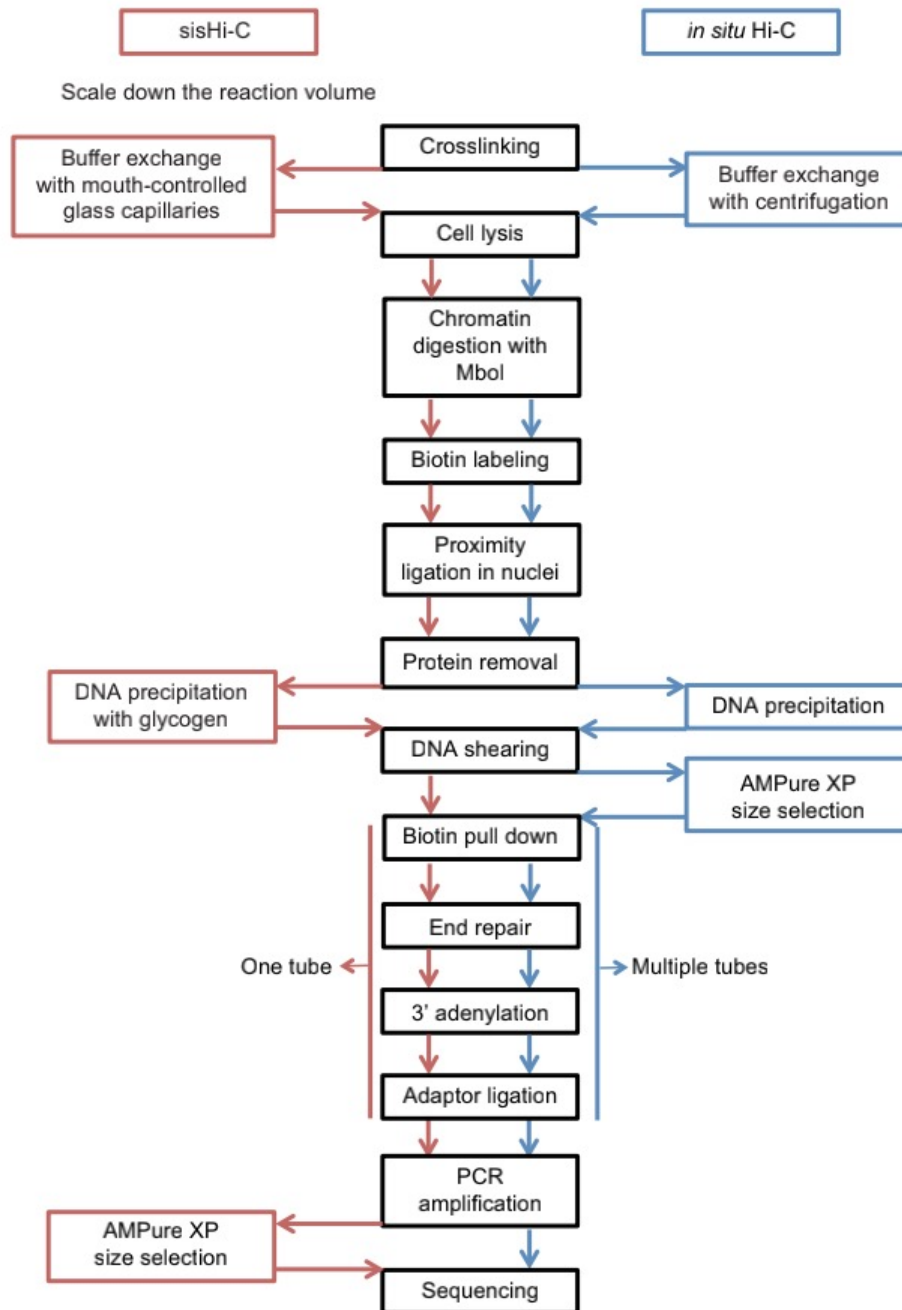


- Low resolution to high resolution
- Reduce cell to single cell

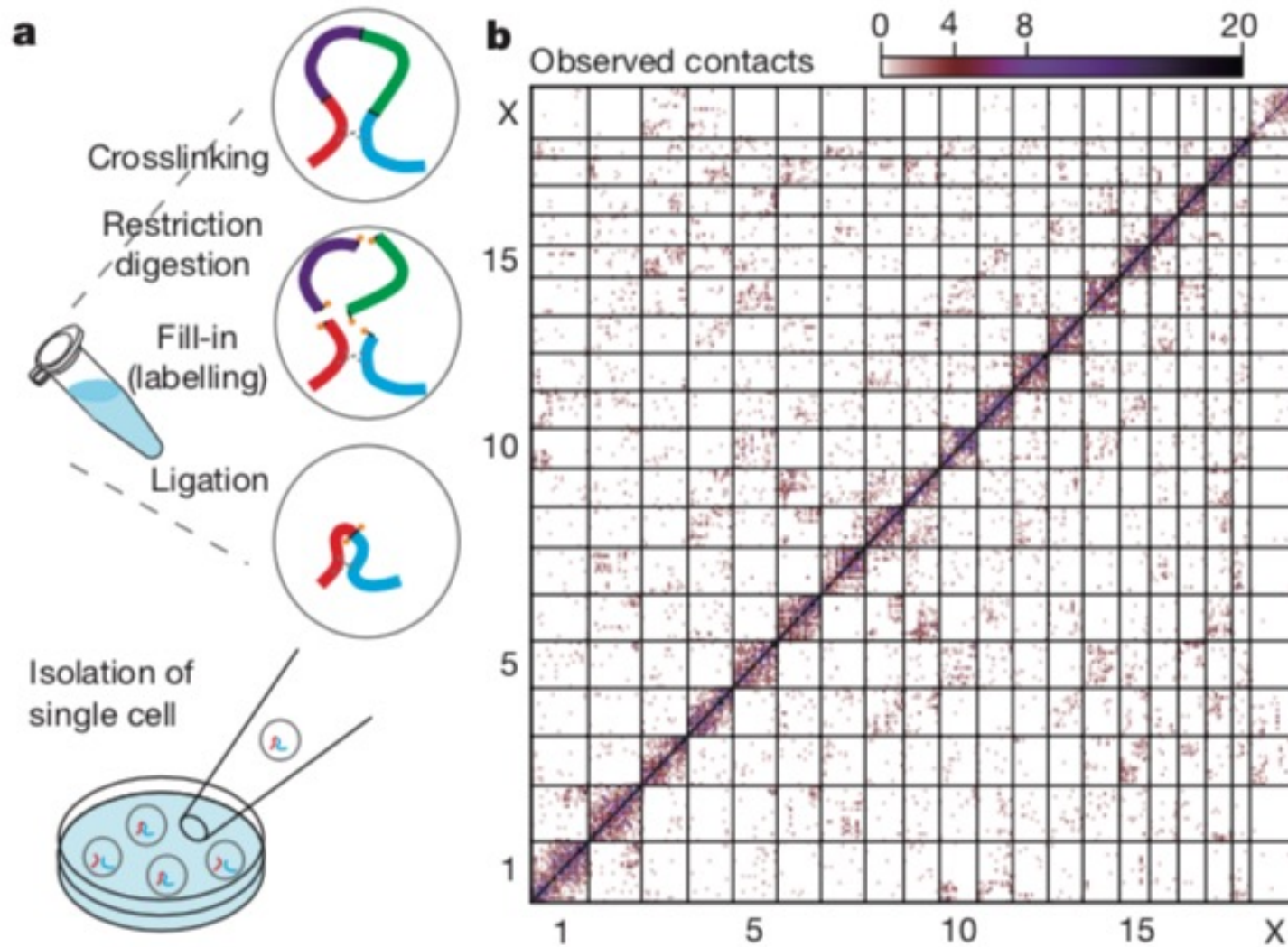
small-scale *in situ* Hi-C

Comparison of Hi-C protocols

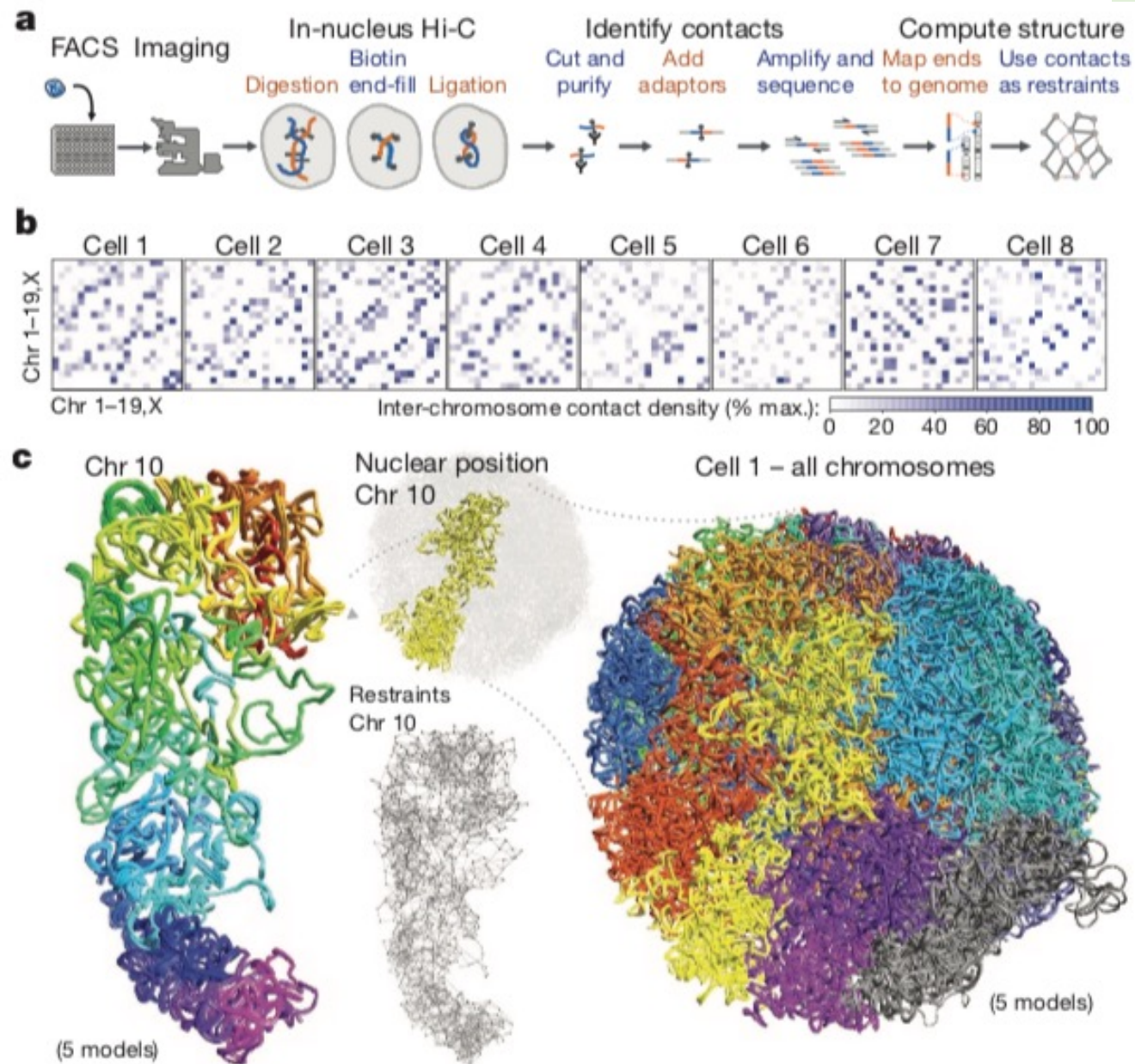
low cell Hi-C



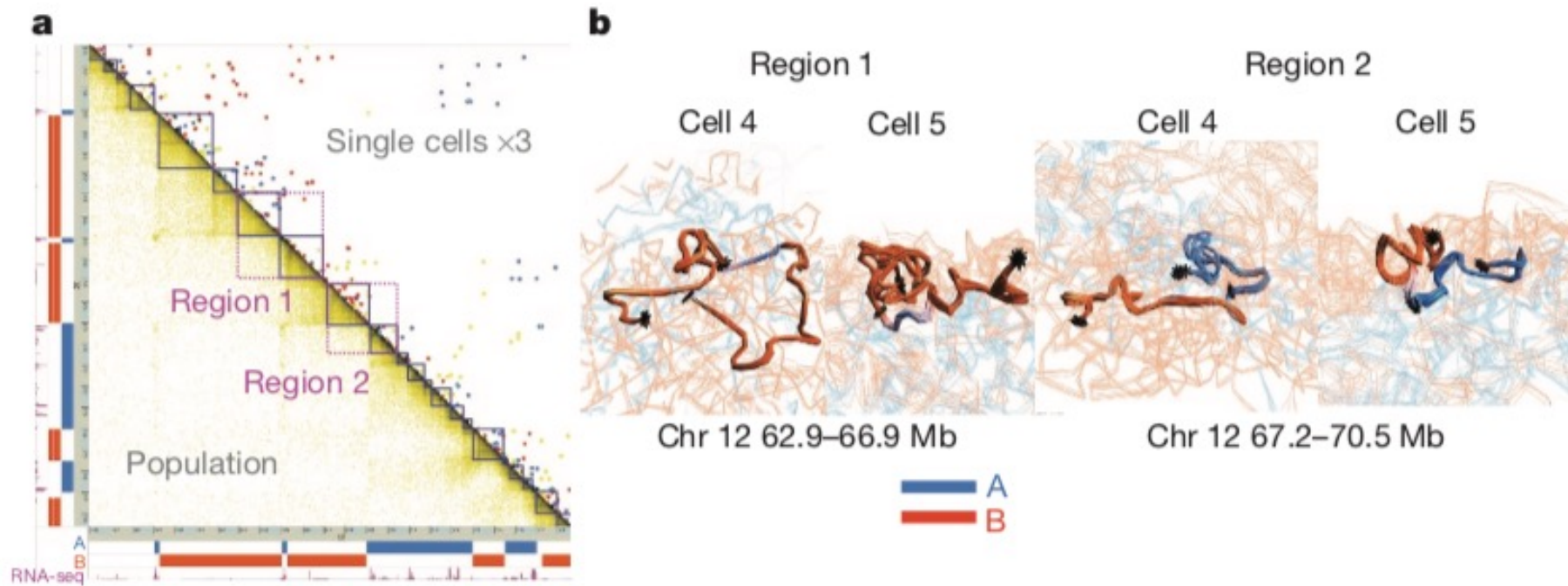
single cell Hi-C



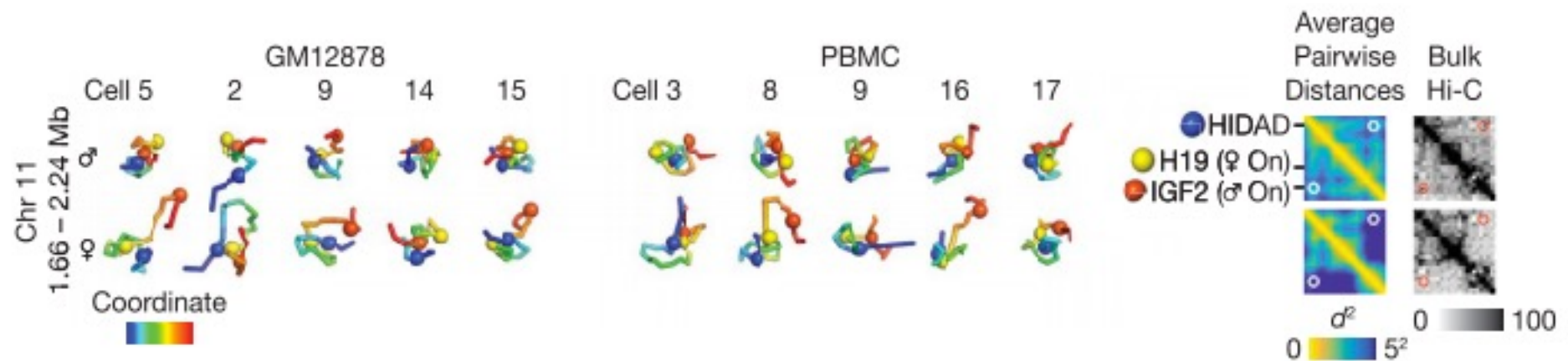
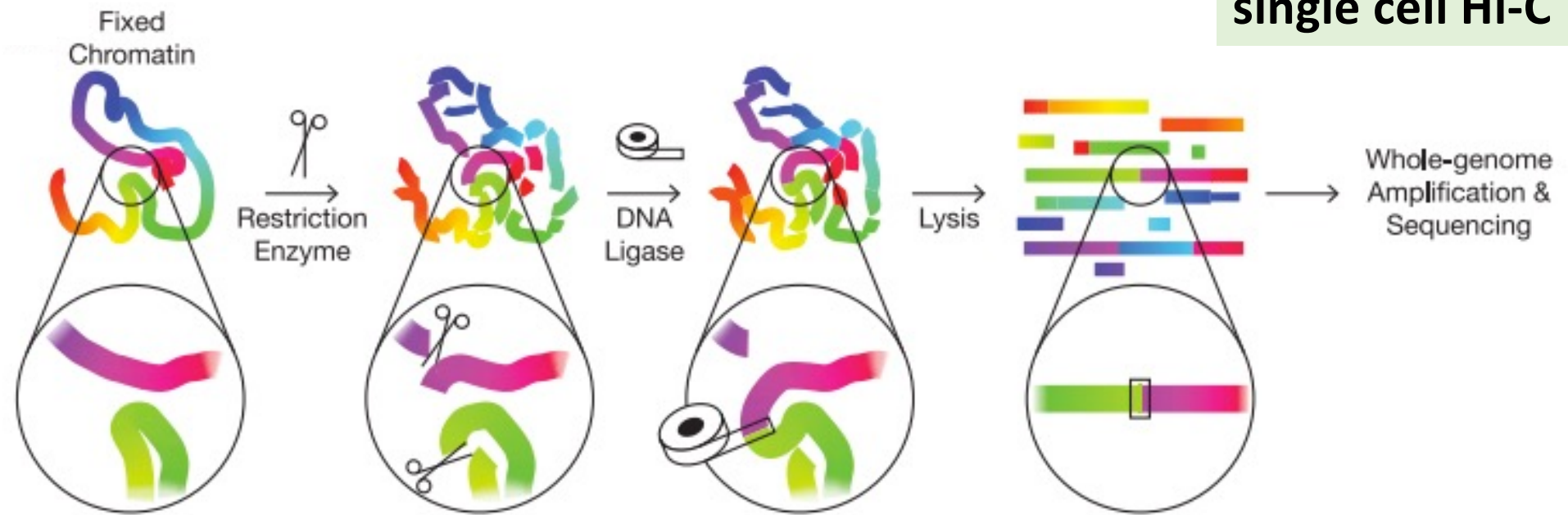
single cell Hi-C



single cell Hi-C



single cell Hi-C





The end